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A neural correlate of individual odor preference in *Drosophila*

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eLife Assessment

What makes one member of the species behave differently from another? This is a core problem in behavioral neuroscience. This **valuable** study seeks an answer for the specific case of the fruit fly expressing preferences for one odor over another. By a combination of behavioral measurements, neurophysiology, and network modeling, the authors find **solid** evidence for at least one locus of individuality in the peripheral olfactory system.

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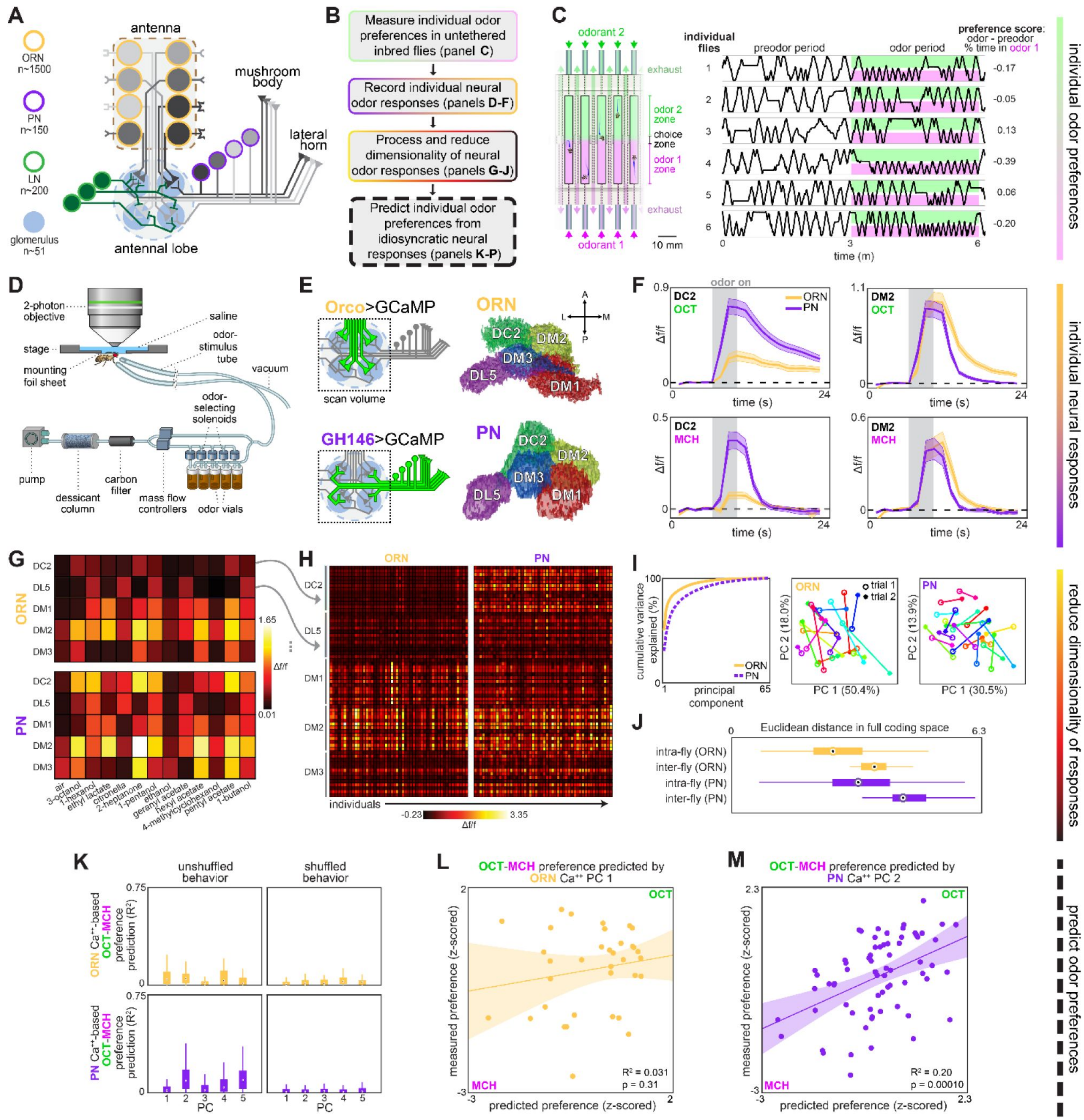
Abstract

Behavior varies even among genetically identical animals raised in the same environment. However, little is known about the circuit or anatomical origins of this individuality. Here, we demonstrate a neural correlate of *Drosophila* odor preference behavior in the olfactory sensory periphery. Namely, idiosyncratic calcium responses in projection neuron (PN) dendrites and densities of the presynaptic protein Bruchpilot in olfactory receptor neuron (ORN) axon terminals correlate with individual preferences in a choice between two aversive odorants. The ORN-PN synapse appears to be a locus of individuality where microscale variation gives rise to idiosyncratic behavior. Simulating microscale stochasticity in ORN-PN synapses of a 3,062 neuron model of the antennal lobe recapitulates patterns of variation in PN calcium responses matching experiments. Conversely, stochasticity in other compartments of this circuit does not recapitulate those patterns. Our results demonstrate how physiological and microscale structural circuit variations can give rise to individual behavior, even when genetics and environment are held constant.

Introduction

Individuality is a fundamental aspect of behavior that is observed even among genetically-identical animals reared in similar environments. We are specifically interested in individuality that is evident as idiosyncratic differences in behavior that persist for much of an animal's lifespan. Such variability is observed across species including round worms (Stern et al., 2017), aphids (Schuett et al., 2011), fish (Laskowski et al., 2022), mice (Freund et al., 2013), and people (Johnson et al., 2010). Small, genetically tractable model species, such as *Drosophila*, are particularly promising for discovering the genetic and neural circuit basis of individual behavior variation. Flies exhibit individuality in many behaviors (Werkhoven et al., 2021), and the mechanistic origins of this variation has been studied for phototactic preference (Kain et al., 2012), temperature preference (Kain et al., 2015), locomotor handedness (Ayroles et al., 2015; Buchanan et al., 2015; de Bivort et al., 2022), object-fixated walking (Linneweber et al., 2020), and odor preference (Honegger et al., 2019). Generally, the neural substrates of individuality are poorly understood, though in a small number of instances nanoscale circuit correlates of individual behavioral biases have been identified (Lillvis et al., 2022; Linneweber et al., 2020; Skutt-Kakaria et al., 2019). We hypothesize that as sensory cues are encoded and transformed to produce motor outputs, their representation in the nervous system becomes increasingly idiosyncratic and predictive of individual behavioral responses. We seek to identify "loci of individuality" – sites at which this idiosyncrasy emerges.

Olfaction in the fruit fly *Drosophila melanogaster* is an amenable sensory system for identifying loci of individuality, as 1) individual odor preferences can be recorded readily, 2) neural representations of odors can be measured via calcium imaging, 3) the circuit elements of the pathway are well-established, and 4) a deep genetic toolkit enables mechanism-probing experiments. The neuroanatomy of the olfactory system, from the antenna through its first central-brain processing neuropil, the antennal lobe (AL), is broadly stereotyped across individuals (Couto et al., 2005; Grabe et al., 2015; Wilson et al., 2004). The AL features ∼50 anatomically identifiable microcircuits called glomeruli (**Figure 1A**). Each glomerulus represents an odor-coding channel and receives axon inputs from olfactory receptor neurons (ORNs) expressing the same olfactory receptor gene (de Bruyne et al., 2001). Uniglomerular projection neurons (PNs) carry odor information from each glomerulus deeper into the brain (Jeanne and Wilson, 2015). AL-intrinsic local neurons (LNs) project among glomeruli (Chou et al., 2010) and modulate odor representations (Wilson and Laurent, 2005). Glomerular organization is a key stereotype of the AL; using glomeruli as landmarks, one can identify comparable ORN axons and PNs across individuals.



A
antenna
ORN n~1500
PN n~150
LN n~200
glomerulus n~51
mushroom body
lateral horn
antennal lobe
B
Measure individual odor preferences in untethered inbred flies (panel C)
Record individual neural odor responses (panels D-F)
Process and reduce dimensionality of neural odor responses (panels G-J)
Predict individual odor preferences from idiosyncratic neural responses (panels K-P)
C
odorant 2
exhaust
odor 2 zone
choice zone
odor 1 zone
odorant 1
10 mm
individual flies
preodor period
odor period
preference score: odor - preodor % time in odor 1
time (m)
individual odor preferences
D
2-photon objective
stage
mounting foil sheet
saline
odor-stimulus tube
vacuum
odor-selecting solenoids
pump
dessicant column
carbon filter
mass flow controllers
odor vials
E
Orco>GCaMP
GH146>GCaMP
scan volume
ORN
PN
DC2
DM2
DM3
DL5
DM1
F
odor on
time (s)
individual neural responses
G
H
individuals
I
cumulative variance explained (%)
principal component
trial 1
trial 2
PC 1 (50.4%)
PC 2 (18.0%)
PC 1 (30.5%)
PC 2 (13.9%)
J
Euclidean distance in full coding space
intra-fly (ORN)
inter-fly (ORN)
intra-fly (PN)
inter-fly (PN)
reduce dimensionality of responses
K
unshuffled behavior
shuffled behavior
ORN Ca++-based OCT-MCH preference prediction (R²)
PN Ca++-based OCT-MCH preference prediction (R²)
PC
L
OCT-MCH preference predicted by ORN Ca++ PC 1
measured preference (z-scored)
predicted preference (z-scored)
R² = 0.031
p = 0.31
M
OCT-MCH preference predicted by PN Ca++ PC 2
R² = 0.20
p = 0.00010
predict odor preferences

Figure 1.

Idiosyncratic calcium dynamics predict individual odor preferences.

(A) Olfactory circuit schematic. Olfactory receptor neurons (ORNs, peach outline) and projection neurons (PNs, plum outline) are comprised of ~51 classes corresponding to odor receptor response channels. ORNs (gray shading) sense odors in the antennae and synapse on dendrites of PNs of the same class in ball-shaped structures called glomeruli located in the antennal lobe (AL). Local neurons (LNs, green outline) mediate interglomerular cross-talk and presynaptic inhibition, amongst other roles (Olsen and Wilson, 2008; Yaksi and Wilson, 2010). Odor signals are normalized and whitened in the AL before being sent to the mushroom body and lateral horn for further processing. Schematic adapted from Honegger et al., 2019 **(B)** Experiment outline. **(C)** Odor preference behavior tracking setup (reproduced from Honegger, et al. 2019) and example individual fly ethograms. OCT (green) and MCH (magenta) were presented for 3 minutes. **(D)** Head-fixed 2-photon calcium imaging and odor delivery setup (reproduced from Honegger et al., 2019) **(E)** Orco and GH146 driver expression profiles (left) and example segmentation masks (right) extracted from 2-photon calcium images for a single fly expressing Orco>GCaMP6m (top, expressed in a subset of all ORN classes) or GH146>GCaMP6m (bottom, expressed in a subset of all PN classes). **(F)** Time-dependent Δf/f for glomerular odor responses in ORNs (peach) and PNs (plum) averaged across all individuals: DC2 to OCT (upper left), DM2 to OCT (upper right), DC2 to MCH (lower left), and DM2 to OCT (lower right). Shaded error bars represent S.E.M. **(G)** Peak Δf/f for each glomerulus-odor pair averaged across all flies. **(H)** Individual neural responses measured in ORNs (left) or PNs (right) for 50 flies each. Columns represent the average of up to 4 odor responses from a single fly. Each row represents one glomerulus-odor response pair. Odors are the same as in panel (G). **(I)** Principal component analysis of individual neural responses. Fraction of variance explained versus principal component number (left). Trial 1 and trial 2 of ORN (middle) and PN (right) responses for 20 individuals (unique colors) embedded in PC 1-2 space. **(J)** Euclidean distances between glomerulus-odor responses within and across flies measured in ORNs (n = 65 flies) and PNs (n = 122 flies). Distances calculated without PCA compression. Points represent the median value, boxes represent the interquartile range, and whiskers the range of the data. **(K)** Bootstrapped R^2 of OCT-MCH preference prediction from each of the first 5 principal components of neural activity measured in ORNs (top, all data) or PNs (bottom, training set). **(L)** Measured OCT-MCH preference versus preference predicted from PC 1 of ORN activity (n = 35 flies). **(M)** Measured OCT-MCH preference versus preference predicted from PC 2 of PN activity in n = 69 flies using a model trained on a training set of n = 47 flies (see **Figure 2 – figure supplement 1C-D** for train/test flies analyzed separately). Shaded regions in L,M are the 95% CIs of the fit estimated by bootstrapping.

Individual flies differ in their PN calcium responses to identical odor stimuli, as well as their odor-vs-odor preference choices (Honegger et al., 2019). Several possible determinants of individual odor preference can already be hypothesized for the fly olfactory circuit (Rihani and Sachse, 2022). The extent of preference variability depends on dopamine and serotonergic modulation (Honegger et al., 2019). Neuromodulation clearly plays a role in the regulation of behavioral individuality (Maloney, 2021), but its effects vary by modulator and behavior (de Bivort et al., 2022; Kain et al., 2012). With respect to wiring variation, the number of ORNs and PNs innervating a given glomerulus varies within hemispheres (Tobin et al., 2017) and across individuals (Grabe et al., 2016; Schlegel et al., 2020), as does the glomerulus-innervation pattern of individual LNs (Chou et al., 2010). Subpopulations of LNs and PNs express variable serotonin receptors (Sizemore and Dacks, 2016), so the effects of neuromodulation and wiring may interact to influence individuality. Little is known about possible molecular or nanoscale correlates of individual behavioral bias. Thus, individual odor preference could have its origins in many potential mechanisms, ranging from circuit wiring to modulation to neuronal intrinsic properties.

Outside the olfactory system, there are a few examples in which microscale circuit variation predicts individual behavioral preference. Wiring asymmetry in an individual fly's dorsal cluster neurons is predictive of the straightness of its object-oriented walking behavior (Linneweber et al., 2020), and left-right asymmetry in the density of presynaptic sites of protocerebral bridge to lateral accessory lobe-projecting neurons predicts an individual fly's idiosyncratic turning bias (Skutt-Kakaria et al., 2019). The number of synaptic connections from the pC2l to pIP10 neurons correlates with male song rate during courtship (Lillvis et al., 2022), and the presence of ectopic branches in neurons of the T2 hemilineage predicts delayed spontaneous flight initiation (Mellert et al., 2016).

In this work, we sought to identify loci of individuality by measuring odor preferences and neural responses to odors in the same individuals and determining the extent to which the latter predicted the former. We found that idiosyncratic calcium responses in PNs were correlated with individual preferences in a choice between two aversive odorants. Examining a molecular component presynaptic to PNs, we found that the density of the scaffolding protein Bruchpilot also predicts odor preference. To unify these results and connect wiring variation to circuit outputs and behavior, we simulated developmental variation in a 3,062-neuron spiking model of the antennal lobe. Simulated stochasticity in the ORN-PN synapse recapitulated our empirical findings. Thus, we identified the ORN-PN synapse as a likely locus of individuality in fly odor preference, demonstrating that behaviorally-relevant variation in neural circuits can be found in the sensory periphery at the nanoscale.

Results

Individual flies encode odors idiosyncratically

Focusing on behavioral variation within a genotype, we used isogenic animals expressing the fluorescent calcium reporter GCamp6m (Chen et al., 2013) in either of the two most peripheral neural subpopulations of the *Drosophila* olfactory circuit, ORNs or PNs (**Figure 1E**). We performed head-fixed 2-photon calcium imaging after measuring odor preference in an untethered assay (Honegger et al., 2019) (**Figure 1B-D**, **Figure 1 – figure supplement 1A**). Individual odor preferences are stable over timescales longer than this experiment (**Figure 1 – figure supplement 1B-E**).

We measured volumetric calcium responses in the antennal lobe (AL), where ORNs synapse onto PNs in ~50 discrete microcircuits called glomeruli (**Figure 1A**) (Couto et al., 2005; Grabe et al., 2015). Flies were stimulated with a panel of 12 odors plus air (**Figure 1D**, **Figure 1 – figure supplement 2**) and *k*-means clustering was used to automatically segment the voxels of 5 glomeruli from the resulting 4-D calcium image stacks (**Figure 1E**, **Figure 1 – figure supplement 5**, Materials and Methods) (Couto et al., 2005). Both ORN and PN odor responses were roughly stereotyped across individuals (**Figure 1G,H**), but also idiosyncratic (Honegger et al., 2019).

Responses in PNs appeared to be more idiosyncratic than ORNs (**Figure 1J**); a logistic linear classifier decoding fly identity from glomerular responses was more accurate when trained on PN than ORN responses (**Figure 1 – figure supplement 6A**). While the responses of single ORNs are known to vary more than those of single PNs (Wilson, 2013), our recordings capture the total response of all ORNs or PNs in a glomerulus. This might explain our observation that ORNs exhibited less idiosyncrasy than PNs. PN responses were more variable within flies, as measured across the left and right hemisphere ALs, compared to ORN responses (**Figure 1 – figure supplement 6C**; $p < 2x10^{-5}$, Mann-Whitney *U* test), suggesting that odor representations become more divergent farther from the sensory periphery.

PN, but not ORN, responses predict odor-vs-odor preference

Next we analyzed the relationship of idiosyncratic coding to odor preference, by asking in which neurons (if any) did calcium responses predict individual preferences of flies choosing between two aversive monomolecular odors: 3-octanol (OCT) and 4-methylcyclohexanol (MCH).

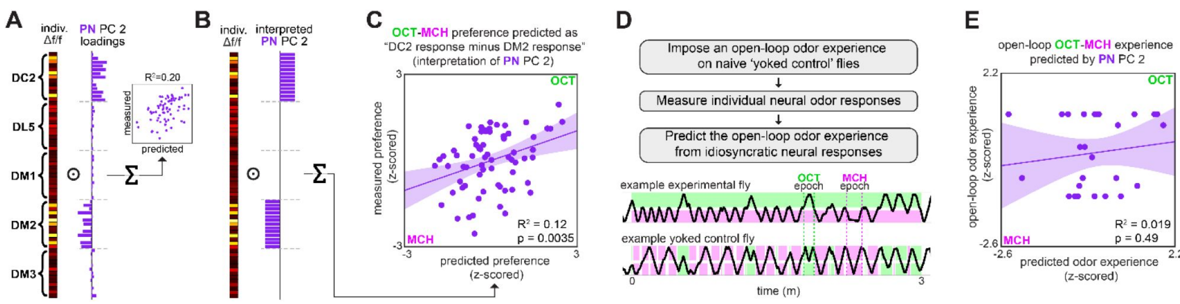


Figure 2.

Variation in relative glomerular responses explains individual odor preference.

(A) PC 2 loadings of PN activity for flies tested for OCT-MCH preference (n = 69 flies). **(B)** Interpreted PN PC 2 loadings. **(C)** Measured OCT-MCH preference versus preference predicted by the average peak PN response in DM2 minus DC2 across all odors (n = 69 flies). **(D)** Yoked-control experiment outline and example behavior traces. Experimental flies are free to move about tunnels permeated with steady state OCT and MCH flowing into either end. Yoked-control flies are delivered the same odor at both ends of the tunnel which matches the odor experienced at the nose of the experimental fly at each moment in time. **(E)** Imposed odor experience versus the odor experience predicted from PC 2 of PN activity (n = 27 flies) evaluated on the model trained from data in **Figure 1M**. Shaded regions in C,E are the 95% CIs of the fit estimated by bootstrapping.

Because we could potentially predict preference (a single value) using numerous glomerular-odor predictors, and had a limited number of observations (dozens), we used dimensionality reduction to hold down the number of comparisons we made. We computed the principal components (PCs) of the glomerulus-odor responses (in either ORNs or PNs) across individuals (**Figure 1G-I**; **Figure 1 – figure supplement 3**, **Figure 1 – figure supplement 8**) and fit linear models to predict the behavior of individual flies from their values on the odor response PCs. No PCs of ORN neural activity could linearly predict OCT-MCH preference beyond the level of shuffled controls (n = 35 flies) (**Figure 1K,L**). The best ORN PC model only predicted odor-vs-odor behavior with a nominal R^2 of 0.031. In contrast, PC 2 of PN activity was a statistically significant predictor of odor preference, accounting for 15% of preference variance in a training set of 47 flies (p = 0.0063; **Figure 2 – figure supplement 1C**) and 31% of preference variance on test data of flies (p = 0.0069; **Figure 2 – figure supplement 1D**). These p-values remaining significant at α = 0.05 following a Bonferroni correction for 5 comparisons. Combined train/test statistics (r = 0.45; p = 0.0001) are presented in **Figure 1K,M**. Thus, idiosyncratic PN calcium predicts odor vs. odor preference.

We conducted a follow-up analysis to contextualize the finding of calcium PCs predicting odor preference with r = ~0.4. This value is lower than 1.0 due to at least two factors: 1) any non-linearity in the relationship between calcium responses and behavior, and 2) sampling error in, and temporal instability of, behavior and calcium responses over the duration of the experiment. A lower bound on the latter can be estimated from the repeatability of behavioral measures over time (**Figure 1 – figure supplement 1B-E**). We performed a statistical analysis to roughly estimate model performance if there were no sampling error or drift in the measurement of behavior and calcium responses (**Figure 1 – figure supplement 9**; Materials and Methods). This analysis suggests that the measured correlation between calcium and behavior (ρ_{signal}) would be 0.68 in the absence of sampling error and temporal instability, but the uncertainty in this estimate is high (90% CI: 0.24-0.95).

We additionally assessed the extent to which idiosyncratic calcium responses in ORNs or PNs could predict preference between air and a single aversive odor (OCT). We found a suggestive correlate: PC 1 of ORN calcium responses explained 23% of preference variance (n = 30 flies, p = 0.0099, **Figure 1 – figure supplement 10B**), but this association was dominated by a single outlier (R^2 of 0.078, p = 0.14 with the outlier removed).

We next sought a biological understanding of the models associating calcium responses with odor preference. The loadings of the ORN and PN PCs indicate that variation across individuals was correlated at the level of glomeruli much more strongly than odorant (**Figure 1H**; **Figure 1 – figure supplements 3**, **8**). This suggests that stochastic variation in the olfactory circuit results in individual-level fluctuations in the responses of glomeruli-specific rather than odor-specific responses. In the odor-vs-odor preference model, the loadings of PC2 of PN calcium responses contrast the responses of the DM2 and DC2 glomeruli with opposing weights (**Figure 2A**), suggesting that the activation of DM2 relative to DC2 predicts the likelihood of a fly preferring OCT to MCH. Indeed, a linear model constructed from the average DM2 minus average DC2 PN response (**Figure 2B**) showed a statistically significant correlation with preference for OCT versus MCH (R^2 = 0.12; p = 0.0035; **Figure 2C**). The model slope coefficient was negative (**Table 1**), indicating that greater activation of DM2 vs DC2 correlates with preference for MCH. With respect to odor-vs-odor behavior, we conclude that the relative responses of DM2 vs DC2 in PNs compactly predict an individual's preference.

Odor experience has been shown to modulate subsequent AL responses (Golovin and Broadie, 2016; Iyengar et al., 2010; Sachse et al., 2007). This raises the possibility that our models were actually predicting individual flies' past odor experiences (i.e., the specific pattern of odor stimulation flies received in the behavioral assay) rather than their preferences. To address this, we imposed the specific odor experiences of previously tracked flies (in the odor-vs-odor assay) on

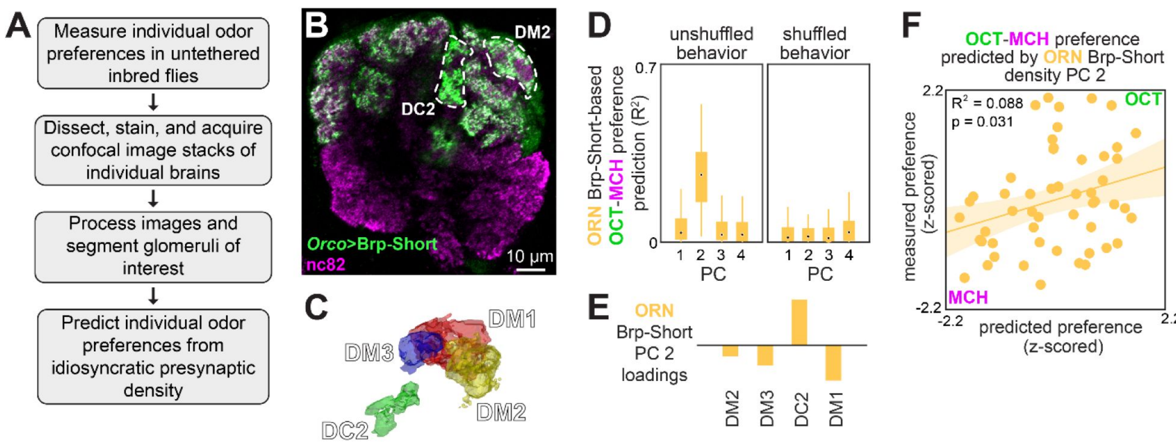


Figure 3.

Idiosyncratic presynaptic marker density in DM2 and DC2 predicts OCT-MCH preference.

(A) Experiment outline. **(B)** Example slice from a z-stack of the antennal lobe expressing Orco>Brp-Short (green) with DC2 and DM2 visible (white dashed outline). nc82 counterstain (magenta). **(C)** Example glomerulus segmentation masks extracted from an individual z-stack. (A) **(D)** Bootstrapped R^2 of OCT-MCH preference prediction from each of the first 4 principal components of Brp-Short density measured in ORNs (training set, n = 22 flies). **(E)** PC 2 loadings of Brp-Short density. **(F)** Measured OCT-MCH preference versus preference predicted from PC 2 of ORN Brp-Short density in n = 53 flies using a model trained on a training set of n = 22 flies (see **Figure 3 – figure supplement 1** for train/test flies analyzed separately).

Behavior Measured	Neural Predictor	Figure Panel	n	β_0	β_1	R^2	p-value
OCT vs. AIR	PN Calcium PC 1	Figure 2 – figure supplement 1A	18	-0.26	-0.079	0.16	0.099
OCT vs. AIR	PN Calcium Average all dimensions	Figure 1 – figure supplement 10I	53	-0.051	-0.38	0.098	0.022
OCT vs. AIR	ORN Calcium PC 1	Figure 1 – figure supplement 10B	30	-0.29	-0.053	0.23	0.007
OCT vs. AIR	ORN Calcium Average all dimensions	Figure 1 – figure supplement 10E	30	-0.032	-0.71	0.25	0.005
OCT vs. MCH	PN Calcium PC 2	Figure 2 – figure supplement 1C	47	-0.058	-0.081	0.15	0.006
OCT vs. MCH	PN Calcium DM2 - DC2 (% difference)	Figure 2I	69	-0.032	-0.0018	0.12	0.004
OCT vs. MCH	ORN Calcium PC 1	Figure 1L	35	-0.14	-0.027	0.031	0.32
OCT vs. MCH	ORN Brp-Short PC 2 (train data only)	Figure 3 – figure supplement 1I	22	-0.087	0.017	0.22	0.028
OCT vs. MCH	ORN Brp-Short PC 2 (all data)	Figure 3F	53	-0.019	0.012	0.088	0.031

Table 1

Calcium & Brp-Short – behavior model statistics

naive “yoked” control flies (**Figure 2D**) and measured PN odor responses of the yoked flies. Applying the PN PC 2 model to the yoked calcium responses did not predict flies’ odor experience (R^2 = 0.019, p = 0.49; **Figure 2E**). This is consistent with PN calcium responses predicting odor preference rather than odor experience.

(Mazor and Laurent, 2005) found that PN response transients, rather than fixed points, contain more odor identity information. We therefore asked at which times during odor presentation an individual’s neural responses could best predict odor preference. Applying our calcium-to-behavior models (PN PC2-odor-vs-odor, as well as ORN PC1-odor-vs-air, PN PC1-odor-vs-air) to the time-varying calcium signals, we found that in all cases, behavior prediction rose during odor delivery (**Figure 2 – figure supplement 2**). In ORNs, the predictive accuracy remained high after odor offset, whereas in PNs it declined. The times during which calcium responses predicted individual behavior generally aligned to the times during which a linear classifier could decode odor identity from neuronal responses (**Figure 2 – figure supplement 2D**), suggesting that idiosyncrasies in odor encoding predict individual preferences.

Variation in a presynaptic scaffolding protein predicts odor preference

We next investigated how structural variation in the nervous system might relate to idiosyncratic behavior. Because PN, but not ORN, calcium responses predicted odor-vs-odor preference, we hypothesized that a circuit element between ORNs to PNs could confer onto PNs behaviorally-relevant physiological idiosyncrasies absent in ORNs. We therefore imaged presynaptic T-bar density in ORNs using transgenic mStrawberry-tagged Brp-Short, immunohistochemistry and confocal microscopy (Mosca and Luo, 2014) after measuring individual preference for OCT versus MCH (**Figure 3A**). Brp-Short density was quantified as total fluorescence intensity / glomerulus volume for 4 of the 5 focus glomeruli (**Figure 3B**, **Figure 3 – figure supplement 1A-F**; DL5 was not readily segmentable in our confocal samples). We chose this metric as we found it could be used to predict individual behavioral biases in a previous study (Skutt-Kakaria et al., 2019). This measure was consistent across hemispheres (**Figure 3 – figure supplement 1C**), while also showing variation among individuals, like calcium responses.

To relate presynaptic structural variation and behavior, we used the same analytical approach as we had for calcium responses. PCs 1 and 2 of Brp-Short density had notable similarities to those of the calcium responses: PC 1 was positive across glomeruli and PC 2 exhibited a sign contrast between DC2 loadings and all other glomerulus loadings (**Figure 3 – figure supplement 1G**). As in the PN calcium response models, PC 2 of Brp-Short density was the best predictor of odor-vs-odor preferences in training data (**Figure 3D-E**, **Figure 3 – figure supplement 1I**, R^2 = 0.22, n = 22 flies, p = 0.028) and for test data (**Figure 3 – figure supplement 1J**, R^2 = 0.078, n = 31 flies, p = 0.13; statistics from combined train and test data: R^2 = 0.088, n = 53 flies, p = 0.031; **Figure 3F**; median ρ_{signal} 0.39, 90% CI 0.03-0.86). To better understand the microstructural basis of our Brp-Short density metric, we performed paired behavior and expansion microscopy (Asano et al., 2018; Gao et al., 2019) in flies expressing Brp-Short specifically in DC2-projecting ORNs (Supplementary Video 3). Expansion yielded a ~4-fold increase in linear resolution, allowing imaging of individual Brp-Short puncta (**Figure 3 – figure supplement 1K**). While the sample size (n = 8) of this imaging pipeline was insufficient for a formal statistical analysis, the trend between Brp-Short density in DC2 (measured as individual puncta / glomerular volume) and odor-vs-odor preference was more consistent with a positive correlation than other metrics, such as median puncta volume (**Figure 3 – figure supplement L,M**).

The best presynaptic density models are less predictive of behavior than the best calcium response models (R^2 = 0.088 vs R^2 = 0.22; ρ_{signal} = 0.39 and 0.68, respectively; **Figure 2 – figure supplement 1C,D** vs **Figure 3 – figure supplement 1I,J**), suggesting that presynaptic density variation is not the full explanation of calcium response variability. Nevertheless, differences in presynaptic inputs to PNs may contribute to variation in the calcium dynamics of those neurons, in turn giving rise to individual preferences for OCT versus MCH.

Developmental stochasticity in a simulated AL recapitulates empirical PN response variation

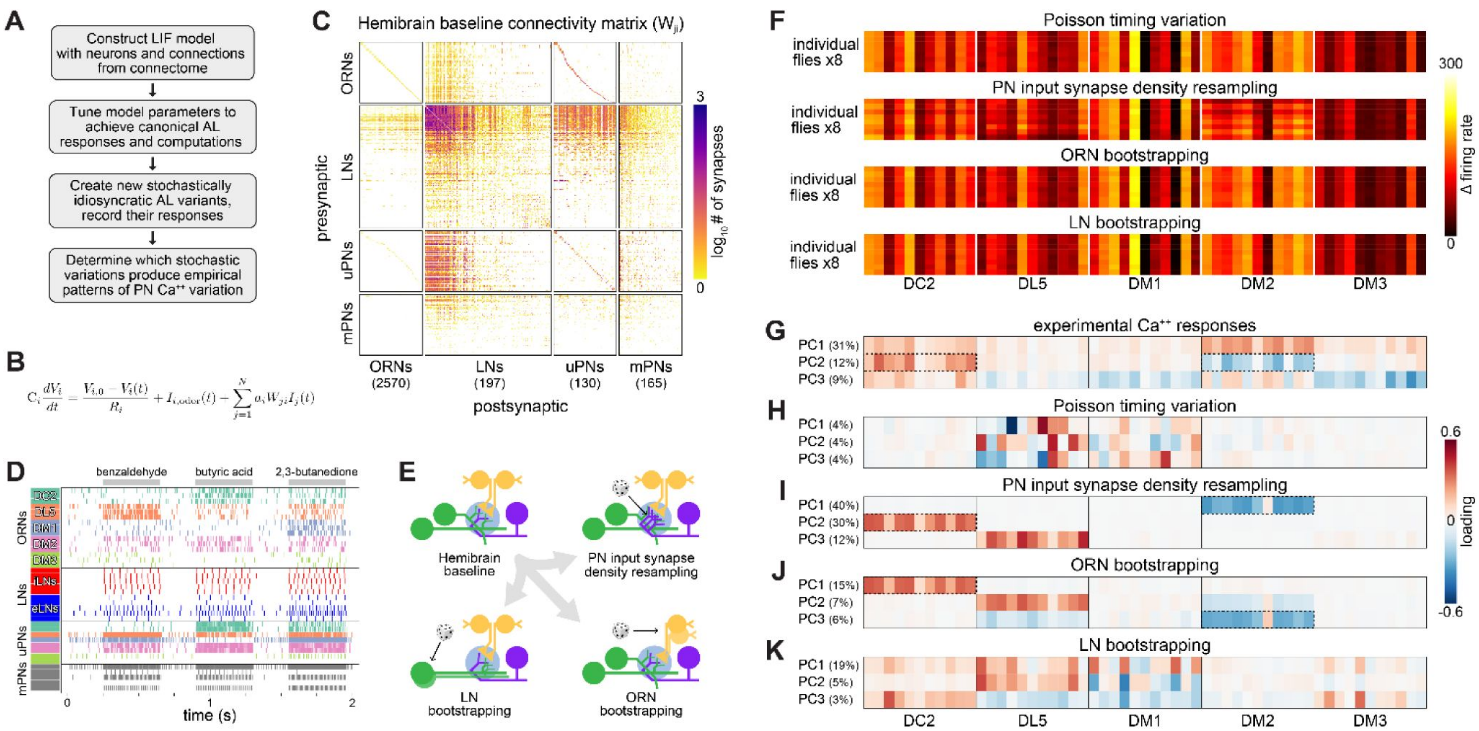
Finally, we sought an integrative understanding of how synaptic variation plays out across the olfactory circuit to produce behaviorally-relevant physiological variation. We developed a leaky-integrate-and-fire model of the entire AL, comprising 3,062 spiking neurons and synaptic connectivity taken directly from the *Drosophila* hemibrain connectome (Scheffer et al., 2020).

After tuning the model to perform canonical AL computations, we introduced different kinds of stochastic variations to the circuit and determined which (if any) would produce the patterns of idiosyncratic PN response variation observed in our calcium imaging experiments (**Figure 4A**). This approach assesses potential mechanisms linking developmental variation in synapses to physiological variation that apparently drives behavioral individuality.

The biophysical properties of neurons in our model (**Figure 4B**, **Table 2**) were determined by published electrophysiological studies (See *Voltage model* in Materials and Methods) and were similar to those used in previous fly models (Kakaria and de Bivort, 2017; Pisokas et al., 2020). The polarity of neurons was determined largely by their cell type (ORNs are excitatory, PNs predominantly excitatory, and LNs predominantly inhibitory – explained further in Materials and Methods). The strength of synaptic connections between any pair of AL neurons was given by the hemibrain connectome (Scheffer et al., 2020) (**Figure 4C**). Odor inputs were simulated by injecting current into ORNs to produce spikes in those neurons at rates that match published ORN-odor recordings (Münch and Galizia, 2016), and the output of the system was recorded as the firing rates of PNs during odor stimulation (**Figure 4D**). At this point, there remained only four free parameters in our model, the relative sensitivity (postsynaptic current per upstream action potential) of each AL cell type (ORNs, PNs, excitatory LNs and inhibitory LNs). We explored this parameter space manually, and identified a configuration in which AL simulation (**Figure 4 – figure supplement 1**) recapitulated four canonical properties seen experimentally (**Figure 4 – figure supplement 2**): 1) typical firing rates at baseline and during odor stimulation (Bhandawat et al., 2007; Dubin and Harris, 1997; Jeanne and Wilson, 2015; Seki et al., 2010), 2) a more uniform distribution of PN firing rates compared to ORN rates (Bhandawat et al., 2007), 3) greater separation of PN odor representations compared to ORN representations (Bhandawat et al., 2007), and 4) a sub-linear transfer function between ORNs and PNs (Bhandawat et al., 2007). Thus, our simulated AL appeared to perform the fundamental computations of real ALs, providing a baseline for assessing the effects of idiosyncratic variation.

We simulated stochastic individuality in the AL circuit in two ways (**Figure 4E**): 1) glomerular-level variation in PN input-synapse density (reflecting a statistical relationship observed between glomerular volume and synapse density in the hemibrain, **Figure 4 – figure supplement 4**), and 2) bootstrapping of neuronal compositions within cell types (reflecting variety in developmental program outcomes for ORNs, PNs, etc.). Supplementary Video 4 shows the diverse connectivity matrices attained under these resampling approaches. We simulated odor responses in thousands of ALs made idiosyncratic by these sources of variation, and in each, recorded the firing rates of PNs when stimulated by the 12 odors from our experimental panel (**Figure 4F**, **Figure 4 – figure supplement 1**).

To determine which sources of variation produced patterns of PN coding variation consistent with our empirical measurements, we compared principal components of PN responses from real idiosyncratic flies to those of simulated idiosyncratic ALs. Empirical PN responses are strongly correlated at the level of glomeruli (**Figure 4G**; **Figure 1 – figure supplement 8**). As a positive



$$C_i \frac{dV_i}{dt} = \frac{V_{i,0} - V_i(t)}{R_i} + I_{i,\mathrm{odor}}(t) - \sum_{j=1}^{N} \alpha_i W_{ji} I_j(t)$$

Figure 4.

Simulation of developmentally stochastic olfactory circuits (A)

AL modeling analysis outline. **(B)** Leaky-integrator dynamics of each simulated neuron. When a neuron's voltage reaches its firing threshold, a templated action potential is inserted, and downstream neurons receive a postsynaptic current. See *Antennal Lobe modeling* in Materials and Methods. **(C)** Synaptic weight connectivity matrix, derived from the hemibrain connectome (Scheffer et al., 2020). **(D)** Spike raster for randomly selected example neurons from each AL cell type. Colors indicate ORN/PN glomerular identity and LN polarity (i = inhibitory, e = excitatory). **(E)** Schematic illustrating sources of developmental stochasticity as implemented in the simulated AL framework. See Supplementary Video 4 for the effects of these resampling methods on the synaptic weight connectivity matrix. **(F)** PN glomerulus-odor response vectors for 8 idiosyncratic ALs subject to Input spike Poisson timing variation, PN input synapse density resampling, and ORN and LN population bootstrapping. **(G)** Loadings of the principal components of PN glomerulus-odor responses as observed across experimental flies (top). Dotted outlines highlight loadings selective for the DC2 and DM2 glomerular responses, which underlie predictions of individual behavioral preference. **(H-K)** As in (G) for simulated PN glomerulus-odor responses subject to Input spike Poisson timing variation, PN input synapse density resampling, and ORN and LN population bootstrapping. See **Figure 4 – figure supplement 5** for additional combinations of idiosyncrasy methods. In (F-K) the sequence of odors within each glomerular block is: OCT, 1-hexanol, ethyl-lactate, 2-heptanone, 1-pentanol, ethanol, geranyl acetate, hexyl acetate, MCH, pentyl acetate and butanol.

Parameter	ORNs	LNs	PNs
Membrane resting potential	-70 mV (Dubin and Harris, 1997)	-50 mV (Seki et al., 2010)	-55 mV (Jeanne and Wilson, 2015)
Action potential threshold	-50 mV (Dubin and Harris, 1997)	-40 mV (Seki et al., 2010)	-40 mV (Jeanne and Wilson, 2015)
Action potential minimum	-70 mV (Cao et al., 2016)	-60 mV (Seki et al., 2010)	-55 mV (Jeanne and Wilson, 2015)
Action potential maximum	0 mV (Dubin and Harris, 1997)	0 mV (Seki et al., 2010)	-30 mV (Wilson and Laurent, 2005)
Action potential duration	2 ms (Jeanne and Wilson, 2015)	4 ms (Seki et al., 2010)	2 ms (Jeanne and Wilson, 2015)
Membrane capacitance	73 pF (assumed = PNs)	64 pF (Huang et al., 2018)	73 pF (Huang et al., 2018)
Membrane resistance	1.8 GOhm (Dubin and Harris, 1997)	1 GOhm (Seki et al., 2010)	0.3 GOhm (Jeanne and Wilson, 2015)

Table 2

Typical electrophysiology features of AL cell types, used as model parameters

control that the model can recapitulate this empirical structure, resampling PN input-synapse density across glomeruli produced PN response correlations strongly organized by glomerulus (**Figure 4I**). As a negative control, variation in PN responses due solely to poisson timing of ORN input spikes (i.e., absent any circuit idiosyncrasy) was not organized at the glomerular level (**Figure 4H**). Strikingly, bootstrapping ORN membership yielded a strong glomerular organization in PN responses (**Figure 4J**). The loadings of the top PCs under ORN bootstrapping are dominated by responses of a single glomerulus to all odors, including DM2 and DC2. This is reminiscent of PC2 of PN calcium responses, with prominent (opposite sign) loadings for DM2 and DC2. Bootstrapping LNs, in contrast, produced much less glomerular organization (**Figure 4K**), with little resemblance to the loadings of the empirical calcium PCs. The PCA loadings for simulated PN responses under all combinations of cell type bootstrapping and PN input-synapse density resampling are given in **Figure 4 – figure supplement 5**.

DM2 and DC2 (also DL5) stand out in the PCA loadings under PN input-synapse density resampling and ORN bootstrapping (**Figure 4I,J**), suggesting that behaviorally-relevant PN coding variation is recapitulated in this modeling framework. To formalize this analysis, for each idiosyncratic AL, we computed a "behavioral preference" by applying the PN PC2 linear model (**Figure 1K,M**) to simulated PN responses. We then determined how accurately a linear classifier could distinguish OCT- vs MCH-preferring ALs in the space of the first 3 PCs of PN responses (**Figure 4 – figure supplement 6**). High accuracy was attained under PN input-synapse density resampling and ORN bootstrapping (sources of circuit variation that produced PN response loadings highlighting DM2 and DC2). Thus, developmental variability in ORN populations may drive patterns of PN physiological variation that in turn drive individuality in odor-vs-odor choice behavior.

Discussion

We found an element of the *Drosophila* olfactory circuit in which patterns of physiological activity predict individual behavioral preferences. This circuit element can be considered a locus of individuality, as it appears to harbor the origins of idiosyncratic preferences among isogenic animals reared in the same environment. Specifically, the difference in the activation of PNs in DC2 and DM2 during odor exposure predicts idiosyncratic OCT-vs-MCH preferences (**Figures 1**, **2**). This circuit element is in the olfactory sensory periphery, suggesting that behavioral idiosyncrasy arises early in the sensorimotor transformation.

Correlating behavior to microscopic circuit features at the individual level is challenging (Koulakov et al., 2005). Measurements of both calcium responses and preference behavior are noisy. Calcium recordings are slow to acquire, making it hard to achieve sample sizes sufficient for machine-learning discovery of correlations with behavior. We conducted three major experiments (paired odor-vs-odor preference and calcium recordings, odor-vs-air preference and calcium recordings, and odor-vs-odor and Brp-Short imaging), each with training and test sets on the scale of 20-60 individuals each. This allowed us to do some limited statistical discovery of correlations, which we restrained by conducting at most five exploratory correlation measurements between circuit and behavioral measures. We were particularly struck by the extent to which PN activity could predict preference between two aversive odors. Importantly, we confirmed this by evaluating the PN calcium-behavior model on a test set of flies measured several weeks after the training flies, finding the same statistically robust trend in both data partitions (training set: R^2 = 0.15, n = 47, p = 0.0063; testing set: R^2 = 0.31, n = 22, p = 0.0069; **Figure 2 – figure supplement 1**).

Previous work has found mammalian peripheral circuit areas are predictive of individual behavior (Britten et al., 1996; Michelson et al., 2017; Newsome et al., 1989; Osborne et al., 2005), but this study is among the first (Linneweber et al., 2020; Mellert et al., 2016; Skutt-Kakaria et al., 2019) to link cellular-level circuit variants and individual behavior in the absence of genetic variation. Another key conclusion is that loci of individuality are likely to vary, even within the sensory periphery, with the specific behavioral paradigm (i.e., odor-vs-odor or odor- vs-air). Our ability to predict behavioral preferences was limited by the repeatability of the behavior itself (**Figure 1 – figure supplement 9**). Low persistence of odor preference may be attributable to factors like internal states or plasticity. It may be fruitful in future studies to map circuit elements whose activity predicts trial-to-trial behavioral fluctuations within individuals.

Seeking insight into the molecular basis of behaviorally-relevant physiological variation, we imaged Brp in the axon terminals of the ORN-PN synapse, using confocal and expansion microscopy. Brp glomerular density was a significant predictor of individual odor-vs-odor preferences (**Figure 3**). The strongest predictor of OCT-MCH preference among principal components of Brp-Short density features contrastive loadings between DM2 and other glomeruli, similar to the DM2 - DC2 contrast present in the model that predicts odor preference from PN calcium. This is consistent with the recent finding of a linear relationship between synaptic density and excitatory postsynaptic potentials (Liu et al., 2022) and another study in which idiosyncratic synaptic density in central complex output neurons predicts individual locomotor behavior (Skutt-Kakaria et al., 2019). The predictive relationship between Brp and behavior was weaker than that of PN calcium responses, suggesting there are other determinants, such as other synaptic proteins, neurite morphology, or the influence of idiosyncratic LNs (Chou et al., 2010) modulating the ORN-PN transformation (Nagel et al., 2015).

To integrate our synaptic and physiological results, we implemented a spiking model with 3,062 neurons and synaptic weights drawn directly from the fly connectome (Scheffer et al., 2020) (**Figure 4**). With light parameter tuning, this model recapitulated canonical AL computations, providing a baseline for assessing the effects of idiosyncratic stochastic variation. The apparent variation in odor responses across simulated individuals (**Figure 4F**) is less than that seen in the empirical calcium responses (**Figure 1H**), likely due to 1) biological phenomena missing from the model, 2) the lack of measurement noise, and 3) the fact that our perturbations are applied to the connectome of a single fly. When examining PCA loadings, however, simulating idiosyncratic ALs by varying PN input synapse density or bootstrapping ORNs produced correlated PN responses across odors in DC2 and DM2, matching our experimental results.

These sources of variation specifically implicate the ORN-PN synapse (like our Brp results) as an important substrate for establishing behaviorally-relevant patterns of PN response variation.

The flies used in our experiments were isogenic and reared in standardized laboratory conditions that produce reduced behavioral individuality compared to enriched environments (Akhund-Zade et al., 2019; Körholz et al., 2018; Zocher et al., 2020). Yet, even these conditions yield substantial behavioral individuality. We do not expect variability in the expression of the flies' transgenes to be a major driver of this individuality, as wildtype flies have a similarly broad distribution of odor preferences (Honegger et al., 2019). The ultimate source of stochasticity in this behavior remains a mystery, with possibilities ranging from thermal fluctuations at the molecular scale to macroscopic, but seemingly irrelevant, variations like the exact fill level of the culture media (Honegger and de Bivort, 2018). Developing nervous systems employ various compensation mechanisms to dampen out the effects of these fluctuations (Marder, 2011; Tobin et al., 2017). Behavioral variation may be beneficial, supporting a bet-hedging strategy (Hopper, 1999) to counter environmental fluctuations (Akhund-Zade et al., 2020; Honegger et al., 2019; Kain et al., 2015; Krams et al., 2021). Empirically, the net effect of dampening and accreted ontological (Gomez-Marin and Ghazanfar, 2019) fluctuations is individuals with diverse behaviors. This process unfolds across all levels of biological regulation. Just as PN response variation appears to be partially rooted in glomerular Brp variation, the latter has its own molecular roots, including, perhaps, stochasticity in gene expression (Li et al., 2017; Raj et al., 2010), itself a predictor of idiosyncratic behavioral biases (Werkhoven et al., 2021). Improved methods to longitudinally assay the fine-scale molecular and anatomical makeup of behaving organisms throughout development and adulthood will be invaluable to further illuminate the mechanistic origins of individuality.

Materials and methods

Data and code availability

All raw data, totaling 600 GB, are available via hard drive from the authors. A smaller (7 GB) repository with partially processed data files and MATLAB/Python scripts sufficient to generate figures and results is available at Zenodo (doi:*10.5281/zenodo.8092971*).

Fly rearing

Experimental flies were reared in a *Drosophila* incubator (Percival Scientific DR-36VL) at 22° C, 40% relative humidity, and 12:12h light:dark cycle. Flies were fed cornmeal/dextrose medium, as previously described (Honegger et al., 2019). Mated female flies aged 3 days post-eclosion were used for behavioral persistence experiments. Mated female flies aged 7 to 15 days post-eclosion were used for all paired behavior-calcium imaging and immunohistochemistry experiments.

Fly stocks

The following stocks were obtained from the Bloomington *Drosophila* Stock Center: P{20XUAS-IVS-GCaMP6m}attP40 (BDSC #42748), w[*]; P{w[+mC]=Or13a-GAL4.F}40.1 (BDSC #9945), w[*]; P{w[+mC]=Or19a-GAL4.F}61.1 (BDSC #9947), w[*]; P{w[+mC]=Or22a-GAL4.7.717}14.2 (BDSC #9951), w[*]; P{w[+mC]=Orco-GAL4.W}11.17; TM2/TM6B, Tb[1] (BDSC #26818). Transgenic lines were outcrossed to the isogenic line isokh11 (Honegger et al., 2019) for at least 5 generations prior to being used in any experiments. GH146-Gal4 was a gift provided by Y. Zhong (Honegger et al., 2019). w; UAS-Brp-Short-mStrawberry; UAS-mCD8-GFP; + was a gift of Timothy Mosca and was not outcrossed to the isokh11 background (Mosca and Luo, 2014).

Odor delivery

Odor delivery during behavioral tracking and neural activity imaging was controlled with isolation valve solenoids (NResearch Inc.) (Honegger et al., 2019). Saturated headspace from 40 ml vials containing 5 ml pure odorant were serially diluted via carbon-filtered air to generate a variably (10-25%) saturated airstream controlled by digital flow controllers (Alicat Scientific) and presented to flies at total flow rates of ~100 mL/min. Dilution on the order of 10% is typical of other odor tunnel assays, as in Claridge-Chang et al. (2009). To yield the greatest signal of individual odor preference, dilution factors for odorants were adjusted on a week-by-week basis to ensure that the mean preference was approximately 50%. The odor panel used for imaging was comprised of the following odorants: 2-heptanone (CAS #110-43-0, Millipore Sigma), 1-pentanol (CAS #71-41-0, Millipore Sigma), 3-octanol (CAS #589-98-0, Millipore Sigma), hexyl-acetate (CAS #142-92-7, Millipore Sigma), 4-methylcyclohexanol (CAS #589-91-3, Millipore Sigma), pentyl acetate (CAS #628-63-7, Millipore Sigma), 1-butanol (CAS #71-36-3, Millipore Sigma), ethyl lactate (CAS #97-64-3, Millipore Sigma), geranyl acetate (CAS #105-87-3, Millipore Sigma), 1-hexanol (CAS #111-27-34, Millipore Sigma), citronella java essential oil (191112, Aura Cacia), and 200 proof ethanol (V1001, Decon Labs).

Odor preference behavior

Odor preference was measured at 25°C and 20% relative humidity. As previously described (Honegger et al., 2019), individual flies confined to custom-fabricated tunnels were illuminated with infrared light and behavior was recorded with a digital camera (Basler) and zoom lens (Pentax). The odor choice tunnels were 50 mm long, 5 mm wide, and 1.3 mm tall. Custom real-time tracking software written in MATLAB was used to track centroid, velocity, and principal body axis angle throughout the behavioral experiment, as previously described (Honegger et al., 2019). After a 3-minute acclimation period, odorants were delivered to either end of the tunnel array for 3 minutes. Odor preference score was calculated as the fraction of time spent in the reference side of the tunnel during odor-on period minus the time spent in the reference side of the tunnel during the pre-odor acclimation period.

Behavioral preference persistence measurements

After measuring odor preference, flies were stored in individual housing fly plates (modified 96-well plates; FlySorter, LLC) on standard food, temperature, humidity, and lighting conditions.

Odor preference of the same individuals was measured 3 and/or 24 hours later. In some cases, fly tunnel position was randomized between measurements. We observed that randomization had little effect on preference persistence.

Calcium imaging

Flies expressing GCaMP6m in defined neural subpopulations were imaged using a custom-built two-photon microscope and ultrafast Ti:Sapphire laser (Spectra-Physics Mai Tai) tuned to 930 nm, at a power of 20 mW out of the objective (Olympus XLUMPlanFL N 20x/1.00 W). For paired behavior and imaging experiments, the time elapsed between behavior measurement and imaging ranged from 15 minutes to 3 hours. Flies were anesthetized on ice and immobilized in an aluminum sheet with a female-fly-sized hole cut in it. The head cuticle between the antennae and ocelli was removed along with the tracheae to expose the ALs from the dorsal side. Volume scanning was performed using a piezoelectric objective mount (Physik Instrumente). ScanImage 2013 software (Vidrio Technologies) was used to coordinate galvanometer laser scanning and image acquisition. Custom Matlab (Mathworks) scripts were used to coordinate image acquisition and control odor delivery. 256 by 192 (x-y) pixel 16-bit tiff images were recorded.

The piezo travel distance was adjusted between 70 and 90 μm so as to cover most of the AL. The number of z-sections in a given odor panel delivery varied between 7 and 12 yielding a volume acquisition rate of 0.833 Hz. Odor delivery occurred from 6-9.6s of each recording.

Each fly experienced up to four deliveries of the odor panel. The antennal lobe being recorded (left or right) was alternated after each successful completion of an odor panel. Odors were delivered in randomized order. In cases where baseline fluorescence was very weak or no obvious odor responses were visible, not all four panels were delivered.

Glomerulus segmentation and labeling

Glomerular segmentation masks were extracted from raw image stacks using a *k*-means clustering algorithm based on time-varying voxel fluorescence intensities, as previously described (Honegger et al., 2019). Each image stack, corresponding to a single odor panel delivery, was processed individually. Time-varying voxel fluorescence values for each odor delivery were concatenated to yield a voxel-by-time matrix consisting of each voxel's recorded value during the course of all 13 odor deliveries of the odor panel. After z-scoring, principal component analysis was performed on this matrix and 75% of the variance was retained. Next, *k*-means ($k = 80$, 50 replicates with random starting seeds) was performed to produce 50 distinct voxel cluster assignment maps which we next used to calculate a consensus map. This approach was more accurate than clustering based on a single *k*-means seed.

Of the 50 generated voxel cluster assignment maps, the top 5 were selected by choosing those maps with the lowest average within-cluster sum of distances, selecting for compact glomeruli. The remaining maps were discarded. Next, all isolated voxel islands in each of the top 5 maps were identified and pruned based on size (minimum size = 100 voxels, maximum size = 10000 voxels). Finally, consensus clusters were calculated by finding voxel islands with significant overlap across all 5 of the pruned maps. Voxels which fell within a given cluster across all 5 pruned maps were added to the consensus cluster. This process was repeated for all clusters until the single consensus cluster map was complete. In some cases we found by manual inspection that some individual glomeruli were clearly split into two discrete clusters. These splits were remedied by automatically merging all consensus clusters whose centroids were separated by a physical distance of less than 30 voxels and whose peak odor response Spearman correlation was greater than 0.8. Finally, glomeruli were manually labeled based on anatomical position, morphology, and size (Grabe et al., 2015). We focused our analysis on 5 glomeruli (DM1, DM2, DM3, DL5, and DC2), which were the only glomeruli that could be observed in all paired behavior-calcium datasets. However, not all 5 glomeruli were identified in all recordings (**Figure 1 – figure supplement 3**). Missing glomerular data was later mean-imputed. Using alternating least squares to impute data (running the pca function with option 'als' to infill missing values 1,000 times and taking the mean infilled matrix – see **Figure 1 – figure supplement 5** of Werkhoven et al., 2021) had negligible effect on the fitted slope and predictive capacity of the PN PC2 OCT-MCH model compared to mean-infilling.

Calcium image data analysis

All data was processed and analyzed in MATLAB 2018a (Mathworks). Calcium responses for each voxel were calculated as $\Delta f/f = [f(t) - F]/F$, where $f(t)$ and F are the instantaneous and average fluorescence, respectively. Each glomerulus' time-dependent calcium response was calculated as the mean $\Delta f/f$ across all voxels falling within the glomerulus' automatically-generated segmentation mask during a single volume acquisition. Time-varying odor responses were normalized to baseline by subtracting the median of pre-odor $\Delta f/f$ from each trace. Peak odor response was calculated as the maximum fluorescence signal from 7.2s to 10.8s (images 6 through 9) of the recording.

To compute principal components of calcium dynamics, each fly's complement of odor panel responses (a 5 glomeruli by 13 odors = 65-dimensional vector) was concatenated. Missing glomerulus-odor response values were filled in with the mean glomerulus-odor pair across all fly recordings for which the data was not missing. After infilling, principal component analysis was carried out with individual odor panel deliveries as observations and glomerulus-odor responses pairs as features.

Inter- and intra-fly distances (**Figure 1J**) were calculated using the projections of each fly's glomerulus-odor responses onto all principal components. For each fly, the average Euclidean distance between response projections 1) among left lobe trials, 2) among right lobe trials, and 3) between left and right lobe trials were averaged together to get a single within-fly distance. Intra-fly distances were computed in a similar fashion (for each fly, taking the average distance of its response projections to those of other flies using only left lobe trials / only right lobe trials / between left-right trials, then averaging these three values to get a single across-fly distance).

In a subset of experiments in which we imaged calcium activity, some solenoids failed to open, resulting in the failure of odor delivery in a small number of trials. In these cases, we identified trials with valve failures by manually recognizing that glomeruli failed to respond during the nominal odor period. These trials were treated as missing data and infilled, as described above. Fewer than ∼10% of flies and 5% of odor trials were affected.

For all predictive models constructed, the average principal component score or glomerulus-odor $\Delta f/f$ response across trials was used per individual; that is, each fly contributed one data point to the relevant model. Linear models were constructed from behavior scores and the relevant predictor (principal component, average $\Delta f/f$ across dimensions, specific glomerulus measurements) as described in the text and **Tables 1**-**2**. All reported linear model p-values are nominal, i.e., unadjusted for multiple hypothesis comparisons. 95% confidence intervals around model regression lines were estimated as +/- 2 standard deviations of the value of the regression line at each x-position across 2000 bootstrap replicates (resampling flies). To predict behavior as a function of time during odor delivery, we analyzed data as described above, but considered only Δf/f at each single time point (**Figure 2 – figure supplement 2A-C**), rather than averaging during the peak response interval.

To decode individual identity from neural responses, we first performed PCA on individual odor panel peak responses. We retained principal component scores constituting specified fractions of variance (**Figure 1 – figure supplement 6A**) and trained a linear logistic classifier to predict individual identity from single odor panel deliveries.

To decode odor identity from neural responses, each of the 5 recorded glomeruli were used as features, and the calcium response of each glomerulus to a specific odor at a specified time point were used as observations (PNs, n = 5317 odor deliveries; ORNs, n = 2704 odor deliveries). A linear logistic classifier was trained to predict the known odor identity using 2-fold cross-validation. That is, a model was trained on half the data and evaluated on the remaining half, and then this process was repeated with the train and test half reversed. The decoding accuracy was quantified as the fraction of odor deliveries in which the predicted odor was correct.

Inference of correlation between latent calcium and behavior states

We performed a simulation-based analysis to infer the strength of the correlation between latent calcium (Brp) and behavior states, given the R^2 of a given linear model. **Figure 1 – figure supplement 9** is a schematic of a possible data generation process that underlies our observed data. We assume that the "true" behavioral and calcium values of the animal are captured by unobserved latent states X_c and X_b, respectively, such that the correlation between X_c and X_b is the biological signal captured by the model, having adjusted for the noise associated with actually measuring behavior and calcium (ρ_{signal}). Our calcium and odor preference scores are subject to measurement error and temporal instability (behavior and neural activity were measured 1-3 hours apart). These effects are both noise with respect to estimating the linear relationship between calcium and behavior. Their magnitude can be estimated using the empirical repeatability of behavior and calcium experiments respectively. Thus, our overall approach was to assume true latent behavior and calcium signals that are correlated at the level ρ_{signal}, add noise to them commensurate with the repeatability of these measures to simulate measured behavior and calcium, and record the simulated empirical R^2 between these measured signals. This was done many times to estimate distributions of empirical R^2 given ρ_{signal}. These distributions could finally be used in the inverse direction to infer ρ_{signal} given the actual model R^2 values computed in our study.

Specifically, we simulated X_c as a set of N standard normal variables (N equalling the number of flies used to compute a correlation between predicted and measured preference) and generated $X_b = \rho_{signal} X_c + (1\text{-}\rho_{signal}{}^2 Z)^{½}$, where Z is a set of N standard normal variables uncorrelated with X_c, a procedure that ensures that $corr(X_c, X_b) = \rho_{signal}$. Next, we simulated observed calcium readouts X_c' and X_c'', such that $corr(X_c, X_c') = corr(X_c, X_c'') = r_c$. Similarly, we simulated noisy observed behavioral assay readouts X_b' and X_b'', such that $corr(X_b, X_b') = corr(X_b, X_b'') = r_b$. The values of r_c and r_b were drawn from the empirical repeatability of calcium ($R_{c,c}{}^2$) and behavior ($R_{b,b}{}^2$) respectively as follows. Since calcium is a multidimensional measure, and our calcium model predictors are based on principal components of glomerulus-odor responses, we used variance explained along the PCs to calculate a single value for the calcium repeatability $R_{c,c}{}^2$. We compared the eigenvalues of the real calcium PCA to those of shuffled calcium data (shuffling glomerulus/odor responses for each individual fly), computing $R_{c,c}{}^2$ by summing the variance explained along the PCs of the calcium data up until the component-wise variance for the calcium data fell below that of the shuffled data, a similar approach as done in Berman et al., 2014 and Werkhoven et al., 2021. That is, we determined which empirical PCs had more variance than their corresponding rank-matched PC in shuffled data, interpreted the remaining PCs as harboring the noise of the experiment, and totaled the variance explained of the non-noise PCs as our measure of the repeatability of the measurement as a whole. $R_{c,c}^2$ was calculated to be 0.77 for the full PN calcium data.

To incorporate uncertainty in calcium-calcium repeatability, we utilized bootstrapping. We resampled the calcium data associated with individual flies 10,000 times, performed PCA and computed $R_{c,c}^2$ for each resampled dataset, then set $r_c = (R_{c,c}^2)^{1/4}$ to ensure $corr(X_c', X_c'')^2 = R_{c,c}^2$. For behavior-behavior uncertainty, we set r_b from the repeatability across odor preference trials in the same flies measured 3h apart ($R_{b,b}^2 = 0.12$ for OCT vs MCH, **Figure 1 – figure supplement 1D** using the full dataset of flies). We also resampled the flies 10,000 times, computed $R_{b,b}^2$ for each resampled dataset, and set $r_b = (R_{b,b}^2)^{1/4}$ to ensure $corr(X_b', X_b'')^2 = R_{b,b}^2$.

We varied ρ_{signal} from 0 to 1 in increments of 0.01, and for each ρ_{signal} and bootstrap iteration we simulated a set of N X_c, and generated X_b, X_c', X_c'', X_b', and X_b'', then we computed a simulated observed calcium-behavior relationship strength $R_{c,b}^2 = corr(X_c', X_b')^2$. We repeated this simulation 10,000 times for each ρ_{signal} and plotted the resultant relationship between ρ_{signal} against $R_{c,b}^2$ (percentiles of $R_{c,b}^2$ are displayed in **Figure 1 – figure supplement 9B**). Then, we inferred ρ_{signal} by first drawing bootstrapped samples of calcium-behavior R^2, then adding together the marginal distributions of ρ_{signal} for each calcium-behavior R^2. We report the median ρ_{signal} and 90% confidence interval as estimated by the 5th-95th quantiles.

The procedure outlined above was done analogously for models using Brp-short relative fluorescence intensity, performing the PCA-based calcium response repeatability step with PCA on the multidimensional Brp-short relative fluorescence intensity (which yielded $R_{brp,brp}^2 = 0.78$).

DoOR data

DoOR data for the glomeruli and odors relevant to our study was downloaded from *http://neuro.uni-konstanz.de/DoOR/default.html* (Münch and Galizia, 2016).

Yoked odor experience experiments

We selected six flies for which both odor preference and neural activity were recorded to serve as the basis for imposed odor experiences for yoked control flies. The experimental flies were chosen to represent a diversity of preference scores. Each experimental fly's odor experience was binned into discrete odor bouts to represent experience of either MCH or OCT based on its location in the tunnel as a function of time (**Figure 2D**). Odor bouts lasting less than 100 ms were omitted due to limitations on odor-switching capabilities of the odor delivery apparatus. To deliver a given experimental fly's odor experience to yoked controls, we set both odor streams (on either end of the tunnel apparatus) to deliver the same odor experienced by the experimental fly at that moment during the odor-on period. No odor was delivered to yoked controls during time points in which the experimental fly resided in the tunnel choice zone (central 5 mm). See **Figure 2D** for an example pair of experimental fly and yoked control behavior and odor experience.

Immunohistochemistry

After measuring odor preference behavior, 7-15 day-old flies were anesthetized on ice and brains were dissected in phosphate buffered saline (PBS). Dissection and immunohistochemistry were carried out as previously reported (Wu and Luo, 2006). The experimenter was blind to the behavioral scores of all individuals throughout dissection, imaging, and analysis. Individual identities were maintained by fixing, washing, and staining each brain in an individual 0.2 mL PCR tube using fluid volumes of 100 uL per brain (Fisher Scientific). Primary incubation solution contained mouse anti-nc82 (1:40, DSHB), chicken anti-GFP (1:1000, Aves Labs), rabbit anti-mStrawberry (1:1000, biorbyt), and 5% normal goat serum (NGS, Invitrogen) in PBT (0.5% Triton X-100 in PBS). Secondary incubation solution contained Atto 647N-conjugated goat anti-mouse (1:250, Millipore Sigma), Alexa Fluor 568-conjugated goat anti-rabbit (1:250), Alexa Fluor 488-conjugated goat anti-chicken (1:250, ThermoFisher), and 5% NGS in PBT. Primary and secondary incubation times were 2 and 3 overnights, respectively, at 4° C. Stained samples were mounted and cleared in Vectashield (H-1000, Vector Laboratories) between two coverslips (12-568B, Fisher Scientific). Two reinforcement labels (5720, Avery) were stacked to create a 0.15 mm spacer.

Expansion microscopy

Immunohistochemistry for expansion microscopy was carried out as described above, with the exception that antibody concentrations were modified as follows: mouse anti-nc82 (1:40), chicken anti-GFP (1:200), rabbit anti-mStrawberry (1:200), Atto 647N-conjugated goat anti-mouse (1:100), Alexa Fluor 568-conjugated goat anti-rabbit (1:100), Alexa Fluor 488-conjugated goat anti-chicken (1:100). Expansion of stained samples was performed as previously described (Asano et al., 2018; Gao et al., 2019). Expanded samples were mounted in coverslip-bottom petri dishes (MatTek Corporation) and anchored by treating the coverslip with poly-l-lysine solution (Millipore Sigma) as previously described (Asano et al., 2018).

Confocal imaging

All confocal imaging was carried out at the Harvard Center for Biological Imaging. Unexpanded samples were imaged on an LSM700 (Zeiss) inverted confocal microscope equipped with a 40x oil-immersion objective (1.3 NA, EC Plan Neofluar, Zeiss). Expanded samples were imaged on an LSM880 (Zeiss) inverted confocal microscope equipped with a 40x water-immersion objective (1.1 NA, LD C-Apochromat, Zeiss). Acquisition of Z-stacks was automated with Zen Black software (Zeiss).

Standard confocal image analysis

We used custom semi-automated code to generate glomerular segmentation masks from confocal z-stacks of unexpanded Orco>Brp-Short brains. Using Matlab, each image channel was median filtered (σ_x, σ_y, σ_z = 11, 11, 1 pixels) and downsampled in x and y by a factor of 11. Next, an ORN mask was generated by multiplying and thresholding the Orco>mCD8 and Orco>Brp-Short channels. Next, a locally normalized nc82 and Orco>mCD8 image stack were multiplied and thresholded, and the ORN mask was applied to remove background and other undesired brain structures. This pipeline resulted in a binary image stack which maximized the contrast of the glomerular structure of the antennal lobe. We then applied a binary distance transform and watershed transform to generate discrete subregions which aimed to represent segmentation masks for each glomerulus tagged by Orco-Gal4.

However, this procedure generally resulted in some degree of under-segmentation; that is, some glomerular segmentation masks were merged. To split each merged segmentation mask, we convolved a ball (whose radius was proportional to the cube root of the volume of the segmentation mask in question) across the mask and thresholded the resulting image. The rationale of this procedure was that 2 merged glomeruli would exhibit a mask shape resembling two touching spheres, and convolving a similarly-sized sphere across this volume followed by thresholding would split the merged object. After ball convolution, we repeated the distance and watershed transform to once more generate discrete subregions representing glomerular segmentation masks. This second watershed step generally resulted in over-segmentation; that is, by visual inspection it was apparent that many glomeruli were split into multiple subregions.

Therefore, we finally manually agglomerated the over-segmented subregions to generate single segmentation masks for each glomerulus of interest. We used a published atlas to aid manual identification of glomeruli (Grabe et al., 2015). The total Brp-Short fluorescence signal within each glomerulus was determined and divided by the volume of the glomerulus' segmentation mask to calculate Brp-Short density values.

Expansion microscopy image analysis

The spots function in Imaris 9.0 (Bitplane) was used to identify individual Brp-Short puncta in expanded sample image stacks of Or13a>Brp-Short samples (Mosca and Luo, 2014). The spot size was set to 0.5 um, background subtraction and region-growing were enabled, and the default spot quality threshold was used for each image stack. Identified spots were used to mask the Brp-Short channel and the resultant image was saved as a new stack. In MATLAB, a glomerular mask was generated by smoothing (σ_x, σ_y, σ_z = 40, 40, 8 pixels) and thresholding (92.5th percentile) the raw Brp-Short image stack. The mask was then applied to the spot image stack to remove background spots. Finally, the masked spot image stack was binarized and spot number and properties were quantified.

Antennal Lobe modeling

We constructed a model of the antennal lobe to test the effect of circuit variation on PN activity variation across individuals. Our general approach to producing realistic circuit activity with the AL model was 1) using experimentally-measured parameters whenever possible (principally the connectome wiring diagram and biophysical parameters measured electrophysiologically), 2) associating free parameters only with biologically plausible categories of elements, while minimizing their number, and 3) tuning the model using those free parameters so that it reproduced high-level patterns of activity considered in the field to represent the canonical operations of the AL. Simulations were run in Python (version 3.6) (van Rossum and Drake, 2011), and model outputs were analyzed using Jupyter notebooks (Kluyver et al., 2016) and Python and Matlab scripts.

AL model neurons

Release 1.2 of the hemibrain connectomics dataset (Scheffer et al., 2020) was used to set the connections in the model. Hemibrain body IDs for ORNs, LNs, and PNs were obtained via the lists of neurons supplied in the supplementary tables in Schlegel et al., 2020. ORNs and PNs of non-olfactory glomeruli (VP1d, VP1l, VP1m, VP2, VP3, VP4, VP5) were ignored, leaving 51 glomeruli. Synaptic connections between the remaining 2574 ORNs, 197 LNs, 166 mPNs, and 130 uPNs were queried from the hemibrain API. All ORNs were assigned to be excitatory (Wilson, 2013). Polarities were assigned to PNs based on the neurotransmitter assignments in Bates et al., 2020. mPNs without neurotransmitter information were randomly assigned an excitatory polarity with probability equal to the fraction of neurotransmitter-identified mPNs that are cholinergic; the same process was performed for uPNs. After confirming that the model's output was qualitatively robust to which mPNs and uPNs were randomly chosen, this random assignment was performed once and then frozen for subsequent analyses.

Of the 197 LNs, we assigned 31 to be excitatory, based on the estimated 1:5.4 ratio of eLNs to iLNs in the AL (Tsai et al., 2018). To account for observations that eLNs broadly innervate the AL (Shang et al., 2007), all LNs were ranked by the number of innervated glomeruli, and the 31 eLNs were chosen uniformly at random from the top 50% of LNs in the list. This produced a distribution of glomerular innervations in eLNs qualitatively similar to that of *krasavietz* LNs in Supplementary Figure 6 of Chou et al., 2010.

Voltage model

We used a single-compartment leaky-integrate-and-fire voltage model for all neurons as in Kakaria and de Bivort, 2017, in which each neuron had a voltage $V_i(t)$ and current $I_i(t)$. When the voltage of neuron i was beneath its threshold $V_{i,\,thr}$, the following dynamics were obeyed:

$$C_i\frac{dV_i}{dt} = \frac{V_{i,0}-V_i(t)}{R_i} + I_{i,odor}(t) + \sum_{j=1}^{N} a_i W_{ji} I_j(t)$$

Each neuron i had electrical properties: membrane capacitance C_i, resistance R_i, and resting membrane potential $V_{i,0}$ with values from electrophysiology measurements (**Table 2**).

When the voltage of a neuron exceeded the threshold $V_{i,\,thr}$, a templated action potential was filled into its voltage time trace, and a templated postsynaptic current was added to all downstream neurons, following the definitions in Kakaria and de Bivort, 2017.

Odor stimuli were simulated by triggering ORNs to spike at frequencies matching known olfactory receptor responses to the desired odor. The timing of odor-evoked spikes was given by a Poisson process, with firing rate *FR* for ORNs of a given glomerulus governed by:

$$FR_{glom,odor}(t) = FR_{max} D_{glom,odor}\left(f_a + (1-f_a)e^{-t/t_a}\right)$$

FR_{max}, the maximum ORN firing rate, was set to 400 Hz. $D_{glom,\,odor}$ is a value between 0 and 1 from the DoOR database, representing the response of an odorant receptor/glomerulus to an odor, estimated from electrophysiology and/or fluorescence data (Münch and Galizia, 2016). ORNs display adaptation to odor stimuli (Wilson, 2013), captured by the final term with timescale t_a = 110 *ms* to 75% of the initial value, as done in Kao and Lo, 2020. Thus, the functional maximum firing rate of an ORN was 75% of 400 Hz = 300 Hz, matching the highest ORN firing rates observed experimentally (Hallem et al., 2004). After determining the times of ORN spikes according to this firing-rate rule, spikes were induced by the addition of 10^6 picoamps in a single time step. This reliably triggered an action potential in the ORN, regardless of currents from other neurons. In the absence of odors, spike times for ORNs were drawn by a Poisson process at 10 Hz, to match reported spontaneous firing rates (de Bruyne et al., 2001).

For odor-glomeruli combinations with missing DoOR values (40% of the dataset), we performed imputation via alternating least squares (using the pca function with option 'als' to infill missing values (*MATLAB documentation*) on the odor x glomerulus matrix 1000 times and taking the mean infilled matrix, which provides a closer match to ground truth missing values than a single run of ALS (**Figure 1 – figure supplement 5** of Werkhoven et al., 2021).

A neuron j presynaptic to i supplies its current $I_j(t)$ scaled by the synapse strength W_{ji}, the number of synapses in the hemibrain dataset from neuron j to i. Rows in W corresponding to neurons with inhibitory polarity (i.e. GABAergic PNs or LNs) were set negative. Finally, post-synaptic neurons (columns of the connectivity matrix) have a class-specific multiplier a_i, a hand-tuned value, described below.

AL model tuning

Class-specific multiplier current multipliers (a_i) were tuned using the panel of 18 odors from Bhandawat et al., 2007 (our source for several experimental observations of high-level AL function): benzaldehyde, butyric acid, 2,3-butanedione, 1-butanol, cyclohexanone, Z3-hexenol, ethyl butyrate, ethyl acetate, geranyl acetate, isopentyl acetate, isoamyl acetate, 4-methylphenol, methyl salicylate, 3-methylthio-1-propanol, octanal, 2-octanone, pentyl acetate, E2-hexenal, trans-2-hexenal, gamma-valerolactone. Odors were “administered” for 400 ms each, with 300 ms odor-free pauses between odor stimuli.

The high-level functions of the AL that represent a baseline, working condition were: (1) firing rates for ORNs, LNs, and PNs matching the literature (listed in **Table 2** and see (Bhandawat et al., 2007; Dubin and Harris, 1997; Jeanne and Wilson, 2015; Seki et al., 2010), (2) a more uniform distribution of PN firing rates during odor stimuli compared to ORN firing rates, (3) greater separation of representations of odors in PN-coding space than in ORN-coding space, and (4) a sublinear transfer function between ORN firing rates and PN firing rates. Features (2) - (4) relate to the role of the AL in enhancing the separability of similar odors (Bhandawat et al., 2007).

To find a parameterization with those functions, we tuned the values of a_i as scalar multipliers on ORN, eLN, iLN, and PN columns of the hemibrain connectivity matrix. Thus, these values represent cell type-specific sensitivities to presynaptic currents, which may be justified by the fact that ORNs/LNs/PNs are genetically distinct cell populations (McLaughlin et al., 2021; Xie et al., 2021). A grid search of the four class-wise sensitivity parameters produced a configuration that reasonably satisfied the above criteria (**Figure 4 – figure supplement 2**). In this configuration, the ORN columns of the hemibrain connectivity matrix are scaled by 0.1, eLNs by 0.04, iLNs by 0.02, and PNs by 0.4. The relatively large multiplier on PNs is potentially consistent with the fact that PNs are sensitive to small differences between weak ORN inputs (Bhandawat et al., 2007). Model outputs were robust over several different sets of a_i, provided iLN sensitivity ≃ eLN < ORN < PN.

We analyzed the sensitivity of the model’s parameters around their baseline values of a_{ORN}, a_{eLN}, a_{iLN}, a_{PN} = (0.1, 0.04, 0.02, 0.4). Each parameter was independently scaled up to 4x or 1/4x of its baseline value (**Figure 4 – figure supplement 3**), and the PN firing rates recorded. Separately, multiple-parameter manipulations were performed by multiplying each parameter by a random log-Normal value with mean 1 and +/-1 standard deviation corresponding to a 2x or 0.5x scaling on each parameter. Mean PN-odor responses were calculated for all manipulated runs and compared to the mean PN-odor responses for the baseline configuration. A manipulation effect size was calculated by cohen’s d ((mean manipulated response - mean baseline response)/(pooled standard deviation)). None of these manipulations reached effect size magnitudes larger than 0.9 (which can be roughly interpreted as the number of standard deviations in the baseline PN responses away from the mean baseline PN response), which signaled that the model was robust to the sensitivity parameters in this range. The most sensitive parameter was, unsurprisingly, a_{PN}.

Notable ways in which the model behavior deviates from experimental recordings (and thus caveats on the interpretation of the model) include: 1) Model LNs appear to have more heterogeneous firing rates than real LNs, with many LNs inactive for this panel of odor stimuli. This likely reflects a lack of plastic/homeostatic mechanisms in the model to regularize LN firing rates given their variable synaptic connectivity (Chou et al., 2010). 2) Some PNs had off-odor rates that are high compared to real PNs, resulting in a distribution of ON-OFF responses that had a lower limit than in real recordings. Qualitatively close matches were achieved between the model and experimental data in the distributions of odor representations in ORN vs PN spaces and the non-linearity of the ORN-PN transfer function.

AL model circuit variation generation

We generated AL circuit variability in two ways: cell-type bootstrapping, and synapse density resampling. These methods assume that the distribution of circuit configurations across individual ALs can be generated by resampling circuit components within a single individual’s AL (neurons and glomerular synaptic densities, respectively, from the hemibrain EM volume).

To test the effect of variation in the developmental complement of neurons of particular types, we bootstrapped populations of interest from the list of hemibrain neurons. Resampling with replacement of ORNs was performed glomerulus-by-glomerulus, i.e., separately among each pool of ORNs expressing a particular *Odorant receptor* gene. The same was done for PNs. For LNs, all 197 LNs were treated as a single pool; there was no finer operation based on LN subtypes or glomerular innervations. This choice reflects the high developmental variability of LNs (Chou et al., 2010). The number of synapses between a pair of bootstrapped neurons was equal to the synapse count between those neurons in the hemibrain connectivity matrix.

In some glomeruli, bootstrapping PNs produced unreasonably high variance in the total PN synapse count. For instance, DP1m, DC4, and DM3 each harbor PNs that differ in total synapse count by a factor of ~10. Since these glomeruli have between two to three PNs each, in a sizable proportion of bootstrap samples, all-highly connected (or all-lowly) connected PNs are chosen in such glomeruli. To remedy this biologically unrealistic outcome, we examined the relationship between total input PN synapses within a glomerulus and glomerular volume (**Figure 4 – figure supplement 4**). In the "synapse density resampling" method, we required that the number of PN input synapses within a glomerulus reflect a draw from the empirical relationship between total input PN synapses and glomerular volume as present in the hemibrain data set. This was achieved by, for each glomerulus, sampling from the following distribution that depends on glomerular volume, then multiplying the number of PN input synapses by a scalar to match that sampled value:

$$log\ S_g\ = log\ (a\ {V_g}^d)\ +\ \varepsilon_g, \varepsilon_g\ \sim N(0, \sigma^2)$$

Here S_g is the PN input synapse count for glomerulus g, V_g is the volume of glomerulus g (in cubic microns), ε is a Gaussian noise variable with standard deviation σ, and a, d are the scaling factor and exponent of the volume term, respectively. The values of these parameters (a = 8.98, d = 0.73, σ = 0.38) were fit using maximum likelihood.

Quantification and statistical analysis

All fly behavior and calcium data was processed and analyzed in MATLAB 2018a (Mathworks). AL simulations were run in Python (version 3.6) (van Rossum and Drake, 2011), and model outputs were analyzed using Jupyter notebooks (Kluyver et al., 2016) and Python scripts. We performed a power analysis prior to the study to determine that recording calcium activity in 20-40 flies would be sufficient to identify moderate calcium-behavior correlations. Sample sizes for expansion microscopy were smaller, as the experimental procedure was more involved – therefore, we did not conduct a formal statistical analysis. Linear models were fit using the fitlm MATLAB function (*https://www.mathworks.com/help/stats/fitlm.html*); coefficients and p-values of models between measured preferences and predicted preferences are listed in **Table 1**. 95% confidence intervals around model regression lines were estimated as +/- 2 standard deviations of the value of the regression line at each x-position across 2000 bootstrap replicates (resampling flies). Boxplots depict the median value (points), interquartile range (boxes), and range of the data (whiskers).

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Additional information

Author contributions

M.C. conducted behavior experiments with assistance from M.S., conducted confocal microscopy, and conducted expansion microscopy with contributions from R. G. and E.B. D.L. implemented the computational AL model. B.L.d.B. supervised the project.

Declaration of interests

E.B. is a co-founder of a company that aims to commercialize expansion microscopy for medical purposes. R.G. and E.B. are co-inventors on multiple patents related to expansion microscopy. The authors declare no other competing interests.

Figure supplements

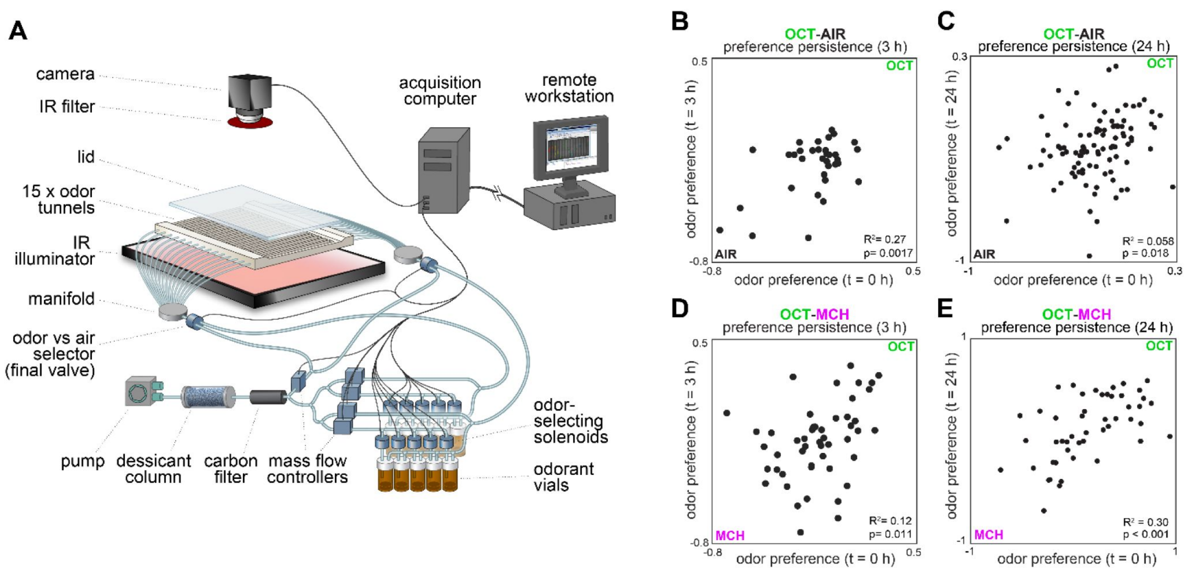

Figure 1 – figure supplement 1.

Behavioral measurements and individual preference persistence.

(A) Behavioral measurement apparatus (adapted from Honegger et al., 2019) **(B)** Odor preference persistence over 3 hours for flies given a choice between 3-octanol and air (n = 34 flies). **(C)** Odor preference persistence over 24 hours for flies given a choice between 3-octanol and air (n = 97 flies). **(D)** Odor preference persistence over 3 hours for flies given a choice between 3-octanol and 4-methylcyclohexanol (n = 51 flies). **(E)** Odor preference persistence over 24 hours for flies given a choice between 3-octanol and 4-methylcyclohexanol (n = 49 flies).

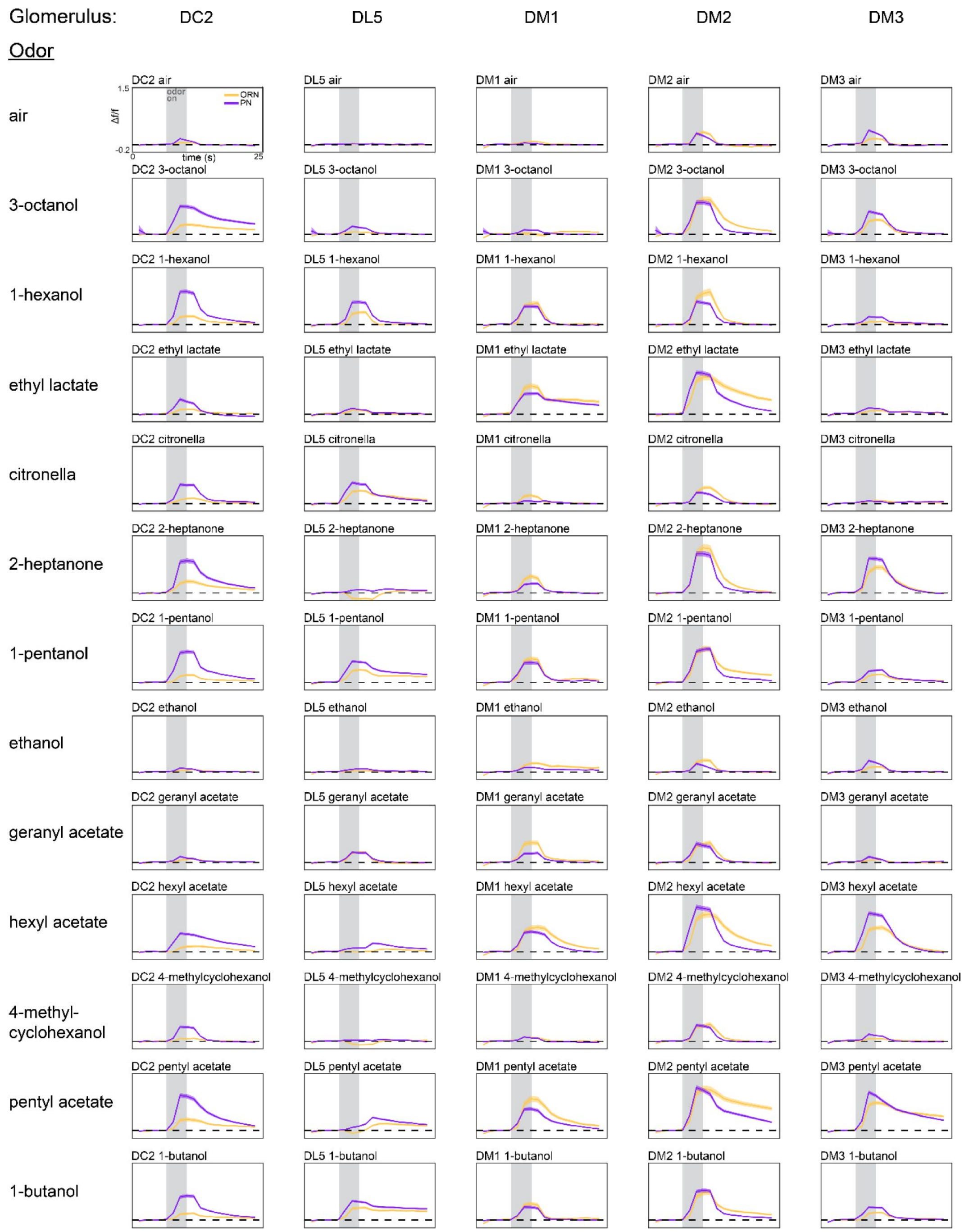

Figure 1 – figure supplement 2.

Average glomerulus-odor time-dependent responses.

Time-dependent responses of each glomerulus identified in our study to the 13 odors in our odor panel. Data represents the average across flies (ORN, peach curves, n = 65 flies; PN, plum curves, n = 122 flies). Shaded error bars represent S.E.M.

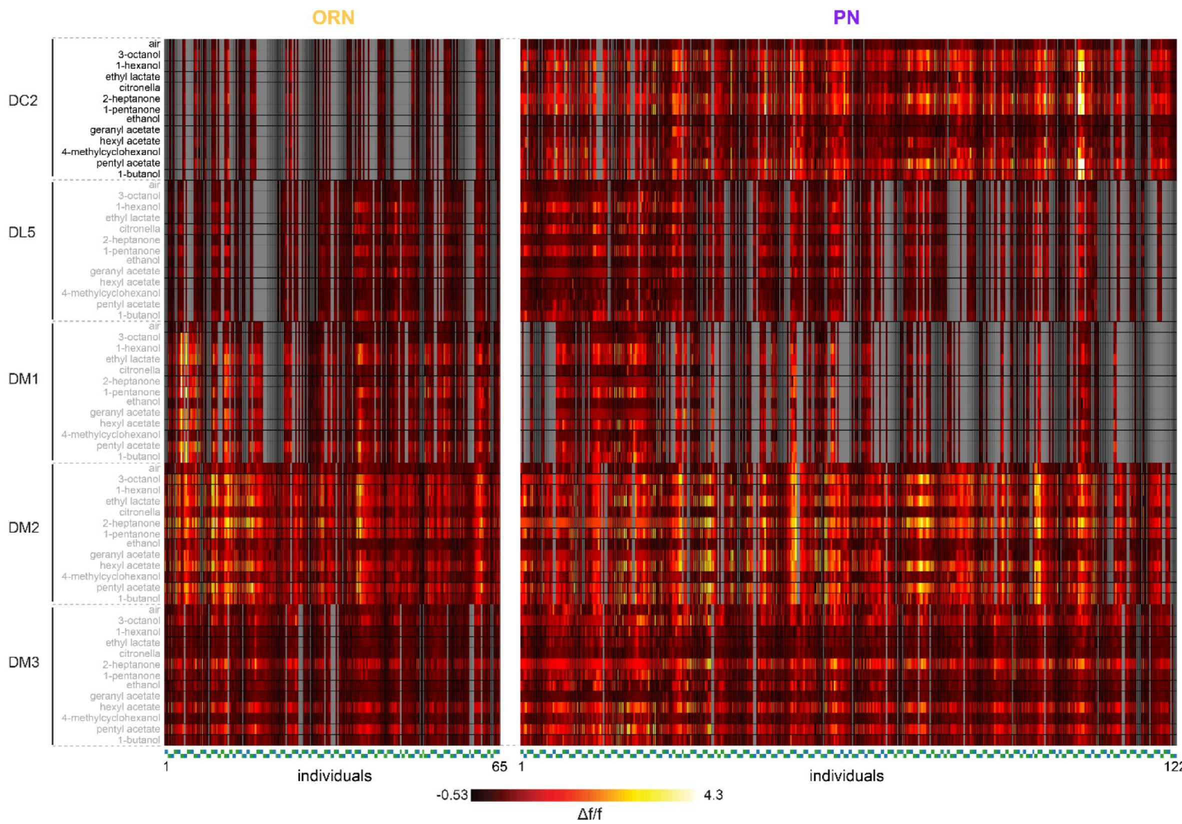

Figure 1 – figure supplement 3.

Individual glomerulus-odor responses.

Idiosyncratic odor coding measured in ORNs (left, 208 recordings across 65 flies) and PNs (right, 406 trials across 122 flies). Each column represents the response (max Δf/f attained over the odor trial) in a single recording from either the left or right lobe of a single fly. Below each heatmap, markers are grouped by individual fly (fly order is arbitrary, markers of adjacent flies alternate in height). Green markers correspond to left lobes, blue markers right lobes. Each row represents a glomerulus-odor response pair. Missing data are indicated in gray.

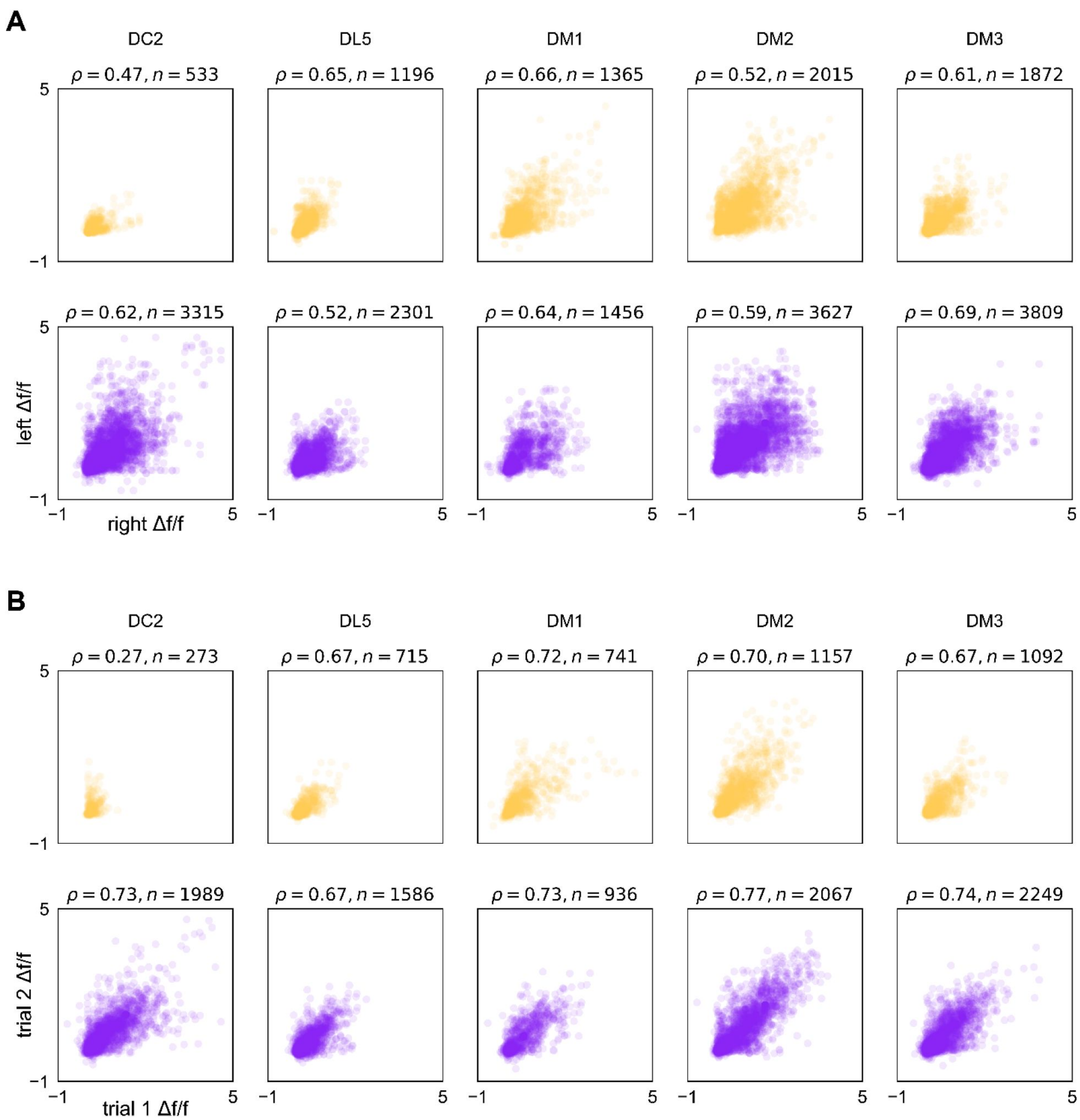

Figure 1 – figure supplement 4.

Correspondence in calcium responses between lobes and trials.

(A) Scatter plots of max Δf/f attained over an odor presentation in a left-lobe recording vs. a right-lobe recording in the same fly (same data as presented in **Figure 1 – figure supplement 3**). Plum points are PN responses and peach points ORNs. ρ is Spearman's rank correlation coefficient, points correspond to fly-odor-trial combinations, and n indicates the number of points within each subplot. **(B)** As in (A), for responses across two trials within the same lobe of the same fly. Points correspond to fly-odor-lobe combinations.

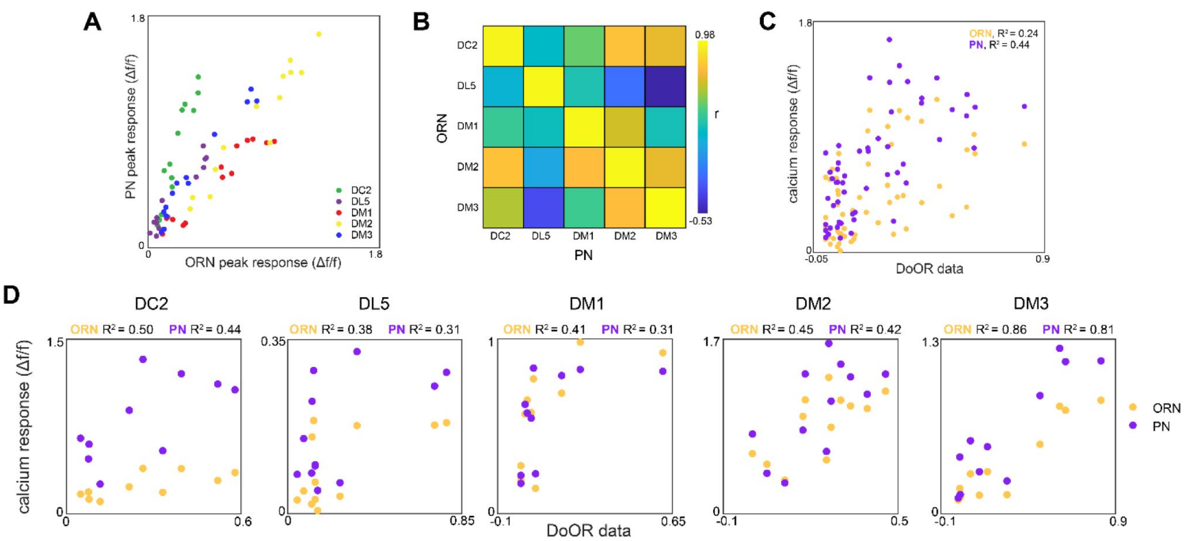

Figure 1 – figure supplement 5.

Glomerulus responses and identification.

(A) Glomerulus odor responses measured in PNs versus those measured in ORNs. Points correspond to the odorants listed in **Figure 1G**. **(B)** Cross-odor trial correlation matrix between glomerular odor responses in ORNs and PNs. **(C)** Peak calcium responses for each glomerulus-odor pair measured in this study plotted against those recorded in the DoOR dataset (Münch and Galizia, 2016). **(D)** Peak calcium responses for each individual glomerulus plotted against those recorded in the DoOR dataset.

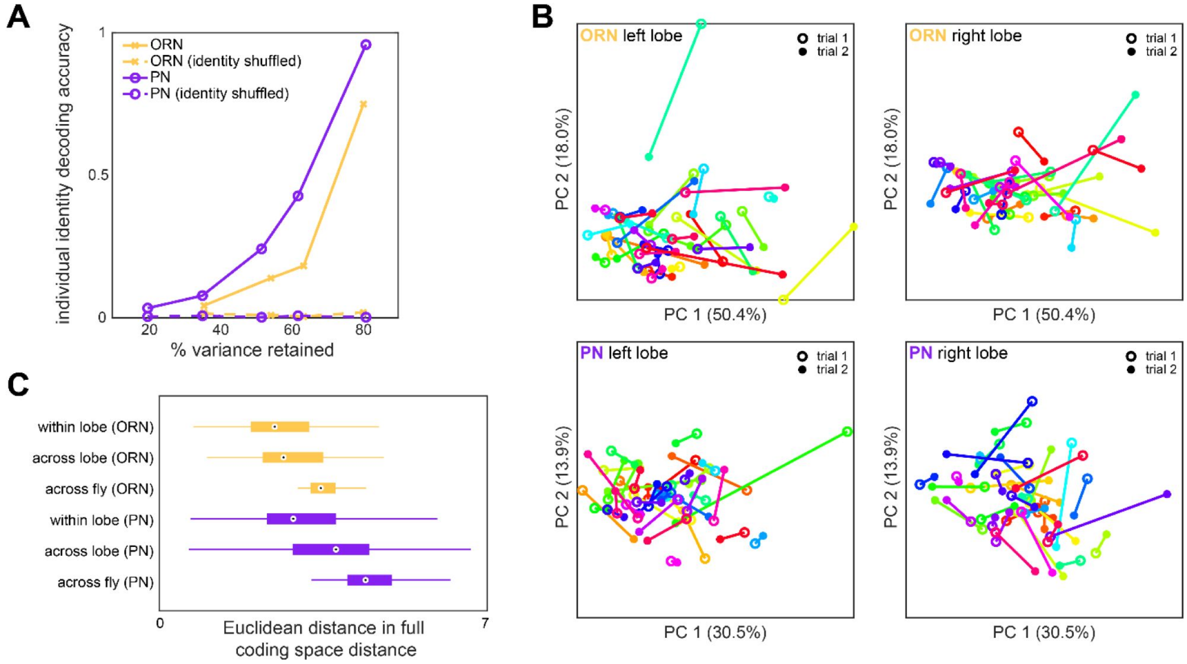

Figure 1 – figure supplement 6.

Idiosyncrasy of ORN and PN responses.

(A) Logistic regression classifier accuracy of decoding individual identity from individual odor panel peak responses. PCA was performed on population responses and the specified fraction of variance (x-axis) was retained. Individual identity can be better decoded from PN responses than ORN responses in all cases. **(B)** Individual trial-to-trial glomerulus-odor responses embedded in PC 1-2 space. Responses for the same flies as **Figure 1I** are shown. Each linked color represents one fly. Trial 1 and trial 2 responses are shown for ORN left lobe (upper left), ORN right lobe (upper right), PN left lobe (lower left), and PN right lobe (lower right). **(C)** Distance in the full glomerulus-odor response space between recordings within a lobe (trial-to-trial), across lobes (within fly), and across flies for ORNs and PNs. Points represent the median value, boxes represent the interquartile range, and whiskers the range of the data.

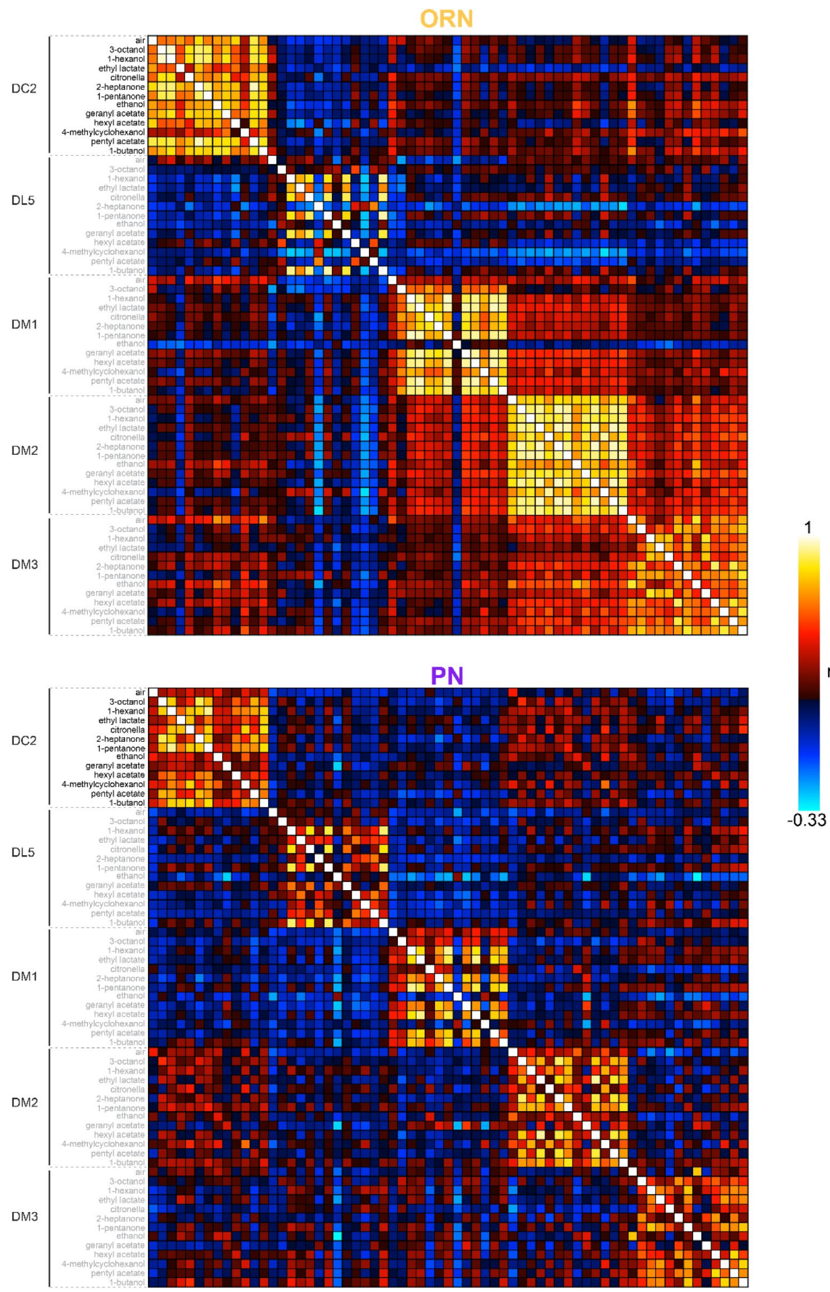

Figure 1 – figure supplement 7.

Calcium response correlation matrices.

Correlation between calcium response dimensions across flies measured in ORNs (top) and PNs (bottom). Glomerulus-odor responses are correlated at the level of glomeruli in both cell types. Inter-glomerulus correlations are more prominent in ORNs than PNs, consistent with known AL transformations that result in decorrelated PN activity (Bhandawat et al., 2007; Luo et al., 2010).

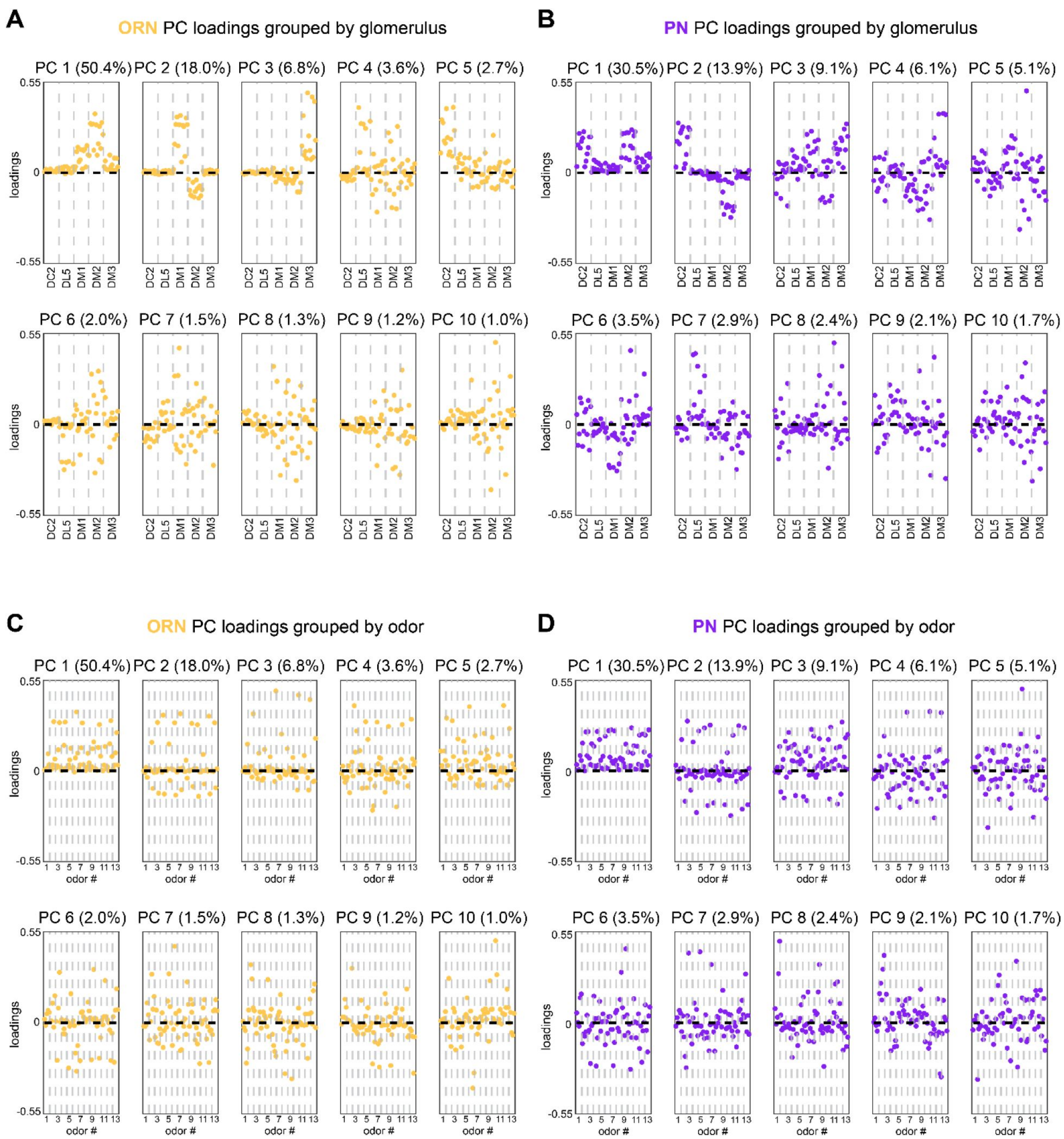

Figure 1 – figure supplement 8.

Calcium imaging principal component loadings.

(A-B) First 10 principal component loadings measured from calcium responses in ORNs (A, n = 65 flies) and PNs (B, n = 122 flies). Loadings are grouped by glomerulus, with each loading within a glomerulus representing the response of that glomerulus to one odor in the odor panel. Odors are the same as those listed in **Figure 1G**. **(C-D)** The same 10 principal component loadings as those shown in panels (A-B) grouped by odor rather than glomerulus. Glomeruli within each odor block are given in the order of panels (A) and (B).

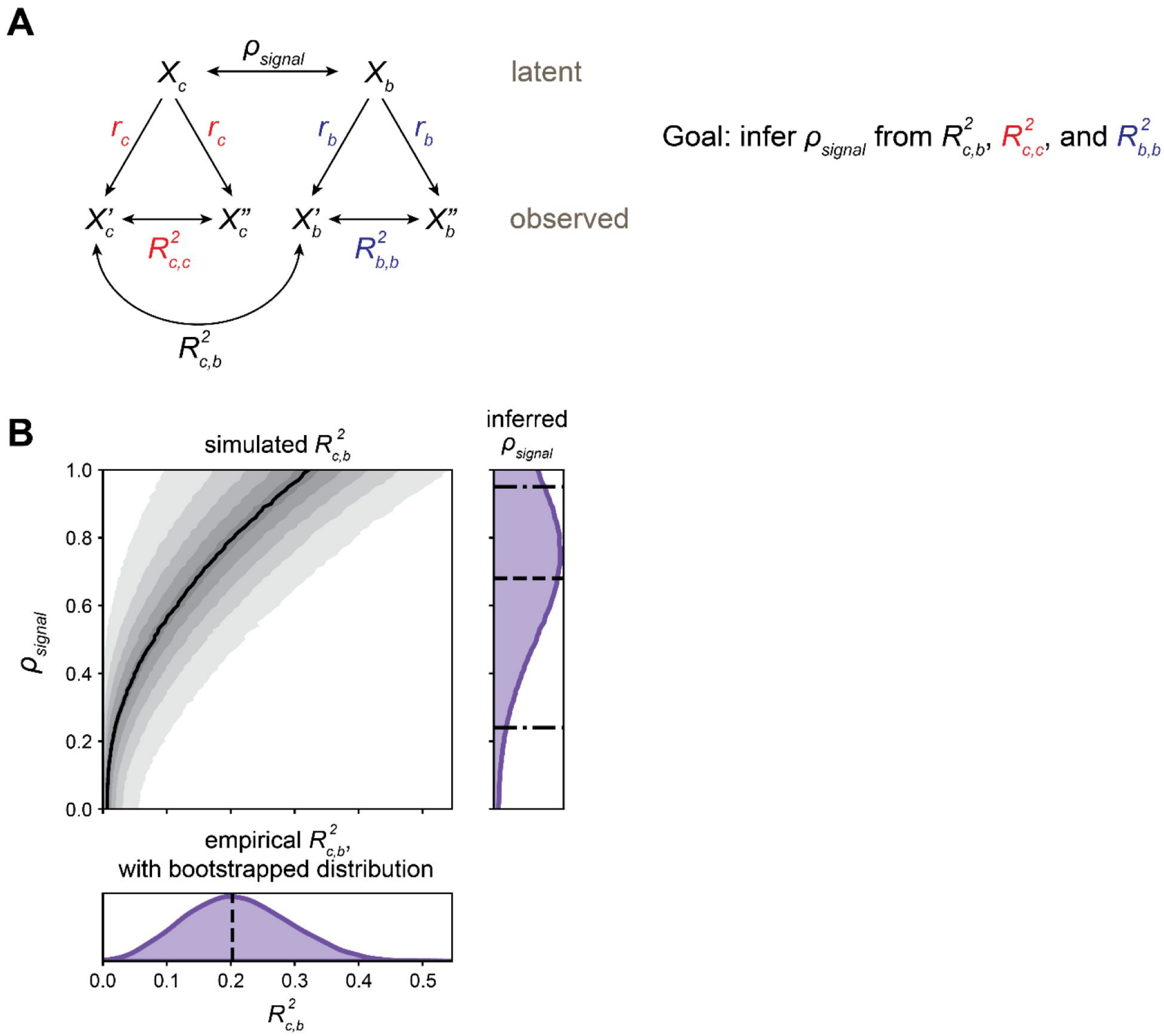

Figure 1 – figure supplement 9.

Estimating latent calcium - behavior correlations.

(A) Schematic of inference approach to estimate the correlation between latent calcium (c) and behavioral (b) states (ρ_{signal}). This method can be applied identically to infer ρ_{signal} between Brp measurements and behavior. **(B)** Demonstration of ρ_{signal} inference for OCT vs MCH model presented in **Figure 1M**: PN calcium PC 2 from trained model applied to train+test data. Bottom subplot: bootstrap distribution of calcium-behavior $R_{c,b}{}^2$ (dashed line: $R_{c,b}{}^2$ = 0.20 for the N = 69 flies). Top left subplot: simulated $R_{c,b}{}^2$ values. Black line indicates median $R_{c,b}{}^2$ among the 10,000 simulations for each ρ_{signal}, shaded areas (from lightest to darkest to lightest) indicate 5- 15th, 15-25th, ..., 85-95th percentile $R_{c,b}{}^2$. Right subplot: inferred distribution for ρ_{signal}, estimated by adding marginal distributions over ρ_{signal} for $R_{c,b}{}^2$ values sampled from the bootstrap $R_{c,b}{}^2$ distribution. The median ρ_{signal} is 0.68 (dashed line), with 90% CI 0.24-0.95 estimated by the 5th-95th percentiles of the marginal distribution (dot-dashed lines).

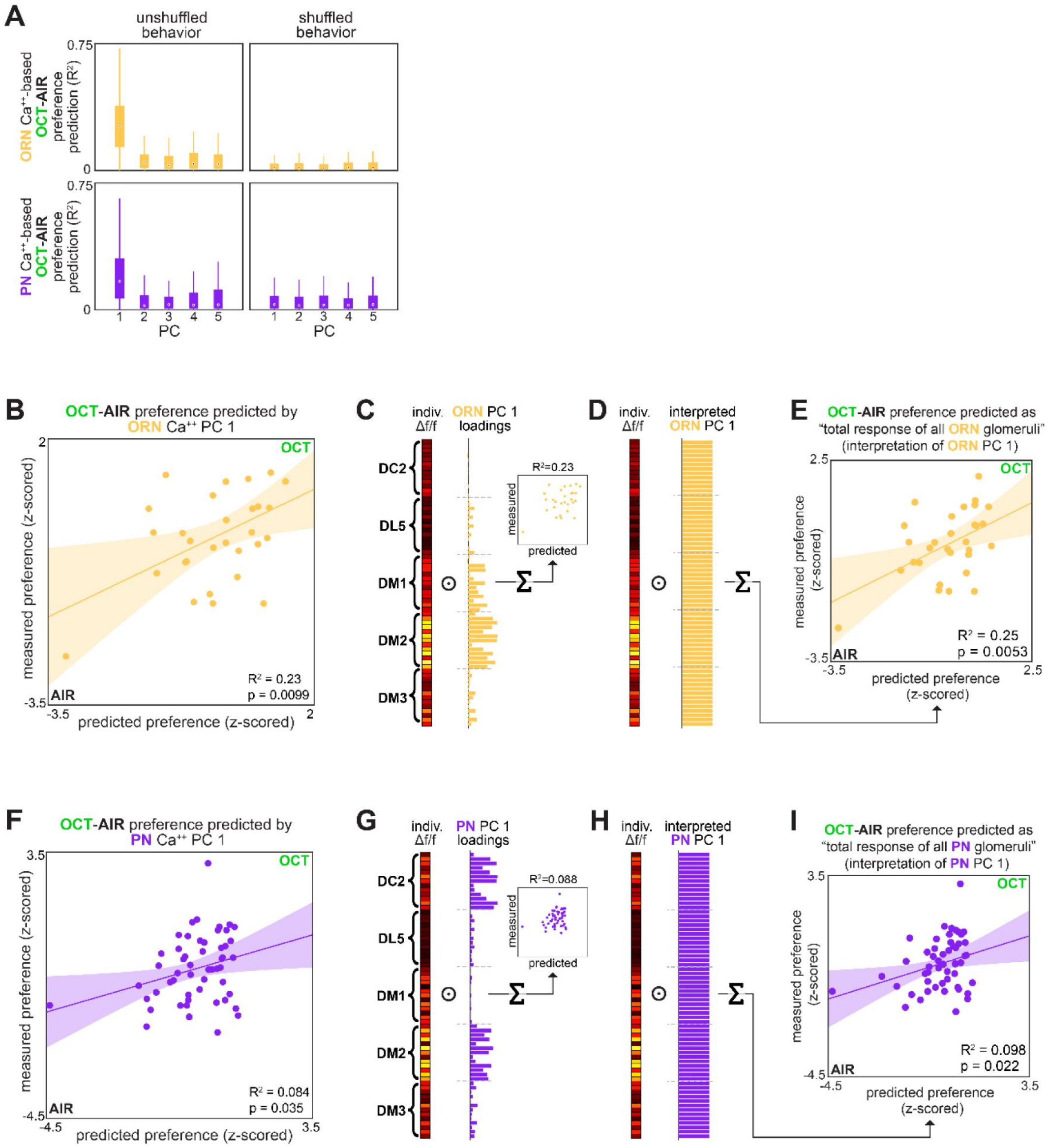

Figure 1 – figure supplement 10.

OCT-AIR preference prediction.

(A) Bootstrapped R^2 of OCT-AIR preference prediction from each of the first 5 principal components of neural activity measured in ORNs (top, all data) or PNs (bottom, training set). (B) Measured OCT-AIR preference versus preference predicted from PC 1 of ORN activity (n = 30 flies). (C) PC 1 loadings of ORN activity for flies in B. (D) Interpreted ORN PC 1 loadings. **(A)** (E) Measured OCT-AIR preference versus preference predicted by the average peak response across all ORN coding dimensions (n = 30 flies). (F) Measured OCT-AIR preference versus preference predicted from PC 1 of PN activity in n = 53 flies using a model trained on a training set of n = 18 flies (see **Figure 2 – figure supplement 1A-B** for train/test flies analyzed separately). **(B)** (G) PC 2 loadings of PN activity for flies in F. (H) Interpreted PN PC2 loadings. (I) Measured OCT-MCH preference versus preference predicted by the average peak PN response in DM2 minus DC2 across all odors (n = 69 flies).

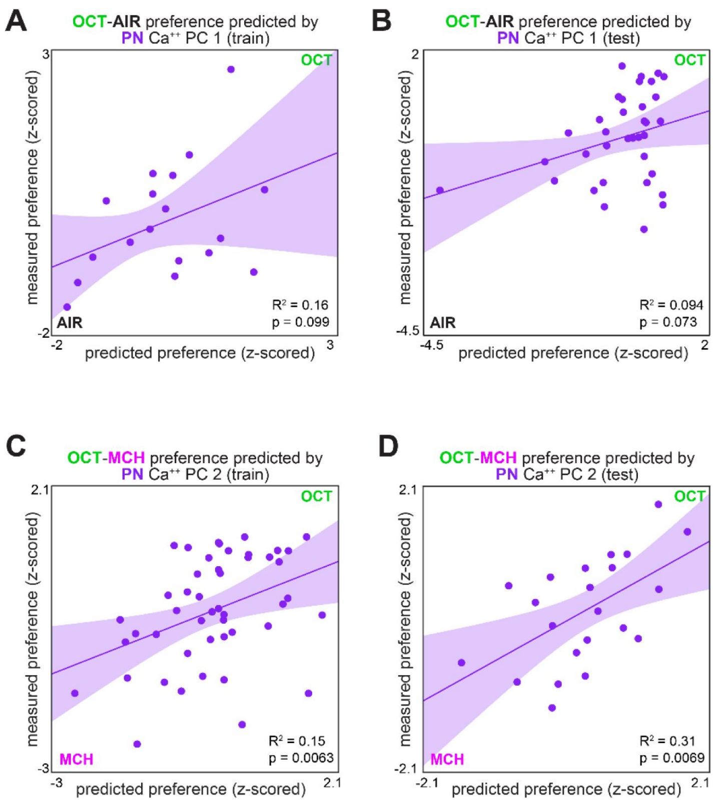

Figure 2 – figure supplement 1.

Measured preference vs. PN activity-based predicted preference, split by training/testing set.

(A) Measured OCT-AIR preference versus preference predicted from PC 1 of PN activity in a training set (n = 18 flies). **(B)** Measured OCT-AIR preference versus preference predicted from PC 1 on PN activity in a test set (n = 35 flies) evaluated on a model trained on data from panel(A). **(C)** Measured OCT-MCH preference versus preference predicted from PC 2 of PN activity in a training set (n = 47 flies). **(D)** Measured OCT-MCH preference versus preference predicted from PC 2 on PN activity in a test set (n = 22 flies) evaluated on a model trained on data from panel (C).

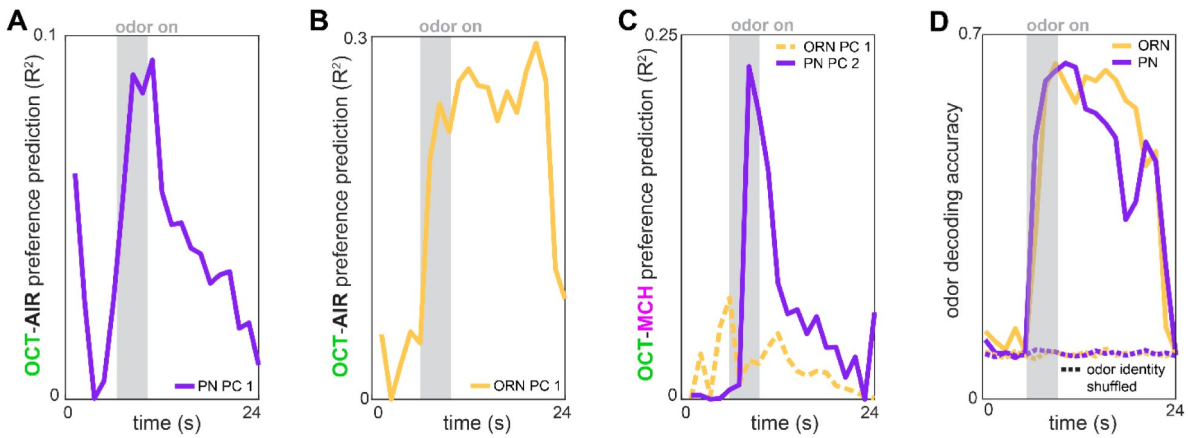

Figure 2 – figure supplement 2.

Time-dependent preference- and odor-decoding.

(A) R^2 of odor-vs-air preference predicted by PC 1 of PN activity as a function of time across trials (n = 53 flies). **(B)** R^2 of odor-vs-air preference predicted by PC 1 of ORN activity as a function of time across trials (n = 30 flies). **(C)** R^2 of odor-vs-odor preference predicted by PC 2 of PN activity (solid plum, n = 69 flies) or PC 1 of ORN activity (dashed peach, n = 35 flies) as a function of time across trials. **(D)** Logistic regression classifier accuracy of decoding odor identity from 5 glomerular responses as a function of time. Dashed curves indicate performance on shuffled data.

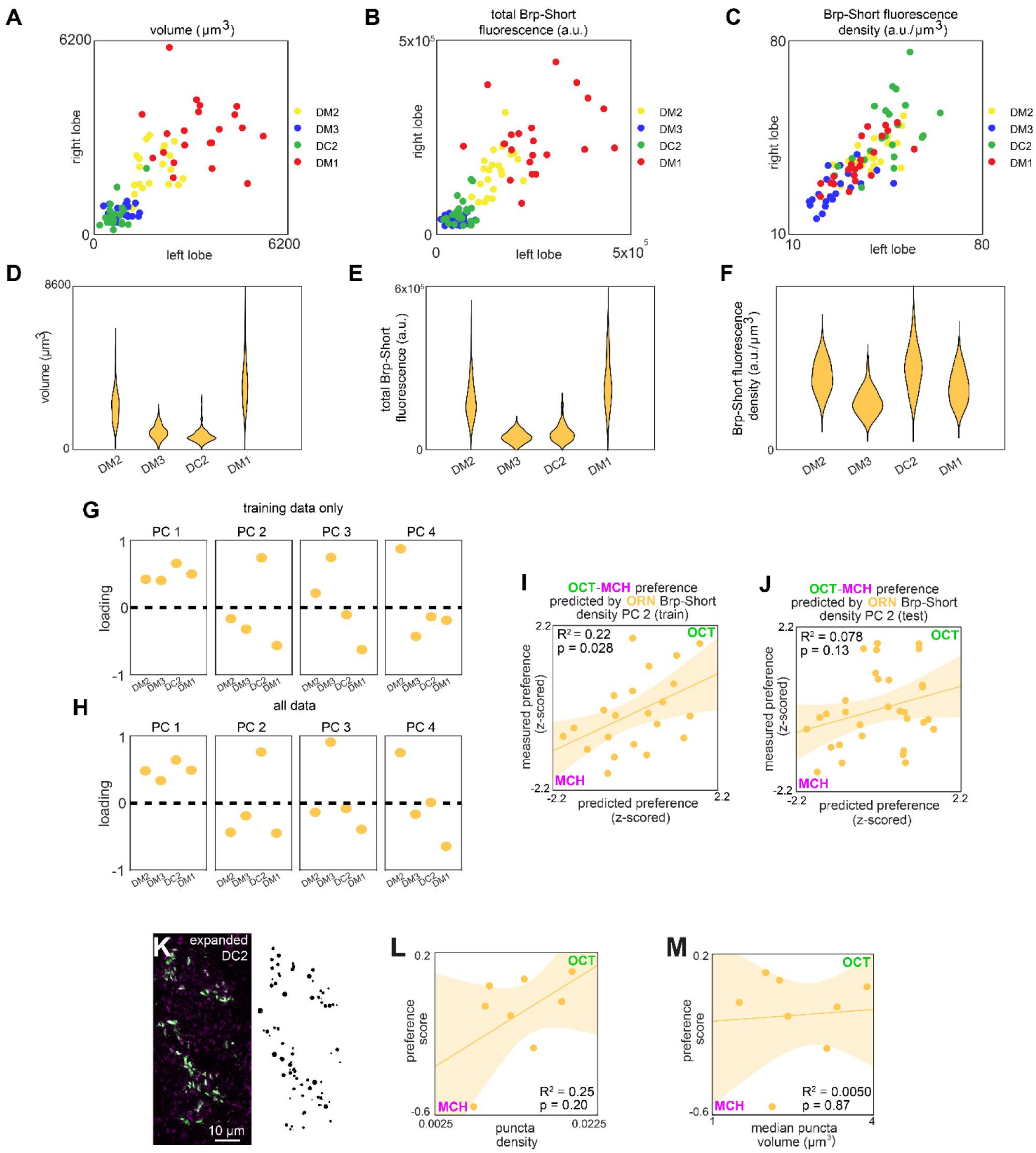

Figure 3 – figure supplement 1.

ORN>Brp-Short characterization and model predictions. (A-C)

Right versus left glomerulus properties measured from z-stacks of stained Orco>Brp- Short samples: (A) Volume, (B) total Brp-Short fluorescence, (C) Brp-Short fluorescence density. **(D-F)** Same data as panels (A-C) represented in violin plots (kernel density estimated). **(G)** Principal component loadings of Brp-Short density calculated using only training data (n = 22 flies). **(H)** Principal component loadings of Brp-Short density calculated using all data (n = 53 flies). **(I)** Measured OCT-MCH preference versus preference predicted from PC 2 of ORN Brp- Short density in a training set (n = 22 flies). **(J)** Measured OCT-MCH preference versus preference predicted from PC 2 on ORN Brp-Short density in a test set (n = 31 flies) evaluated on a model trained on data from panel (I). **(K)** Example expanded AL expressing Or13a>Brp- Short (left) and Imaris-identified puncta from that sample (right). **(L)** OCT-MCH preference score plotted against Brp-Short puncta density in expanded Or13a>Brp-Short samples (n = 8 flies). **(M)** OCT-MCH preference score plotted against Brp-Short median puncta volume in expanded Or13a>Brp-Short samples (n = 8 flies). Shaded regions in I,J,L,M are the 95% CI of the fit estimated by bootstrapping.

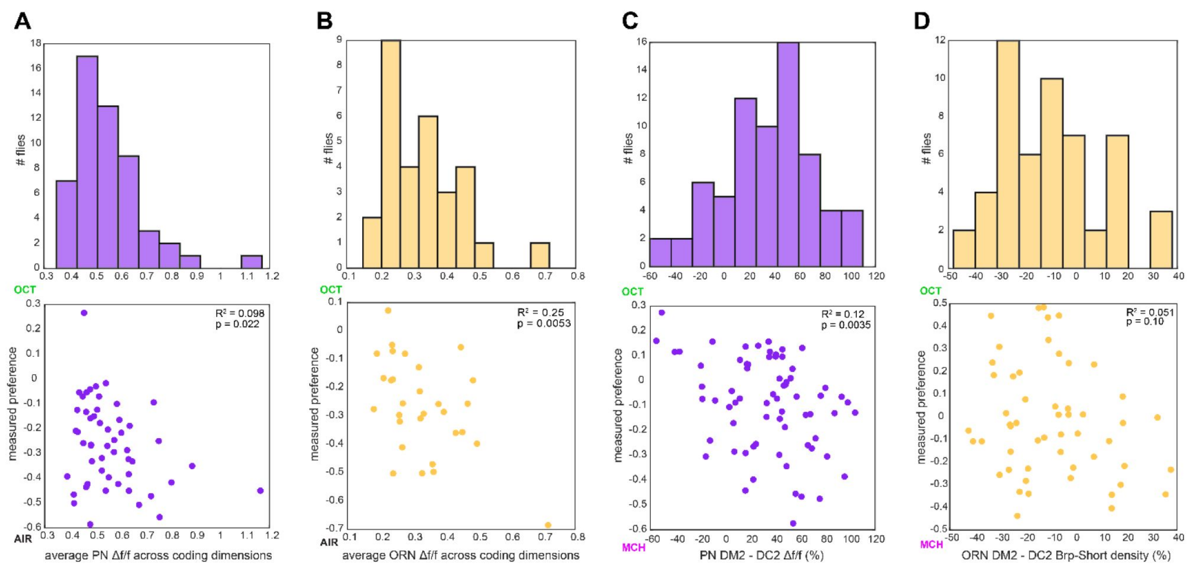

Figure 3 – figure supplement 2.

Calcium and Brp-Short predictor variation.

(A) Histogram of average PN Δf/f across all coding dimensions in flies in which OCT-AIR preference was measured (top) and OCT-AIR preference versus average PN Δf/f (n = 53 flies) (bottom). **(B)** Similar to (A) for ORN Δf/f and OCT-AIR preference (n = 30 flies). **(C)** Similar to (A) for Δf/f difference between DM2 and DC2 PN responses and OCT-MCH preference (n = 69 flies). **(D)** Similar to (A) for % Brp-Short density difference between DM2 and DC2 ORNs and OCT-MCH (n = 53 flies).

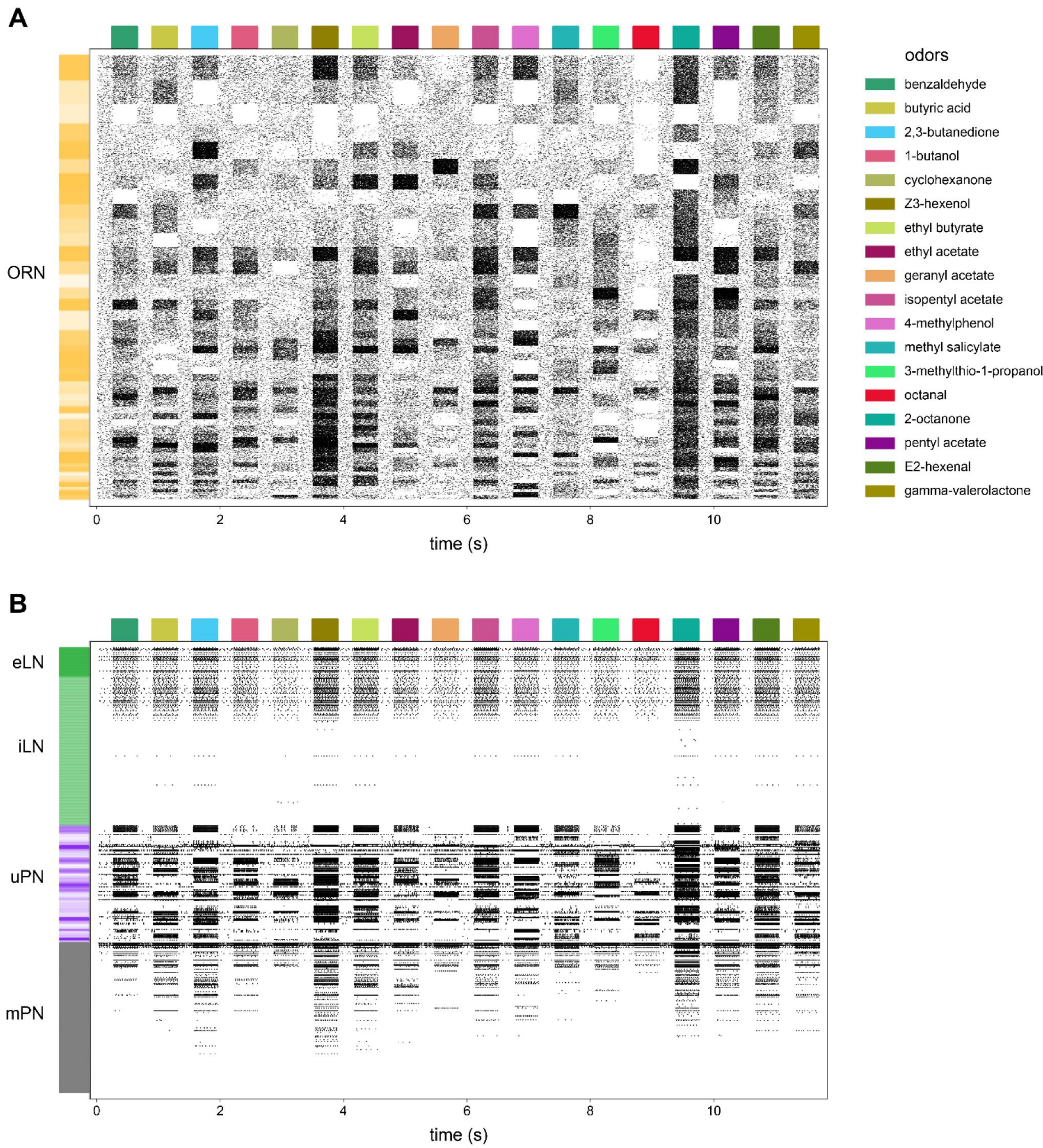

Figure 4 – figure supplement 1.

AL model raster plot.

(A) Action potential raster plot of ORNs in the baseline simulated AL. Rows are individual ORNs, black ticks indicate action potentials. Random shades of gold at left indicate blocks of ORN rows projecting to the same glomerulus. **(B)** The remaining neurons in the model. Shades of green indicate excitatory vs inhibitory LNs and shades of purple indicate PNs with dendrites in the same glomeruli.

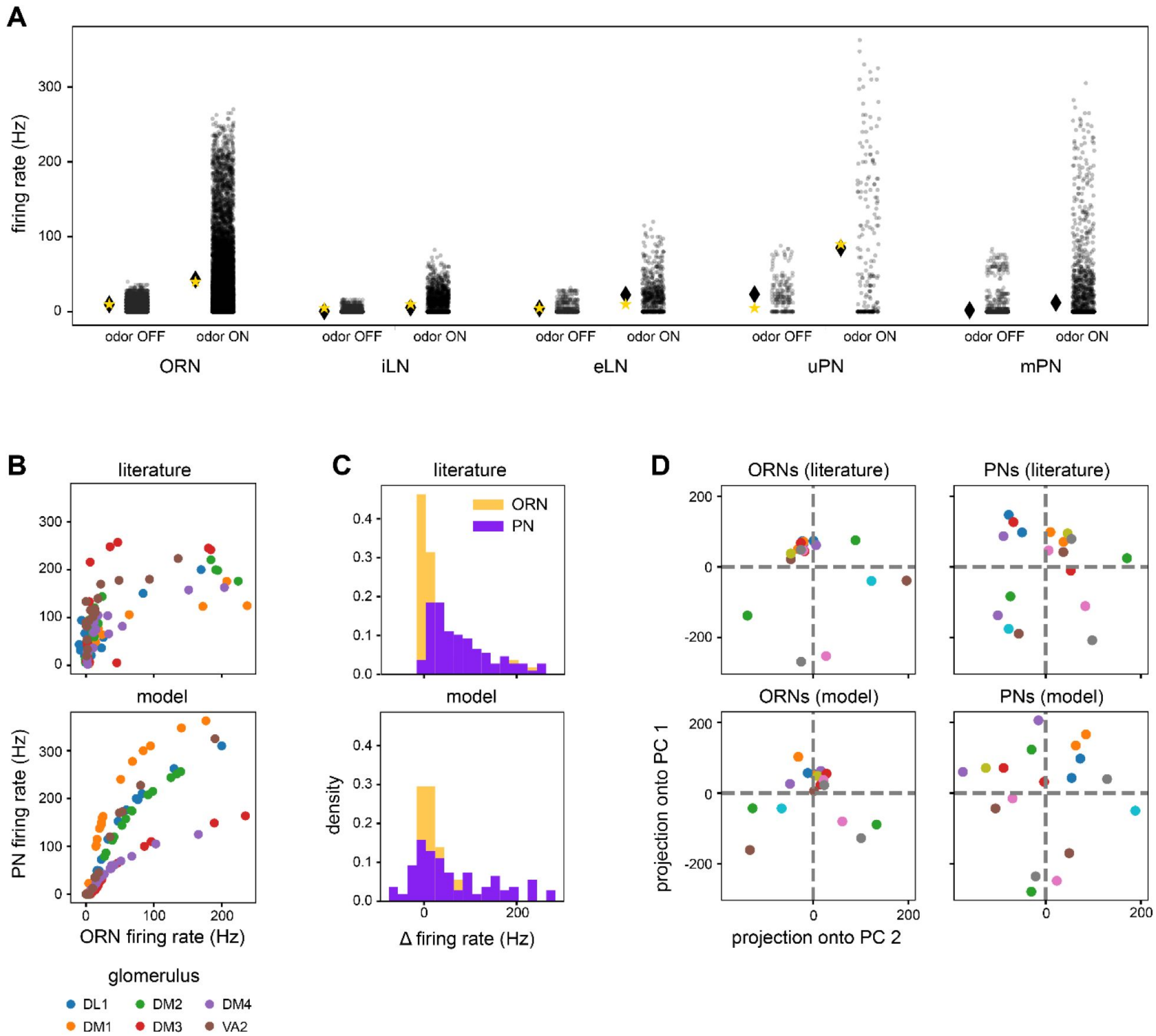

Figure 4 – figure supplement 2.

AL model baseline outputs compared to experimental data.

(A) Distributions of model neuron firing rates by cell type across odors (transparent black points are individual neuron-odor combinations). Black lozenge symbols indicate the mean firing rate of the points to the right. Yellow stars indicate the comparable experimental values reported in (Chou et al., 2010; de Bruyne et al., 2001; Nagel et al., 2015; Wilson, 2004). **(B)** Scatter plots of average PN firing rate vs ORN firing rate during odor stimuli in the model vs experimental values (Bhandawat et al., 2007). Points are odors, colors are glomeruli. **(C)** Histograms of ON odor minus OFF odor glomerulus-average PN and ORN firing rates in the model vs experimental values (Bhandawat et al., 2007), showing flatter distributions in PNs. **(D)** Odor representations in the first 2 PCs of glomerulus-average ORN responses and PN responses in the model and experimental results (Bhandawat et al., 2007). Points are odors. Pairwise distances between PN representations are more uniform than in ORNs in both the model and experimental data. Panels (B)-(D) use glomerulus-average PN and ORN firing rates from six of the seven glomeruli in Bhandawat et al., 2007, as VM2 is significantly truncated in the hemibrain (Scheffer et al., 2020). Literature features in panels (B)-(D) were extracted from Bhandawat et al., 2007 using WebPlotDigitizer (Rohatgi, 2021).

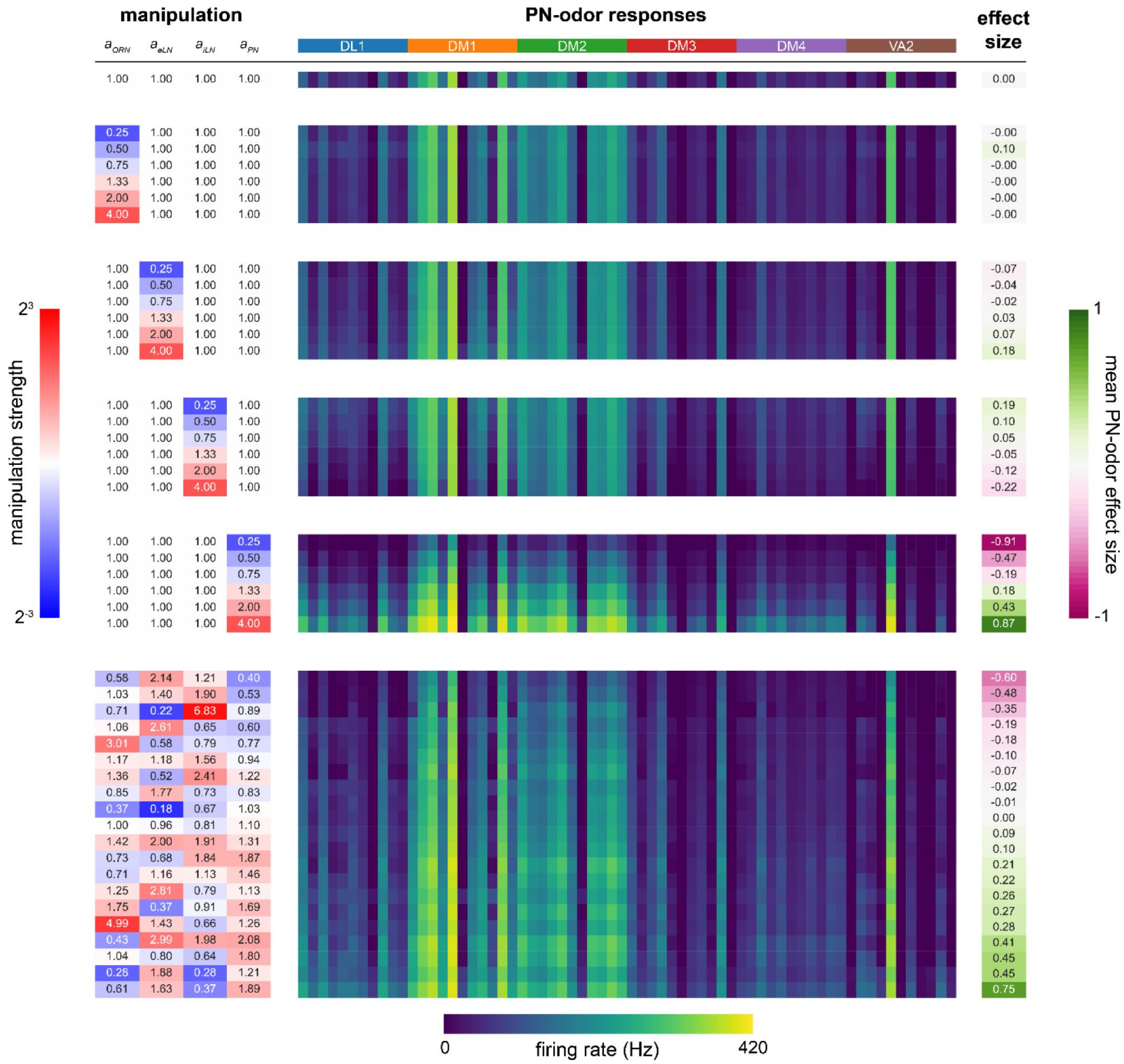

Figure 4 – figure supplement 3.

Sensitivity analysis of a_{ORN}, a_{eLN}, a_{iLN}, a_{PN} parameters.

(Left, blue to red colormap): magnitude of parameter manipulation. (Center, dark blue to yellow colormap): mean glomerular firing rate (Hz) responses of PNs (DL1, DM1, DM2, DM3, DM4, VA2) to 11 odors (order within each glomerulus (colored bands at top): 3-octanol, 1-hexanol, ethyl lactate, 2-heptanone, 1-pentanol, ethanol, geranyl acetate, hexyl acetate, 4-methylcyclohexanol, pentyl acetate, 1-butanol, 3-octanol). (Right, pink to green colormap): manipulation effect size on mean PN-odor responses (Cohen's *d*). (Top): baseline parameter set. (Middle): single-parameter manipulations from 1/4x to 4x. (Bottom): multiple-parameter manipulations. For further detail see *AL model tuning* in Materials and Methods. No manipulations yielded effect sizes larger than 0.9; a_{PN} is the most sensitive parameter.

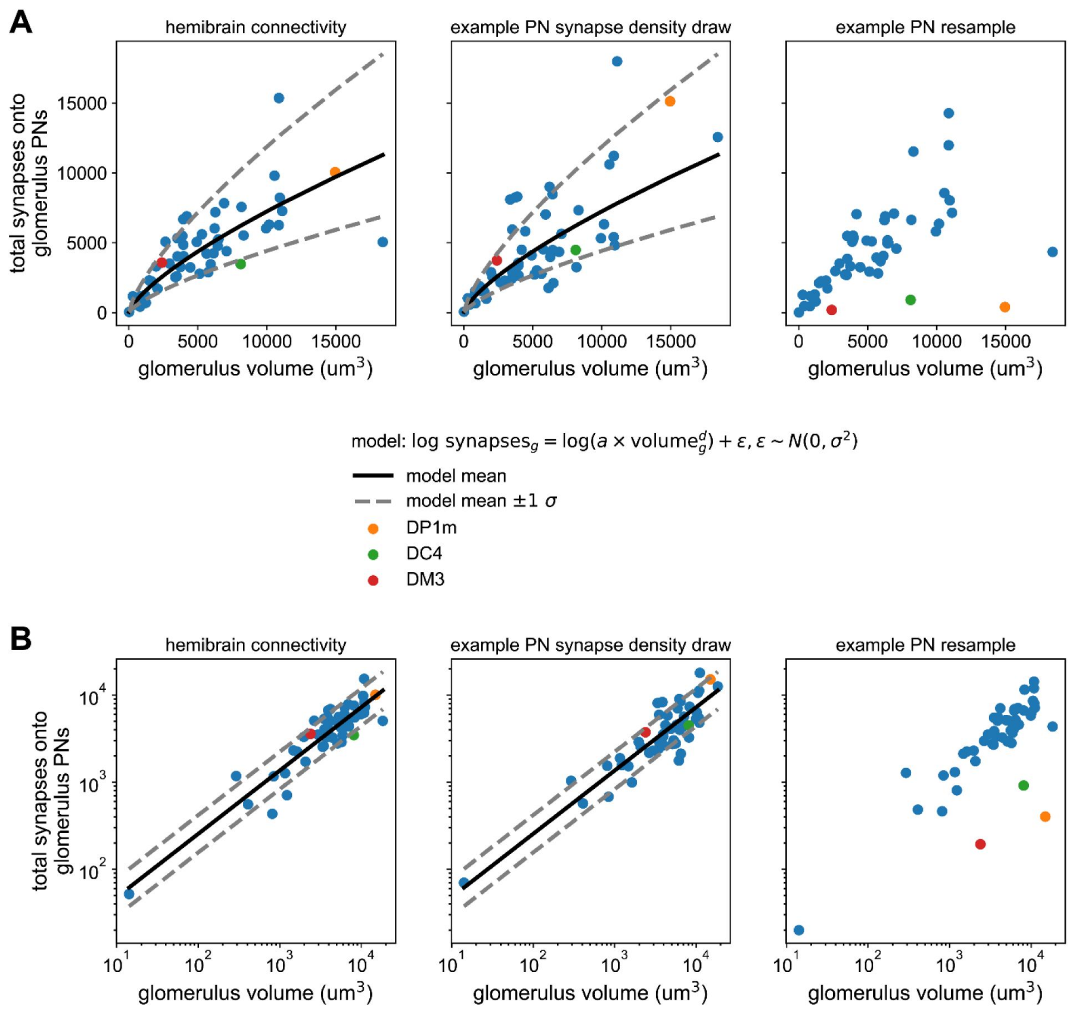

Figure 4 – figure supplement 4.

Synapse counts vs glomerular volume in the hemibrain and AL model.

(A) Left) Scatter plot of total PN input synapses within a glomerulus vs that glomerulus' volume from the hemibrain data set. Solid line represents the maximum likelihood-fit mean synapse count vs glomerular volume, and dashed lines the fit +/-1 standard deviation. Middle) As (left) but for a single sample from the parameterized distribution of PN input synapses vs glomerular volume. Right) As in previous for a single bootstrap resample of PNs. Color-highlighted glomeruli illustrate that when PNs within a glomerulus have highly asymmetrical synapse counts, bootstrapping them alone can result in apparent synapse densities that lie outside the empirical distribution (left). **(B)** As in (A) but on log-log axes, showing the linear relationship between synapse density and glomerular volume after this transformation, and bootstrapped densities falling outside this distribution at right.

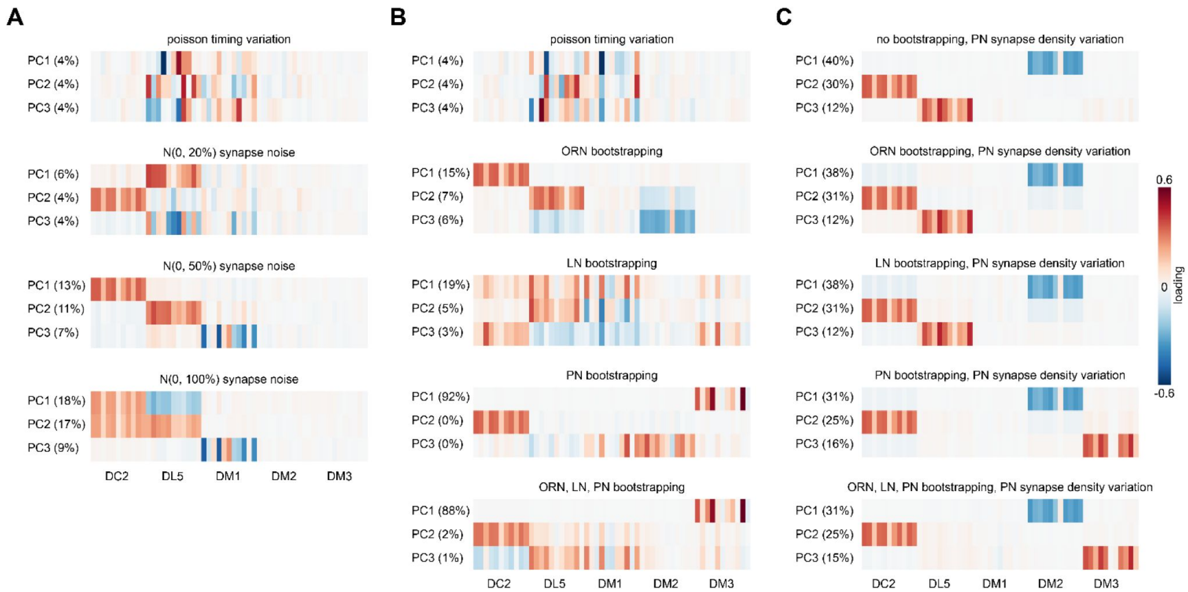

Figure 4 – figure supplement 5.

PN response PCA loadings under various sources of circuit idiosyncrasy.

(A) Loadings of the principal components of PN glomerulus-odor responses as simulated across AL models where Gaussian noise with a standard deviation equal to 0, 20, 50, and 100% of each synapse weight was added to each synaptic weight in the hemibrain data set. **(B)** circuit variation coming from bootstrapping of each major AL cell type or all three simultaneously. **(C)** circuit variation coming from bootstrap resampling of different cell-type combinations in addition to PN input synapse density resampling as illustrated in **Figure 4 – figure supplement 4**.

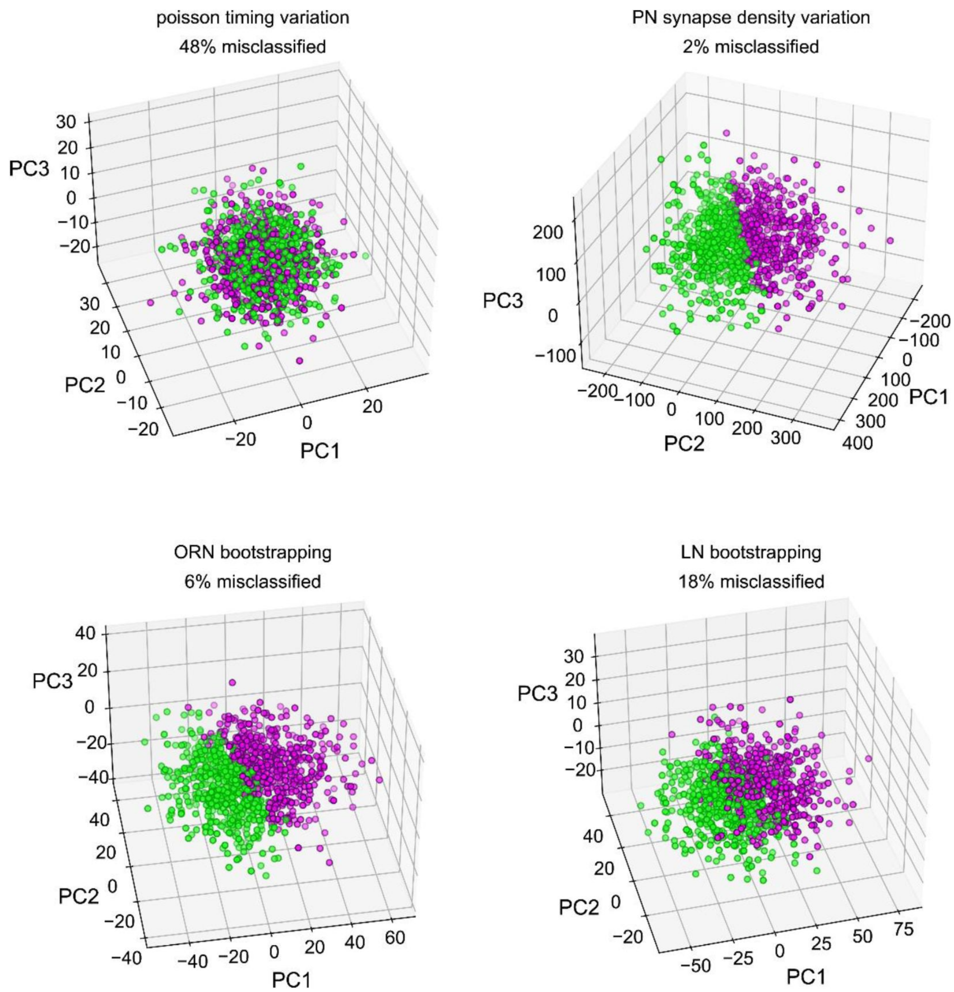

Figure 4 – figure supplement 6.

Classifiability of simulated idiosyncratic behavior under different sources of circuit idiosyncrasy.

Simulated PN odor-glomerulus firing rates projected into their first 3 principal components. Individual points represent single runs of resampled AL models, under four different sources of idiosyncratic variation. PN responses in all odor-glomerulus dimensions were used to calculate simulated behavior scores for each resampled AL, by applying the PN calcium-odor-vs-odor linear model (**Figure 2A**). Magenta points represent flies with simulated preference for MCH in the top 50%, and green OCT preference. % Misclassification refers to 100% – the accuracy of a linear classifier trained on MCH-vs-OCT preference in the space of the first three PCs. This measures how much of the variance along the PN calcium-odor-vs-odor linear model lies outside the first three PCs of simulated PN variation.

References

Akhund-Zade J, Ho S, O'Leary C, de Bivort B. (2019) **The effect of environmental enrichment on behavioral variability depends on genotype, behavior, and type of enrichment** *J Exp Biol* **222** https://doi.org/10.1242/jeb.202234

Akhund-Zade J *et al.* (2020) **Wild flies hedge their thermal preference bets in response to seasonal fluctuations** *bioRxiv* https://doi.org/10.1101/2020.09.16.300731

Asano SM, Gao R, Wassie AT, Tillberg PW, Chen F, Boyden ES (2018) **Expansion Microscopy: Protocols for Imaging Proteins and RNA in Cells and Tissues** *Curr Protoc Cell Biol* **80** https://doi.org/10.1002/cpcb.56

Ayroles JF, Buchanan SM, O'Leary C, Skutt-Kakaria K, Grenier JK, Clark AG, Hartl DL, De Bivort BL. (2015) **Behavioral idiosyncrasy reveals genetic control of phenotypic variability** *Proceedings of the National Academy of Sciences* **112**:6706–6711

Bates AS *et al.* (2020) **Complete Connectomic Reconstruction of Olfactory Projection Neurons in the Fly Brain** *Curr Biol* https://doi.org/10.1016/j.cub.2020.06.042

Berman GJ, Choi DM, Bialek W, Shaevitz JW (2014) **Mapping the stereotyped behaviour of freely moving fruit flies** *J R Soc Interface* **11** https://doi.org/10.1098/rsif.2014.0672

Bhandawat V, Olsen SR, Gouwens NW, Schlief ML, Wilson RI (2007) **Sensory processing in the Drosophila antennal lobe increases reliability and separability of ensemble odor representations** *Nat Neurosci* **10**:1474–1482 https://doi.org/10.1038/nn1976

Britten KH, Newsome WT, Shadlen MN, Celebrini S, Movshon JA (1996) **A relationship between behavioral choice and the visual responses of neurons in macaque MT** *Vis Neurosci* **13**:87–100 https://doi.org/10.1017/s095252380000715x

Buchanan SM, Kain JS, de Bivort BL. (2015) **Neuronal control of locomotor handedness in Drosophila** *Proc Natl Acad Sci U S A* **112**:6700–6705 https://doi.org/10.1073/pnas.1500804112

Cao L-H, Jing B-Y, Yang D, Zeng X, Shen Y, Tu Y, Luo D-G (2016) **Distinct signaling of Drosophila chemoreceptors in olfactory sensory neurons** *Proc Natl Acad Sci U S A* **113**:E902–11 https://doi.org/10.1073/pnas.1518329113

Chen T-W *et al.* (2013) **Ultrasensitive fluorescent proteins for imaging neuronal activity** *Nature* **499**:295–300 https://doi.org/10.1038/nature12354

Chou Y-H, Spletter ML, Yaksi E, Leong JCS, Wilson RI, Luo L (2010) **Diversity and wiring variability of olfactory local interneurons in the Drosophila antennal lobe** *Nat Neurosci* **13**:439–449 https://doi.org/10.1038/nn.2489

Claridge-Chang A, Roorda RD, Vrontou E, Sjulson L, Li H, Hirsh J, Miesenböck G (2009) **Writing memories with light-addressable reinforcement circuitry** *Cell* **139**:405–415 https://doi.org/10.1016/j.cell.2009.08.034

Couto A, Alenius M, Dickson BJ (2005) **Molecular, anatomical, and functional organization of the Drosophila olfactory system** *Curr Biol* **15**:1535–1547 https://doi.org/10.1016/j.cub.2005.07.034

de Bivort B *et al.* (2022) **Precise Quantification of Behavioral Individuality From 80 Million Decisions Across 183,000 Flies** *Front Behav Neurosci* **16** https://doi.org/10.3389/fnbeh.2022.836626

de Bruyne M, Foster K, Carlson JR. (2001) **Odor coding in the Drosophila antenna** *Neuron* **30**:537–552 https://doi.org/10.1016/s0896-6273(01)00289-6

Dubin AE, Harris GL (1997) **Voltage-activated and odor-modulated conductances in olfactory neurons of Drosophila melanogaster** *Journal of Neurobiology* **32**:123–137 https://doi.org/10.1002/(SICI)1097-4695(199701)32:1<123::AID-NEU11>3.0.CO;2-L

Freund J, Brandmaier AM, Lewejohann L, Kirste I, Kritzler M, Krüger A, Sachser N, Lindenberger U, Kempermann G (2013) **Emergence of individuality in genetically identical mice** *Science* **340**:756–759 https://doi.org/10.1126/science.1235294

Gao R *et al.* (2019) **Cortical column and whole-brain imaging with molecular contrast and nanoscale resolution** *Science* **363** https://doi.org/10.1126/science.aau8302

Golovin RM, Broadie K (2016) **Developmental experience-dependent plasticity in the first synapse of the Drosophila olfactory circuit** *J Neurophysiol* **116**:2730–2738 https://doi.org/10.1152/jn.00616.2016

Gomez-Marin A, Ghazanfar AA (2019) **The Life of Behavior** *Neuron* **104**:25–36 https://doi.org/10.1016/j.neuron.2019.09.017

Grabe V, Baschwitz A, Dweck HKM, Lavista-Llanos S, Hansson BS, Sachse S (2016) **Elucidating the Neuronal Architecture of Olfactory Glomeruli in the Drosophila Antennal Lobe** *Cell Rep* **16**:3401–3413 https://doi.org/10.1016/j.celrep.2016.08.063

Grabe V, Strutz A, Baschwitz A, Hansson BS, Sachse S (2015) **Digital in vivo 3D atlas of the antennal lobe of Drosophila melanogaster** *J Comp Neurol* **523**:530–544 https://doi.org/10.1002/cne.23697

Hallem EA, Ho MG, Carlson JR (2004) **The molecular basis of odor coding in the Drosophila antenna** *Cell* **117**:965–979 https://doi.org/10.1016/j.cell.2004.05.012

Honegger K, de Bivort B. (2018) **Stochasticity, individuality and behavior** *Curr Biol* **28**:R8–R12 https://doi.org/10.1016/j.cub.2017.11.058

Honegger KS, Smith MA-Y, Churgin MA, Turner GC, de Bivort BL. (2019) **Idiosyncratic neural coding and neuromodulation of olfactory individuality in Drosophila** *Proc Natl Acad Sci U S A* https://doi.org/10.1073/pnas.1901623116

Hopper KR (1999) **Risk-spreading and bet-hedging in insect population biology** *Annu Rev Entomol* **44**:535–560 https://doi.org/10.1146/annurev.ento.44.1.535

Huang Y-C, Wang C-T, Su T-S, Kao K-W, Lin Y-J, Chuang C-C, Chiang A-S, Lo C-C (2018) **A Single-Cell Level and Connectome-Derived Computational Model of the Drosophila Brain** *Front Neuroinform* **12** https://doi.org/10.3389/fninf.2018.00099

Iyengar A, Chakraborty TS, Goswami SP, Wu C-F, Siddiqi O (2010) **Post-eclosion odor experience modifies olfactory receptor neuron coding in Drosophila** *Proc Natl Acad Sci U S A* **107**:9855–9860 https://doi.org/10.1073/pnas.1003856107

Jeanne JM, Wilson RI (2015) **Convergence, Divergence, and Reconvergence in a Feedforward Network Improves Neural Speed and Accuracy** *Neuron* **88**:1014–1026 https://doi.org/10.1016/j.neuron.2015.10.018

Johnson W, Turkheimer E, Gottesman II, Jr Bouchard TJ (2010) **Beyond Heritability: Twin Studies in Behavioral Research** *Curr Dir Psychol Sci* **18**:217–220 https://doi.org/10.1111/j.1467-8721.2009.01639.x

Kain JS, Stokes C, de Bivort BL. (2012) **Phototactic personality in fruit flies and its suppression by serotonin and white** *Proc Natl Acad Sci U S A* **109**:19834–19839 https://doi.org/10.1073/pnas.1211988109

Kain JS, Zhang S, Akhund-Zade J, Samuel ADT, Klein M, de Bivort BL. (2015) **Variability in thermal and phototactic preferences in Drosophila may reflect an adaptive bet-hedging strategy** *Evolution* **69**:3171–3185 https://doi.org/10.1111/evo.12813

Kakaria KS, de Bivort BL. (2017) **Ring Attractor Dynamics Emerge from a Spiking Model of the Entire Protocerebral Bridge** *Front Behav Neurosci* **11** https://doi.org/10.3389/fnbeh.2017.00008

Kao K-W, Lo C-C (2020) **Short term depression, presynaptic inhibition and local neuron diversity play key functional roles in the insect antennal lobe** *J Comput Neurosci* https://doi.org/10.1007/s10827-020-00747-4

Kluyver T *et al.* (2016) **Jupyter Notebooks -- a publishing format for reproducible computational workflows In: Loizides F, Schmidt B, editors. Positioning and Power in Academic Publishing: Players, Agents and Agendas** IOS Press :87–90

Körholz JC, Zocher S, Grzyb AN, Morisse B, Poetzsch A, Ehret F, Schmied C, Kempermann G (2018) **Selective increases in inter-individual variability in response to environmental enrichment in female mice** *eLife* **7** https://doi.org/10.7554/eLife.35690

Koulakov AA, Rinberg DA, Tsigankov DN (2005) **How to find decision makers in neural networks** *Biol Cybern* **93**:447–462 https://doi.org/10.1007/s00422-005-0022-z

Krams IA *et al.* (2021) **Serotoninergic Modulation of Phototactic Variability Underpins a Bet-Hedging Strategy in Drosophila melanogaster** *Front Behav Neurosci* **15** https://doi.org/10.3389/fnbeh.2021.659331

Laskowski KL, Bierbach D, Jolles JW, Doran C, Wolf M (2022) **The emergence and development of behavioral individuality in clonal fish** *Nat Commun* **13**:1–9 https://doi.org/10.1038/s41467-022-34113-y

Li H, Horns F, Wu B, Xie Q, Li J, Li T, Luginbuhl DJ, Quake SR, Luo L (2017) **Classifying Drosophila Olfactory Projection Neuron Subtypes by Single-Cell RNA Sequencing** *Cell* **171**:1206–1220 https://doi.org/10.1016/j.cell.2017.10.019

Lillvis JL *et al.* (2022) **Rapid reconstruction of neural circuits using tissue expansion and light sheet microscopy** *eLife* **11** https://doi.org/10.7554/eLife.81248

Linneweber GA *et al.* (2020) **A neurodevelopmental origin of behavioral individuality in the Drosophila visual system** *Science* **367**:1112–1119 https://doi.org/10.1126/science.aaw7182

Liu TX, Davoudian PA, Lizbinski KM, Jeanne JM (2022) **Connectomic features underlying diverse synaptic connection strengths and subcellular computation** *Curr Biol* **32**:559–569 https://doi.org/10.1016/j.cub.2021.11.056

Luo SX, Axel R, Abbott LF (2010) **Generating sparse and selective third-order responses in the olfactory system of the fly** *Proc Natl Acad Sci U S A* **107**:10713–10718 https://doi.org/10.1073/pnas.1005635107

Maloney RT (2021) **Neuromodulation and Individuality** *Front Behav Neurosci* **15** https://doi.org/10.3389/fnbeh.2021.777873

Marder E (2011) **Variability, compensation, and modulation in neurons and circuits** *Proc Natl Acad Sci U S A* **108**:15542–15548 https://doi.org/10.1073/pnas.1010674108

The MathWorks Inc **MATLAB pca documentation** MathWorks

Mazor O, Laurent G (2005) **Transient dynamics versus fixed points in odor representations by locust antennal lobe projection neurons** *Neuron* **48**:661–673 https://doi.org/10.1016/j.neuron.2005.09.032

McLaughlin CN *et al.* (2021) **Single-cell transcriptomes of developing and adult olfactory receptor neurons in Drosophila** *eLife* **10** https://doi.org/10.7554/eLife.63856

Mellert DJ, Williamson WR, Shirangi TR, Card GM, Truman JW (2016) **Genetic and Environmental Control of Neurodevelopmental Robustness in Drosophila** *PLoS One* **11** https://doi.org/10.1371/journal.pone.0155957

Michelson CA, Pillow JW, Seidemann E. (2017) **Majority of choice-related variability in perceptual decisions is present in early sensory cortex** *bioRxiv* https://doi.org/10.1101/207357

Mosca TJ, Luo L (2014) **Synaptic organization of the Drosophila antennal lobe and its regulation by the Teneurins** *eLife* https://doi.org/10.7554/elife.03726

Münch D, Galizia CG (2016) **DoOR 2.0--Comprehensive Mapping of Drosophila melanogaster Odorant Responses** *Sci Rep* **6** https://doi.org/10.1038/srep21841

Nagel KI, Hong EJ, Wilson RI (2015) **Synaptic and circuit mechanisms promoting broadband transmission of olfactory stimulus dynamics** *Nat Neurosci* **18**:56–65 https://doi.org/10.1038/nn.3895

Newsome WT, Britten KH, Anthony Movshon J (1989) **Neuronal correlates of a perceptual decision** *Nature* https://doi.org/10.1038/341052a0

Olsen SR, Wilson RI (2008) **Lateral presynaptic inhibition mediates gain control in an olfactory circuit** *Nature* **452**:956–960 https://doi.org/10.1038/nature06864

Osborne LC, Lisberger SG, Bialek W (2005) **A sensory source for motor variation** *Nature* **437**:412–416 https://doi.org/10.1038/nature03961

Pisokas I, Heinze S, Webb B (2020) **The head direction circuit of two insect species** *eLife* **9** https://doi.org/10.7554/eLife.53985

Raj A, Rifkin SA, Andersen E, van Oudenaarden A. (2010) **Variability in gene expression underlies incomplete penetrance** *Nature* **463**:913–918 https://doi.org/10.1038/nature08781

Rihani K, Sachse S (2022) **Shedding Light on Inter-Individual Variability of Olfactory Circuits in Drosophila** *Front Behav Neurosci* **16** https://doi.org/10.3389/fnbeh.2022.835680

Rohatgi A. (2021) **Webplotdigitizer** *automeris.io*

Sachse S, Rueckert E, Keller A, Okada R, Tanaka NK, Ito K, Vosshall LB (2007) **Activity-dependent plasticity in an olfactory circuit** *Neuron* **56**:838–850 https://doi.org/10.1016/j.neuron.2007.10.035

Scheffer LK *et al.* (2020) **A connectome and analysis of the adult Drosophila central brain** *eLife* **9** https://doi.org/10.7554/eLife.57443

Schlegel P *et al.* (2020) **Information flow, cell types and stereotypy in a full olfactory connectome** *bioRxiv* https://doi.org/10.1101/2020.12.15.401257

Schuett W, Dall SRX, Baeumer J, Kloesener MH, Nakagawa S, Beinlich F, Eggers T (2011) **Personality variation in a clonal insect: the pea aphid, Acyrthosiphon pisum** *Dev Psychobiol* **53**:631–640 https://doi.org/10.1002/dev.20538

Seki Y, Rybak J, Wicher D, Sachse S, Hansson BS (2010) **Physiological and morphological characterization of local interneurons in the Drosophila antennal lobe** *J Neurophysiol* **104**:1007–1019 https://doi.org/10.1152/jn.00249.2010

Shang Y, Claridge-Chang A, Sjulson L, Pypaert M, Miesenböck G (2007) **Excitatory local circuits and their implications for olfactory processing in the fly antennal lobe** *Cell* **128**:601–612 https://doi.org/10.1016/j.cell.2006.12.034

Sizemore TR, Dacks AM (2016) **Serotonergic Modulation Differentially Targets Distinct Network Elements within the Antennal Lobe of Drosophila melanogaster** *Sci Rep* **6** https://doi.org/10.1038/srep37119

Skutt-Kakaria K, Reimers P, Currier TA, Werkhoven Z, de Bivort BL. (2019) **A neural circuit basis for context-modulation of individual locomotor behavior** *bioRxiv* https://doi.org/10.1101/797126

Stern S, Kirst C, Bargmann CI (2017) **Neuromodulatory Control of Long-Term Behavioral Patterns and Individuality across Development** *Cell* **171**:1649–1662 https://doi.org/10.1016/j.cell.2017.10.041

Tobin WF, Wilson RI, Lee W-CA (2017) **Wiring variations that enable and constrain neural computation in a sensory microcircuit** *eLife* **6** https://doi.org/10.7554/eLife.24838

Tsai K-T, Hu C-K, Li K-W, Hwang W-L, Chou Y-H (2018) **Circuit variability interacts with excitatory-inhibitory diversity of interneurons to regulate network encoding capacity** *Sci Rep* **8** https://doi.org/10.1038/s41598-018-26286-8

van Rossum G, Drake FL. (2011) **The Python Language Reference Manual** *Network Theory Ltd*

Werkhoven Z, Bravin A, Skutt-Kakaria K, Reimers P, Pallares LF, Ayroles J, de Bivort BL. (2021) **The structure of behavioral variation within a genotype** *eLife* **10** https://doi.org/10.7554/eLife.64988

Wilson RI (2013) **Early Olfactory Processing in Drosophila: Mechanisms and Principles** *Annual Review of Neuroscience* https://doi.org/10.1146/annurev-neuro-062111-150533

Wilson RI (2004) **Transformation of Olfactory Representations in the Drosophila Antennal Lobe** *Science* https://doi.org/10.1126/science.1090782

Wilson RI, Laurent G (2005) **Role of GABAergic inhibition in shaping odor-evoked spatiotemporal patterns in the Drosophila antennal lobe** *J Neurosci* **25**:9069–9079 https://doi.org/10.1523/JNEUROSCI.2070-05.2005

Wilson RI, Turner GC, Laurent G (2004) **Transformation of olfactory representations in the Drosophila antennal lobe** *Science* **303**:366–370 https://doi.org/10.1126/science.1090782

Wu JS, Luo L (2006) **A protocol for dissecting Drosophila melanogaster brains for live imaging or immunostaining** *Nat Protoc* **1**:2110–2115 https://doi.org/10.1038/nprot.2006.336

Xie Q *et al.* (2021) **Temporal evolution of single-cell transcriptomes of Drosophila olfactory projection neurons** *eLife* **10** https://doi.org/10.7554/eLife.63450

Yaksi E, Wilson RI (2010) **Electrical coupling between olfactory glomeruli** *Neuron* **67**:1034–1047 https://doi.org/10.1016/j.neuron.2010.08.041

Zocher S, Schilling S, Grzyb AN, Adusumilli VS, Bogado Lopes J, Günther S, Overall RW, Winter Y, Kempermann G (2020) **Early-life environmental enrichment generates persistent individualized behavior in mice** *Sci Adv* **6** https://doi.org/10.1126/sciadv.abb1478

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Joint Public Review:

Summary:

The authors aimed to identify the neural sources of behavioral variation in fruit flies deciding between odor and air, or between two odors.

Strengths:

- The question is of fundamental importance.
- The behavioral studies are automated, and high-throughput.
- The data analyses are sophisticated and appropriate.
- The paper is clear and well-written aside from some initially strong wording.
- The figures beautifully illustrate their results.
- The modeling efforts mechanistically ground observed data correlations.

Weaknesses:

-The correlations between behavioral variations and neural activity/synapse morphology are relatively weak, and sometimes overstated in the wording that describes them.

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Author response:

The following is the authors’ response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *The authors seek to establish what aspects of nervous system structure and function may explain behavioral differences across individual fruit flies. The behavior in question is a preference for one odor or another in a choice assay. The variables related to neural function are odor responses in olfactory receptor neurons or in the second-order projection neurons, measured via calcium imaging. A different variable related to neural structure is the density of a presynaptic protein BRP. The authors measure these variables in the same fly along with the behavioral bias in the odor assays. Then they look for correlations across flies between the structure-function data and the behavior.*
>
> *Strengths:*
>
> *Where behavioral biases originate is a question of fundamental interest in the field. In an earlier paper (Honegger 2019) this group showed that flies do vary with regard to odor preference, and that there exists neural variation in olfactory circuits, but did not connect the two in the same animal. Here they do, which is a categorical advance, and opens the door to establishing a correlation. The authors inspect many such possible correlations. The underlying experiments reflect a great deal of work, and appear to be done carefully. The reporting is clear and transparent: All the data underlying the conclusions are shown, and associated code is available online.*

We are glad to hear the reviewer is supportive of the general question and approach.

> *Weaknesses:*
>
> *The results are overstated. The correlations reported here are uniformly small, and don't inspire confidence that there is any causal connection. The main problems are*

Our revision overhauls the interpretation of the results to prioritize the results we have high confidence in (specifically, PC 2 of our Ca++ data as a predictor of OCT-MCH preference) versus results that are suggestive but not definitive (such as PC 1 of Ca++ data as a predictor of Air-OCT preference).

It’s true that the correlations are small, with R2 values typically in the 0.1-0.2 range. That said, we would call it a victory if we could explain 10 to 20% of the variance of a behavior measure, captured in a 3 minute experiment, with a circuit correlate. This is particularly true because, as the reviewer notes, the behavioral measurement is noisy.

> *(1) The target effect to be explained is itself very weak. Odor preference of a given fly varies considerably across time. The systematic bias distinguishing one fly from another is small compared to the variability. Because the neural measurements are by necessity separated in time from the behavior, this noise places serious limits on any correlation between the two.*

This is broadly correct, though to quibble, it’s our *measurement of odor preference* which varies considerably over time. We are reasonably confident that more variance in our measurements can be attributed to sampling error than changes to true preference over time. As evidence, the correlation in sequential measures of individual odor preference, with delays of 3 hours or 24 hours, are not obviously different. We are separately working on methodological improvements to get more precise estimates of persistent individual odor preference, using averages of multiple, spaced measurements. This is promising, but beyond the scope of this study.

> *(2) The correlations reported here are uniformly weak and not robust. In several of the key figures, the elimination of one or two outlier flies completely abolishes the relationship. The confidence bounds on the claimed correlations are very broad. These uncertainties propagate to undermine the eventual claims for a correspondence between neural and behavioral measures.*

We are broadly receptive to this criticism. The lack of robustness of some results comes from the fundamental challenge of this work: measuring behavior is noisy at the individual level. Measuring Ca++ is also somewhat noisy. Correlating the two will be underpowered unless the sample size is huge (which is impractical, as each data point requires a dissection and live imaging session) or the effect size is large (which is generally not the case in biology). In the current version we tried in some sense to avoid discussing these challenges head-on, instead trying to focus on what we thought were the conclusions justified by our experiments with sample sizes ranging from 20 to 60. Our revision is more candid about these challenges.

That said, we believe the result we view as the most exciting — that PC2 of Ca++ responses predicts OCT-MCH preference — is robust. 1) It is based on a training set with 47 individuals and a test set composed of 22 individuals. The p-value is sufficiently low in each of these sets (0.0063 and 0.0069, respectively) to pass an overly stringent Bonferroni correction for the 5 tests (each PC) in this analysis. 2) The BRP immunohistochemistry provides independent evidence that is consistent with this result — PC2 that predicts behavior ($p = 0.03$ from only one test) and has loadings that contrast DC2 and DM2. Taken together, these results are well above the field-standard bar of statistical robustness.

In our revision, we are explicit that this is the (one) result we have high confidence in. We believe this result convincingly links Ca++ and behavior, and warrants spotlighting. We have less confidence in other results, and say so, and we hope this addresses concerns about overstating our results.

> *(3) Some aspects of the statistical treatment are unusual. Typically a model is proposed for the relationship between neuronal signals and behavior, and the model predictions are correlated with the actual behavioral data. The normal practice is to train the model on part of the data and test it on another part. But here the training set at times includes the testing set, which tends to give high correlations from overfitting. Other times the testing set gives much higher correlations than the training set, and then the results from the testing set are reported. Where the authors explored many possible relationships, it is unclear whether the significance tests account for the many tested hypotheses. The main text quotes the key results without confidence limits.*

Our primary analyses are exactly what the reviewer describes, scatter plots and correlations of actual behavioral measures against predicted measures. We produced test data in separate experiments, conducted weeks to months after models were fit on training data. This is more rigorous than splitting into training and test sets data collected in a single session, as batch/environmental effects reduce the independence of data collected within a single session.

We only collected a test set when our training set produced a promising correlation between predicted and actual behavioral measures. We never used data from test sets to train models.

In our main figures, we showed scatter plots that combined test and training data, as the training and test partitions had similar correlations.

We are unsure what the reviewer means by instances where we explored many possible relationships. The greatest number of comparisons that could lead to the rejection of a null hypothesis was 5 (corresponding to the top 5 PCs of Ca++ response variation or Brp signal). We were explicit that the p-values reported were nominal. As mentioned above, applying a Bonferroni correction for n=5 comparisons to either the training or test correlations from the Ca++ to OCT-MCH preference model remains significant at alpha=0.05.

Our revision includes confidence intervals around ρ*signal* for the PN PC2 OCT-MCH model, and for the ORN Brp-Short PC2 OCT-MCH model (lines 170-172, 238)

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *The authors aimed to identify the neural sources of behavioral variation in a decision between odor and air, or between two odors.*
>
> *Strengths:*
>
> *-The question is of fundamental importance.*
>
> *-The behavioral studies are automated, and high-throughput.*
>
> *-The data analyses are sophisticated and appropriate.*
>
> *-The paper is clear and well-written aside from some strong wording.*
>
> *-The figures beautifully illustrate their results.*
>
> *-The modeling efforts mechanistically ground observed data correlations.*

We are glad to read that the reviewer sees these strengths in the study. We hope the current revision addresses the strong wording.

> *Weaknesses:*
>
> *-The correlations between behavioral variations and neural activity/synapse morphology are (i) relatively weak, (ii) framed using the inappropriate words "predict", "link", and "explain", and (iii) sometimes non-intuitive (e.g., PC 1 of neural activity).*

Taking each of these points in turn:

i) It would indeed be nicer if our empirical correlations are higher. One quibble: we primarily report relatively weak correlations between *measurements* of behavior and Ca++/Brp. This could be the case even when the correlation between true behavior and Ca++/Brp is higher. Our analysis of the potential correlation between latent behavioral and Ca++ signals was an attempt to tease these relationships apart. The analysis suggests that there could, in fact, be a high underlying correlation between behavior and these circuit features (though the error bars on these inferences are wide).

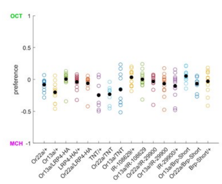
ii) We worked to ensure such words are used appropriately. “Predict” can often be appropriate in this context, as a model predicts true data values. Explain can also be appropriate, as X “explaining” a portion of the variance of Y is synonymous with X and Y being correlated. We cannot think of formal uses of “link,” and have revised the manuscript to resolve any inappropriate word choice.

iii) If the underlying biology is rooted in non-intuitive relationships, there’s unfortunately not much we can do about it. We chose to use PCs of our Ca++/Brp data as predictors to deal with the challenge of having many potential predictors (odor-glomerular responses) and relatively few output variables (behavioral bias). Thus, using PCs is a conservative approach to deal with multiple comparisons. Because PCs are just linear transformations of the original data, interpreting them is relatively easy, and in interpreting PC1 and PC2, we were able to identify simple interpretations (total activity and the difference between DC2 and DM2 activation, respectively). All in all, we remain satisfied with this approach as a means to both 1) limit multiple comparisons and 2) interpret simple meanings from predictive PCs.

> *No attempts were made to perturb the relevant circuits to establish a causal relationship between behavioral variations and functional/morphological variations.*

We did conduct such experiments, but we did not report them because they had negative results that we could not definitively interpret. We used constitutive and inducible effectors to alter the physiology of ORNs projecting to DC2 and DM2. We also used UAS-LRP4 and UAS-LRP4-RNAi to attempt to increase and decrease the extent of Brp puncta in ORNs projecting to DC2 and DM2. None of these manipulations had a significant effect on mean odor preference in the OCT-MCH choice, which was the behavioral focus of these experiments. We were unable to determine if the effectors had the intended effects in the targeted Gal4 lines, particularly in the LRP experiments, so we could not rule out that our negative finding reflected a technical failure.

Author response image 1.



We believe that even if these negative results are not technical failures, they are not necessarily inconsistent with the analyses correlating features of DC2 and DM2 to behavior. Specifically, we suspect that there are correlated fluctuations in glomerular Ca++ responses and Brp across individuals, due to fluctuations in the developmental spatial patterning of the antennal lobe. Thus, the DC2-DM2 predictor may represent a slice/subset of predictors distributed across the antennal lobe. This would also explain how we “got lucky” to find two glomeruli as predictors of behavior, when we were only able to image a small portion of the glomeruli.

> ***Reviewer #3 (Public Review):***
>
> *Churgin et. al. seeks to understand the neural substrates of individual odor preference in the Drosophila antennal lobe, using paired behavioral testing and calcium imaging from ORNs and PNs in the same flies, and testing whether ORN and PN odor responses can predict behavioral preference. The manuscript's main claims are that ORN activity in response to a panel of odors is predictive of the individual's preference for 3-octanol (3-OCT) relative to clean air, and that activity in the projection neurons is predictive of both 3-OCT vs. air preference and 3-OCT vs. 4-methylcyclohexanol (MCH). They find that the difference in density of fluorescently-tagged brp (a presynaptic marker) in two glomeruli (DC2 and DM2) trends towards predicting behavioral preference between 3-oct vs. MCH. Implementing a model of the antennal lobe based on the available connectome data,*

> *they find that glomerulus-level variation in response reminiscent of the variation that they observe can be generated by resampling variables associated with the glomeruli, such as ORN identity and glomerular synapse density.*
>
> *Strengths:*
>
> *The authors investigate a highly significant and impactful problem of interest to all experimental biologists, nearly all of whom must often conduct their measurements in many different individuals and so have a vested interest in understanding this problem. The manuscript represents a lot of work, with challenging paired behavioral and neural measurements.*
>
> *Weaknesses:*
>
> *The overall impression is that the authors are attempting to explain complex, highly variable behavioral output with a comparatively limited set of neural measurements.*

We would say that we are attempting to explain a simple, highly variable behavioral measure with a comparatively limited set of neural measurements, i.e. we make no claims to explain the complex behavioral components of odor choice, like locomotion, reversals at the odor boundary, etc.

> *Given the degree of behavioral variability they observe within an individual (Figure 1-supp 1) which implies temporal/state/measurement variation in behavior, it's unclear that their degree of sampling can resolve true individual variability (what they call "idiosyncrasy") in neural responses, given the additional temporal/state/measurement variation in neural responses.*

We are confident that different Ca++ recordings are statistically different. This is borne out in the analysis of repeated Ca++ recordings in this study, which finds that the significant PCs of Ca++ variation contain 77% of the variation in that data. That this variation is persistent over time and across hemispheres was assessed in Honegger & Smith, et al., 2019. We are thus confident that there is true individuality in neural responses (Note, we prefer not to call it "individual variability" as this could refer to variability within individuals, not variability across individuals.) It is a separate question of whether individual differences in neural responses bear some relation to individual differences in behavioral biases. That was the focus of this study, and our finding of a robust correlation between PC 2 of Ca++ responses and OCT-MCH preference indicates a relation. Because behavior and Ca++ were collected with an hours-to-day long gap, this implies that there are latent versions of both behavioral bias and Ca++ response that are stable on timescales at least that long.

> *The statistical analyses in the manuscript are underdeveloped, and it's unclear the degree to which the correlations reported have explanatory (causative) power in accounting for organismal behavior.*

With respect, we do not think our statistical analyses are underdeveloped, though we acknowledge that the detailed reviewer suggestions included the helpful suggestion to include uncertainty in the estimation of confidence intervals around the point estimate of the strength of correlation between latent behavioral and Ca++ response states – we have added these for the PN PC2 linear model (lines 170-172).

It is indeed a separate question whether the correlations we observed represent causal links from Ca++ to behavior (though our yoked experiment suggests there is not a behavior-to-Ca++ causal relationship — at least one where odor experience through behavior is an upstream cause). We attempted to be precise in indicating that our observations are correlations. That is why we used that word in the title, as an example. In the revision, we worked to ensure this is appropriately reflected in all word choice across the paper.

> ***Recommendations for the Authors:***
>
> ***Reviewer #1 (Recommendations for the Authors):***
>
> *Detailed comments: Many of the problems can be identified starting from Figure 4, which summarizes the main claims. I will focus on that figure and its tributaries.*

Acknowledging that the strength of several of our inferences are weak compared to what we consider the main result (the relationship between PC2 of Ca++ and OCT-MCH preference),we have removed Figure 4. This makes the focus of the paper much clearer and appropriately puts focus on the results that have strong statistical support.

> *(1) The process of "inferring" correlation among the unobserved latent states for neural sensitivity and behavioral bias is unconventional and risky. The larger the assumed noise linking the latent to the observed variables (i.e. the smaller r_b and r_c) the bigger the inferred correlation rho from a given observed correlation R^2_cb. In this situation, the value of the inferred rho becomes highly dependent on what model one assumes that links latent to observed states. But the specific model drawn in Fig 4 suppl 1 is just one of many possible guesses. For example, models with nonlinear interactions could produce different inference.*

We agree with the reviewer's notes of caution. To be clear, we do not intend for this analysis to be the main takeaway of the paper and have revised it to make this clear. The signal we are most confident in is the simple correlation between measured Ca++ PC2 and measured behavior. We have added more careful language saying that the attempt to infer the correlation between latent signals is one attempt at describing the data generation process (lines 166-172), and one possible estimate of an "underlying" correlation.

> *(2) If one still wanted to go through with this inference process and set confidence bounds on rho, one needs to include all the uncertainties. Here the authors only include uncertainty in the value of R^2_c,b and they peg that at +/-20% (Line 1367). In addition there is plenty of uncertainty associated also with R^2_c,c and R^2_b,b. This will propagate into a wider confidence interval on rho.*

We have replaced the arbitrary +/- 20% window with bootstrapping the pairs of (predicted preference by PN PC2, measured preference) points and getting a bootstrap distribution of R2c,b, which is, not surprisingly, considerably wider. Still, we think there is some value in this analysis as the 90% CI of ϱsignal under this model is 0.24-0.95. That is, including uncertainty about the R2b,b and R2c,c in the model still implies a significant relationship between latent calcium and behavior signals.

> *(2.1) The uncertainty in R^2_cb is much greater than +/-20%. Take for example the highest correlation quoted in Fig 4: R^2=0.23 in the top row of panel A. This relationship refers to Fig 1L. Based on bootstrapping from this data set, I find a 90% confidence interval of CI=[0.002, 0.527]. That's an uncertainty of -100/+140%, not +/-20%. Moreover, this correlation is due entirely to the lone outlier on the bottom left. Removing that single fly abolishes any correlation in the data (R^2=0.04, p>0.3). With that the correlation of rho=0.64, the second-largest effect in Fig 4, disappears.*

We acknowledge that removal of the outlier in Fig 1L abolishes the correlation between predicted and measured OCT-AIR preference. We have thus moved that subfigure to the supplement (now Figure 1 – figure supplement 10B), note that we do not have robust statistical support of ORN PC1 predicting OCT-AIR preference in the results (lines 177-178), and place our emphasis on PN PC2's capacity to predict OCT-MCH preference throughout the text.

> *(2.2) Similarly with the bottom line of Fig 4A, which relies on Fig 1M. With the data as plotted, the confidence interval on R^2 is CI=[0.007, 0.201], again an uncertainty of -100/+140%. There are two clear outlier points, and if one removes those, the correlation disappears entirely (R^2=0.06, p=0.09).*

We acknowledge that removal of the two outliers in Fig 1M between predicted and measured OCT-AIR preference abolishes the correlation. We have also moved that subfigure to the supplement (now Figure 1 – figure supplement 10F) and do not claim to have robust statistical support of PN PC1 predicting OCT-AIR preference.

> *(2.3) Similarly, the correlation R^2_bb of behavior with itself is weak and comes with great uncertainty (Fig 1 Suppl 1, panels B-E). For example, panel D figures prominently in computing the large inferred correlation of 0.75 between PN responses and OCT-MCH choice (Line 171ff). That correlation is weak and has a very wide confidence interval CI=[0.018, 0.329]. This uncertainty about R^2_bb should be taken into account when computing the likelihood of rho.*

We now include bootstrapping of the 3 hour OCT-MCH persistence data in our inference of ϱsignal.

> *(2.4) The correlation R^2_cc for the empirical repeatability of Ca signals seems to be obtained by a different method. Fig 4 suppl 1 focuses on the repeatability of calcium recording at two different time points. But Line 625ff suggests the correlation R^2_cc=0.77 all derives from one time point. It is unclear how these are related.*

Because our calcium model predictors utilize principal components of the glomerulus-odor responses (the mean Δf/f in the odor presentation window), we compute R2c,c through adding variance explained along the PCs, up to the point in which the component-wise variance explained does not exceed that of shuffled data (lines 609-620 in Materials and Methods). In this revision we now bootstrap the calcium data on the level of individual flies to get a bootstrap distribution of R2c,c, and propagate the uncertainty forward in the inference of ϱsignal.

> *(2.5) To summarize, two of the key relationships in Fig 1 are due entirely to one or two outlier points. These should not even be used for further analysis, yet they underlie two of the claims in Fig 4. The other correlations are weak, and come with great uncertainty, as confirmed by resampling. Those uncertainties should be propagated through the inference procedure described in Fig 4. It seems possible that the result will be entirely uninformative, leaving rho with a confidence interval that spans the entire available range [0,1]. Until that analysis is done, the claims of neuron-to-behavior correlation in this manuscript are not convincing.*

It is important to note that we never thought our analysis of the relationship between latent behavior and calcium signals should be interpreted as the main finding. Instead, the observed correlation between measured behavior and calcium is the take-away result. Importantly, it is also conservative compared to the inferred latent relationship, which in our minds was always a "bonus" analysis. Our revisions are now focused on highlighting the correlations between measured signals that have strong statistical support.

As a response to these specific concerns, we have propagated uncertainty in all R2's (calcium-calcium, behavior-behavior, calcium-behavior) in our new inference for ϱsignal, yielding a new median estimate for PN PC 2 underlying OCT-MCH preference of 0.68, with a 90% CI of 0.24-0.95. (Lines 171-172 in results, *Inference of correlation between latent calcium and behavior states* section in *Materials and Methods*).

> *(3) Other statistical methods:*
>
> *(3.1) The caption of Fig 4 refers to "model applied to train+test data". Does that mean the training data were included in the correlation measurement? Depending on the number of degrees of freedom in the model, this could have led to overfitting.*

We have removed Figure 4 and emphasize the key results in Figure 1 and 2 that we see statistically robust signal of PN PC 2 explaining OCT-MCH preference variation in both a training set and a testing set of flies (Fig 2 – figure supplement 1C-D).

> *(3.2) Line 180 describes a model that performed twice as well on test data (31% EV) as it did on training data (15%). What would explain such an outcome? And how does that affect one's confidence in the 31% number?*

The test set recordings were conducted several weeks after the training set recordings, which were used to establish PN PC 2 as a correlate of OCT-MCH preference. The fact that the test data had a higher R2 likely reflects sampling error (these two correlation coefficients are not significantly different). Ultimately this gives us more confidence in our model, as the predictive capacity is maintained in a totally separate set of flies.

> *(3.340 Multiple models get compared in performance before settling on one. For example, sometimes the first PC is used, sometimes the second. Different weighting schemes appear in Fig 2. Do the quoted p-values for the correlation plots reflect a correction for multiple hypothesis testing?*

For all calcium-behavior models, we restricted our analysis to 5 PCs, as the proportion of calcium variance explained by each of these PCs was higher than that explained by the respective PC of shuffled data — i.e., there were at most five significant PCs in that data. We thus performed at most 5 hypothesis tests for a given model. PN PC 2 explained 15% of OCT-MCH preference variation, with a p-value of 0.0063 – this p-value is robust to a conservative Bonferroni correction to the 5 hypotheses considered at alpha=0.05.

The weight schemes in Figure 2 and Figure 1 – figure supplement 10 reflect our interpretations of the salient features of the PCs and are follow-up analysis of the single principal component hypothesis tests. Thus they do not constitute additional tests that should be corrected. We now state in the methods explicitly that all reported p-values are nominal (line 563).

> *(3.4) Line 165 ff: Quoting rho without giving the confidence interval is misleading. For example, the rho for the presynaptic density model is quoted as 0.51, which would be a sizeable correlation. But in fact, the posterior on rho is almost flat, see caption of Fig 4 suppl 1, which lists the CI as [0.11, 0.85]. That means the experiments place virtually no constraint on rho. If the authors had taken no data at all, the posterior on rho would be uniform, and give a median of 0.5.*

We now provide a confidence interval around ϱsignal for the PN PC 2 model (lines 170-172). But per above, and consistent with the new focus of this revision, we view the ϱsignal inference as secondary to the simple, significant correlation between PN PC 2 and OCT-MCH preference.

> *(4) As it stands now, this paper illustrates how difficult it is to come to a strong conclusion in this domain. This may be worth some discussion. This group is probably in a better position than any to identify what are the limiting factors for this kind of research.*

We thank the reviewer for this suggestion and have added discussion of the difficulties in detecting signals for this kind of problem. That said, we are confident in stating that there is a meaningful correlation between PC 2 of PN Ca++ responses and OCT-MCH behavior given our model's performance in predicting preference in a test set of flies, and in the consistent signal in ORN Bruchpilot.

> ***Reviewer #3 (Recommendations for the Authors):***
>
> *Two major concerns, one experimental/technical and one conceptual:*
>
> *(1) I appreciate the difficulty of the experimental design and problem. However, the correlations reported throughout are based on neural measurements in only 5 glomeruli (~10% of the olfactory system) at early stages of olfactory processing.*

We acknowledge that only imaging 5 glomeruli is regrettable. We worked hard to develop image analysis pipelines that could reliably segment as many glomeruli as possible from almost all individual flies. In the end, we concluded that it was better to focus our analysis on a (small) core set of glomeruli for which we had high confidence in the segmentation. Increasing the number of analyzed glomeruli is high on the list of improvements for subsequent studies. Happily, we are confident that we are capturing a significant, biologically meaningful correlation between PC 2 of PN calcium (dominated by the responses in DC2 and DM2) and OCT-MCH preference.

> *3-OCT and MCH activate many glomeruli in addition to the five studied, especially at the concentrations used. There is also limited odor-specificity in their response matrix: notably responses are more correlated in all glomeruli within an individual, compared to responses across individuals (they note this in lines 194-198, though I don't quite understand the specific point they make here). This is a sign of high experimental variability (typically the dynamic range of odor response within an individual is similar to the range across individuals) and makes it even more difficult to resolve underlying individual variation.*

We respectfully disagree with the reviewer's interpretation here. There is substantial odor-specificity in our response matrix. This is evident in both the ORN and PN response matrices (and especially the PN matrix) as variation in the brightness across rows. Columns, which correspond to individuals, are more similar than rows, which correspond to odor-glomerulus pairs. The dynamic range within an individual (within a column, across rows) is indeed greater than the variation among individuals (within a row, across columns).

> *As an (important) aside, the odor stimuli are very unusual in this study. Odors are delivered at extremely high concentrations (variably 10-25% sv, line 464, not exactly sure what "variably' means- is the stimulus intensity not constant?) as compared to even the highest concentrations used in >95% of other studies (usually <~0.1% sv delivered).*

We used these concentrations for a variety of reasons. First, following the protocol of Honegger and Smith (2020), we found that dilutions in this range produce a linear input-output relationship, i.e. doubling or halving one odorant yields proportionate changes in odor-choice behavior metrics. Second, such fold dilutions are standard for tunnel assays of the kind we used. Claridge-Chang et al. (2009) used 14% and 11% for MCH and OCT respectively, for instance. Finally, the specific dilution factor (i.e., within the range of 10-25%) was adjusted on a week-by-week basis to ensure that in an OCT-MCH choice, the mean preference was approximately 50%. This yields the greatest signal of individual odor preference. We have added this last point to the methods section where the range of dilutions is described (lines 442-445).

> *A parsimonious interpretation of their results is that the strongest correlation they see (ORN PC1 predicts OCT v. air preference) arises because intensity/strength of ORN responses across all odors (e.g. overall excitability of ORNs) partially predicts behavioral avoidance of 3-OCT. However, the degree to which variation in odor-specific glomerular activation patterns can explain behavioral preference (3-OCT v. MCH) seems much less clear, and correspondingly the correlations are weaker and p-values larger for the 3-OCT v. MCH result.*

With respect, we disagree with this analysis. The correlation between ORN PC 1 and OCT v. air preference (R2 = 0.23) is quite similar to that of PN PC 2 and OCT vs MCH preference (R2 = 0.20). However, the former is dependent on a single outlying point, whereas the latter is not. The latter relationship is also backed up by the BRP imaging and modeling. Therefore in the revision we have de-emphasized the OCT v. air preference model and emphasized the OCT v. MCH preference models.

> *(2) There is a broader conceptual concern about the degree of logical consistency in the authors' interpretation of how neural variability maps to behavioral variability. For instance, the two odors they focus on, 3-OCT and MCH, barely activate ORNs in 4 of the 5 glomeruli they study. Most of the correlation of ORN PC1 vs. behavioral choice for 3-OCT vs. air, then, must be driven by overall glomerular activation by other odors (but remains predictive since responses across odors appear correlated within an individual). This gives pause to the interpretation that 3-OCT-evoked ORN activity in these five glomeruli is the neural substrate for variability in the behavioral response to 3-OCT.*

Our interpretation of the ORN PC1 linear model is not that 3-OCT-evoked ORN activity is the neural substrate for variability – instead, it is the general responsiveness of an individual's AL across multiple odors (this is our interpretation of the the uniformly positive loadings in ORN PC1). It is true that OCT and MCH do not activate ORNs as strongly as other odorants – our analysis rests on the loadings of the PCs that capture all odor/glomerulus combinations available in our data. All that said, since a single outlier in Figure 1L dominates the relationship, therefore we have de-emphasized these particular results in our revision.

> *This leads to the most significant concern, which is that the paper does not provide strong evidence that odor-specific patterns of glomerular activation in ORNs and PNs underlie individual behavioral preference between different odors (that each drive significant levels of activity, e.g. 3-OCT v. MCH), or that the ORN-PN synapse is a major driver of individual behavioral variability. Lines 26-31 of the abstract are not well supported, and the language should be softened.*

We have modified the abstract to emphasize our confidence in PN calcium correlating with odor-vs-odor preference (removing the ORN & odor-vs-air language).

> *Their conclusions come primarily from having correlated many parameters reduced from the ORN and PN response matrices against the behavioral data. Several claims are made that a given PC is predictive of an odor preference while others are not, however it does not appear that the statistical tests to support this are shown in the figures or text.*

For each linear model of calcium dynamics predicting preference, we restricted our analysis to the first 5 principal components. Thus, we do not feel that we correlated **many** parameters against the behavioral data. As mentioned below, the correlations identified by this approach comfortably survive a conservative Bonferroni correction. In this revision, a linear model with a single predictor – the projection onto PC 2 of PN calcium – is the result we emphasize in the text, and we report R2 between measured and predicted preference for both a training set of flies and for a test set of flies (Figure 1M and Figure 2 – figure supplement 1).

> *That is, it appears that the correlation of models based on each component is calculated, then the component with the highest correlation is selected, and a correlation and p-value computed based on that component alone, without a statistical comparison between the predictive values of each component, or to account for effectively performing multiple comparisons. (Figure 1, k l m n o p, Figure 3, d f, and associated analyses).*

To reiterate, this was our process: 1) Collect a training data set of paired Ca++ recordings and behavioral preference scores. 2) Compute the first five PCs of the Ca++ data, and measure the correlation of each to behavior. 3) Identify the PC with the best correlation. 4) Collect a test data set with new experimental recordings. 5) Apply the model identified in step 3. For some downstream analyses, we combined test and training data, but only after confirming the separate significance of the training and test correlations.

The p-values associated with the PN PC 2 model predicting OCT-MCH preference are sufficiently low in each of the training and testing sets (0.0063 and 0.0069, respectively) to pass a conservative Bonferroni multiple hypothesis correction (one hypothesis for each of the 5 PCs) at an alpha of 0.05.

> *Additionally, the statistical model presented in Figure 4 needs significantly more explanation or should be removed- it's unclear how they "infer" the correlation, and the conclusions appears inconsistent with Figure 3 - Figure Supplement 2.*

We have removed Figure 4 and have improved upon our approach of inferring the strength of the correlation between latent calcium and behavior in the *Methods*, incorporating bootstrapping of all sources of data used for the inference (lines 622-628). At the same time, we now emphasize that this analysis is a bonus of sorts, and that the simple correlation between Ca++ and behavior is the main result.

> *Suggestions:*
>
> *(1) If the authors want to make the claim that individual variation in ORN or PN odor representations (e.g. glomerular activation patterns) underlie differences in odor preference (MCH v. OCT), they should generalize the weak correlation between ORN/PN activity and behavior to additional glomeruli and pair of odors, where both odors drive significant activity. Otherwise, the claims in the abstract should be tempered.*

We have modified the abstract to focus on the effect we have the highest confidence in: contrasting PN calcium activation of DM2 and DC2 predicting OCT-MCH preference.

> *(2) One of the most valuable contributions a study like this could provide is to carefully quantify the amount of measurement variation (across trials, across hemispheres) in neural responses relative to the amount of individual variation (across individuals). Beyond the degree of variation in the amplitude of odor responses, the rank ordering of odor response strength between repeated measurements (to try to establish conditions that account for adaptation, etc.), between hemispheres, and between individuals is*

> *important. Establishing this information is foundational to this entire field of study. The authors take a good first step towards this in Figure 1J and Figure 1, supplement 5C, but the plots do not directly show variance, and the comparison is flawed because more comparisons go into the individual-individual crunch (as evidenced by the consistently smaller range of quartiles). The proper way to do this is by resampling.*

We do not know what the reviewer means by “individual-individual crunch,” unfortunately. Thus, it is difficult to determine why they think the analysis is flawed. We are also uncertain about the role of resampling in this analysis. The medians, interquartile ranges and whiskers in the panels referenced by the reviewer are not confidence intervals as might be determined by bootstrap resampling. Rather, these are direct statistics on the coding distances as measured – the raw values associated with these plots are visualized in Figure 1H.

In our revision we updated the heatmaps in Figure 1 – figure supplement 3 to include recordings across the lobes and trials of each individual fly, and we have added a new supplementary figure, Figure 1 – figure supplement 4, to show the correspondence between recordings across lobes or trials, with associated rank-order correlation coefficients. Since the focus of this study was whether measured individual differences predict individual behavioral preference, a full characterization of the statistics of variation in calcium responses was not the focus, though it was the focus of a previous study (Honegger & Smith et al., 2019).

> *To help the reader understand the data, we would encourage displaying data prior to dimensionality reduction - why not show direct plots of the mean and variance of the neural responses in each glomerulus across repeats, hemispheres, individuals?*

We added a new supplementary figure, Figure 1 – figure supplement 4, to show the correspondence between recordings across lobes or trials.

> *A careful analysis of this point would allow the authors to support their currently unfounded assertion that odor responses become more "idiosyncratic" farther from the periphery (line 135-36); presumably they mean beyond just noise introduced by synaptic transmission, e.g. "idiosyncrasy" is reproducible within an individual. This is a strong statement that is not well-supported at present - it requires showing the degree of similarity in the representation between hemispheres is more similar within a fly than between flies in PNs compared to ORNs (see Hige... Turner, 2015).*

Here are the lines in question: “PN responses were more variable within flies, as measured across the left and right hemisphere ALs, compared to ORN responses (Figure 1 – figure supplement 5C), consistent with the hypothesis that odor representations become more idiosyncratic farther from the sensory periphery.”

That responses are more idiosyncratic farther from the periphery is therefore not an “unfounded assertion.” It is clearly laid out as a hypothesis for which we can assess consistency in the data. We stand by our original interpretation: that several observations are consistent with this finding, including greater distance in coding space in PNs compared to ORNs, particularly across lobes and across flies. In addition, higher accuracy in decoding individual identity from PN responses compared to ORN responses (now appearing as Figure 1 – figure supplement 6A) is also consistent with this hypothesis.

Still, to make confusion at this sentence less likely, we have reworded it as “suggesting that odor representations become more divergent farther from the sensory periphery.” (lines 139-140)

> *(3) Figure 3 is difficult to interpret. Again, the variability of the measurement itself within and across individuals is not established up front. Expression of exogenous tagged brp in ORNs is also not guaranteed to reflect endogenous brp levels, so there is an additional assumption at that level.*

Figure 3 – figure supplement 1 Panels A-C display the variability of measurements (Brp volume, total fluorescence and fluorescence density) both within (left/right lobes) and across individuals (the different data points). We agree that exogenous tagged Brp levels will not be identical to endogenous levels. The relationship appears significant despite this caveat.

> *Again there are statistical concerns with the correlations. For instance, the claim that "Higher Brp in DM2 predicted stronger MCH preference... " on line 389 is not statistically supported with p<0.05 in the ms (see Figure 3 G as the closest test, but even that is a test of the difference of DM2 and DC2, not DM2 alone).*

We have changed the language to focus on the pattern of the loadings in PC 2 of Brp-Short density and replaced "predict." (lines 366-369).

> *Can the authors also discuss what additional information is gained from the expansion microscopy in the figure supplement, and how it compares to brp density in DC2 using conventional methods?*

The expansion microscopy analysis was an attempt to determine what specific aspect of Brp expression was predictive of behavior, on the level of individual Brp puncta, as a finer look compared to the glomerulus-wide fluorescence signal in the conventional microscopy approach. Since this method did not yield a large sample size, at best we can say it provided evidence consistent with the observation from confocal imaging that Brp fluorescent density was the best measure in terms of predicting behavior.

> *I would prefer to see the calcium and behavioral datasets strengthened to better establish the relationship between ORN/PN responses and behavior, and to set aside the anatomical dataset for a future work that investigates mechanisms.*

We are satisfied that our revisions put appropriate emphasis on a robust result relating calcium and behavior measurements: the relationship between OCT-MCH preference and idiosyncratic PN calcium responses. Finding that idiosyncratic Brp density has similar PC 2 loadings that also significantly predict behavior is an important finding that increases confidence in the calcium-behavior finding. We agree with the reviewer that these anatomical findings are secondary to the calcium-behavior analyses, but think they warrant a place in the main findings of the study. As the reviewer suggests, we are conducting follow-on studies that focus on the relationship between neuroanatomical measures and odor preference.

> *(4) The mean imputation of missing data may have an effect on the conclusions that it is possible to draw from this dataset. In particular, as shown in Figure 1, supplemental figure 3, there is a relatively large amount of missing data, which is unevenly distributed across glomeruli and between the cell types recorded from. Strikingly, DC2 is missing in a large fraction of ORN recordings, while it is present in nearly all the PN recordings. Because DC2 is one of the glomeruli implicated in predicting MCH-OCT preference, this lack of data may be particularly likely to effect the evaluation of whether this preference can be predicted from the ORN data. Overall, mean imputation of glomerulus activity prior to PCA will artificially reduce the amount of variance contributed by the glomerulus.*

> *It would be useful to see an evaluation of which results of this paper are robust to different treatments of this missing data.*

We confirmed that the linear model of predicted OCT-MCH using PN PC2 calcium was minimally altered when we performed imputation via alternating least squares using the pca function with option 'als' to infill missing values on the calcium matrix 1000 times and taking the mean infilled matrix (see MATLAB documentation and Figure 1 – figure supplement 5 of Werkhoven et al., 2021). Fitted slope value for model using mean-infilled data presented in article: -0.0806 (SE = 0.028, model R2 = 0.15), fitted slope value using ALS-imputed model: -0.0806 (SE 0.026, model R2 = 0.17).

> *Additional comments:*
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> *(1) On line 255 there is an unnecessary condition: "non-negative positive".*

Thank you – non-negative has been removed.

> *(2) In Figure 4 and the associated analysis, selection of +/- 20% interval around the observed R^2 appears arbitrary. This could be based on the actual confidence interval, or established by bootstrapping.*

We have replaced the +/- 20% rule by bootstrapping the calculation of behavior-behavior R2, calcium-calcium R2, and calcium-behavior R2 and propagating the uncertainties forward (Inference of correlation between latent calcium and behavior states section in Materials and Methods).

> *(3) On line 409 the claim is made "These sources of variation specifically implicate the ORN-PN synapse..." While the model recapitulates the glomerulus specific variation of activity under PN synapse density variation, it also occurs under ORN identity variation, which calls into question whether the synapse distribution itself is specifically implicated, or if any variation that is expected to be glomerulus specific would be equally implicated.*

We agree with this observation. We found that varying either the ORNs or the PNs that project to each glomeruli can produce patterns of PN response variation similar to what is measured experimentally. This is consistent with the idea that the ORN-PN synapse is a key site of behaviorally-relevant variation.

> *(4) Line 214 "... we conclude that the relative responses of DM2 vs DC2 in PNs largely explains an individual's preference." is too strong of a claim, based on the fact that using the PC2 explains much more of the variance, while using the stated hypothesis noticeable decreases the predictive power (R^2 = 0.2 vs R^2 = 0.12)*

We have changed the wording here to "we conclude that the relative responses of DM2 vs DC2 in PNs compactly predict an individual's preference." (lines 192-193)

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